TOPIC 4 · NAVIER-STOKES EQUATIONS

Solved Problems: Official Collection NS1 to NS12

Worked solutions to the Navier-Stokes collection of the Department of Thermal and Fluid Engineering, Universidad Carlos III de Madrid. Every problem restates the official statement word for word, then follows four phases: (1) hypotheses, data, unknowns and degrees of freedom; (2) coordinate system, form of the fields and boundary conditions; (3) the derivation, with continuity and each Navier-Stokes component simplified term by term, the integrations written out and every constant fixed with its boundary condition; (4) interpretation, limiting cases and the dimensional check. Where a computed result differs from the handwritten official solution, the discrepancy is flagged in the problem; anything beyond the course material is marked as an extension. References to Notes.pdf are to Chapter 4 (The Navier-Stokes equations) and Chapter 3 (Conservation laws).

PROBLEM NS1

Porous Wall with Suction

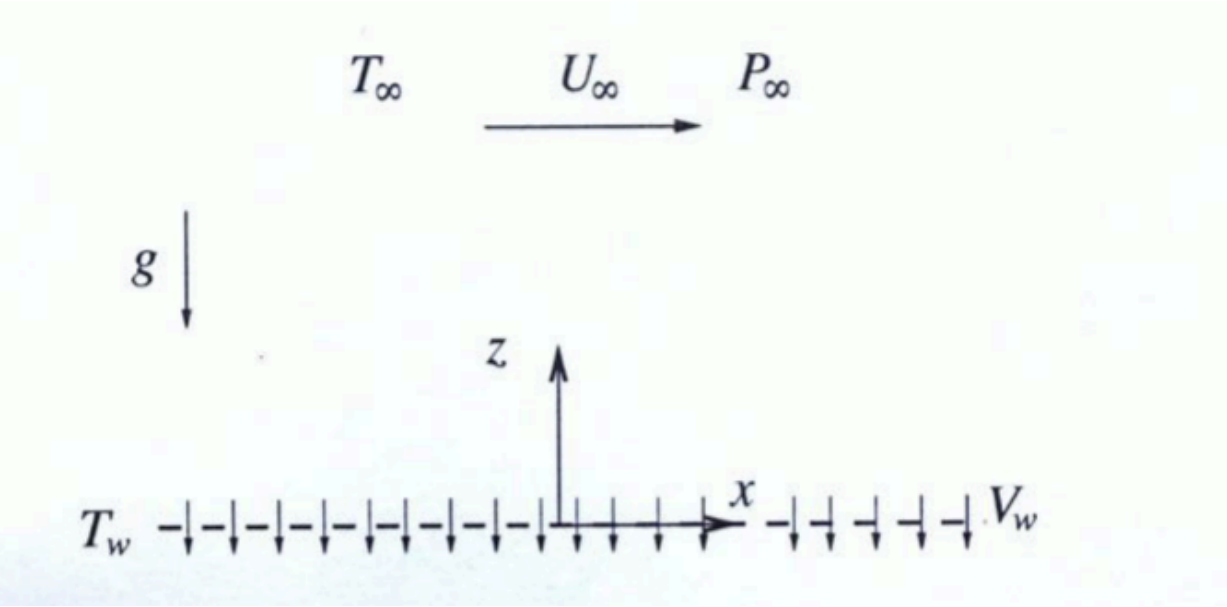
Original statement · NS1.pdf

Unidirectional flow | Porous wall | Viscous dissipation | Streamline

OFFICIAL PROBLEM STATEMENT

Consider the steady planar motion of a perfect liquid of density ρ, viscosity μ, thermal conductivity k and specific heat c in the vicinity of a horizontal porous wall. Far from the wall, the horizontal velocity component is $v_x = U_\infty$, the pressure is given by $p + \rho g z = P_\infty$ and the temperature is $T = T_\infty$, different from the value T_w at the wall. The liquid is sucked through the pores of the wall, resulting in a transverse velocity $v_z = -V_w$ at the wall surface. In the analysis, assume that the velocity and temperature fields are only a function of the distance to the wall z.

1. Determine the velocity components $v_x(z)$ and $v_z(z)$ and well as the pressure field $p(x, z)$.
2. Obtain the temperature field $T(z)$.
3. Calculate the streamline that crosses $x = x_o$ and $z = 0$.
4. Compute the viscous stress on the wall.
5. Determine the heat flux from the fluid to the wall.



Statement figure (crop of NS1.pdf, p. 1): horizontal porous wall at temperature T_w with suction velocity V_w, free stream U_∞, T_∞, P_∞, gravity g and axes x, z

`Source: NS1.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.7, incompressible momentum Eq. 4.12, perfect-liquid energy Eq. 4.42, no-slip and far-field conditions (Section on initial and boundary conditions).`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady planar flow of an incompressible liquid with constant ρ, μ, k, c (perfect liquid, Notes.pdf, Eq. 4.38): the equations to use are (4.40), (4.41) and (4.42).
- Gravity acts along $-\bar{e}_z$ (the far-field pressure is $p + \rho g z = P_\infty$, so $\bar{f}_m = -g\,\bar{e}_z$).
- Given: the fields depend only on the distance to the wall, $\partial/\partial x = \partial/\partial y = \partial/\partial t = 0$ for every variable, except that the pressure is allowed to depend on x (it is determined below).
- No chemical reaction and no radiation: $Q_c = Q_r = 0$.
- The wall is at rest and porous: the tangential velocity is zero (no-slip) and the normal velocity is the suction velocity.
- Data: $[\rho,\ \mu,\ k,\ c,\ U_\infty,\ P_\infty,\ T_\infty,\ T_w,\ V_w,\ g]$ with $V_w > 0$. Unknowns: $v_x(z), v_z(z)$, $p(x,z), T(z)$, the streamline, the wall stress and the wall heat flux.

- Degrees of freedom: two velocity components, the pressure and the temperature, each a function of one coordinate (plus x for the pressure): four unknown fields against continuity, two momentum components and the energy equation.

Symbol	Meaning	SI unit
$\rho, \mu, \nu = \mu/\rho$	density, dynamic and kinematic viscosity	kg/m^3, Pa s, m^2/s
$k, c, \alpha = k/(\rho c)$	thermal conductivity, specific heat, thermal diffusivity	W/(m K), J/(kg K), m^2/s
$\mathrm{Pr} = \nu/\alpha = \mu c/k$	Prandtl number	1
U_∞, V_w	free-stream velocity, suction velocity	m/s
P_∞, T_∞, T_w	far-field reduced pressure, far-field and wall temperatures	Pa, K, K

PHASE 2

Frames, Field Form and Boundary Conditions

- Cartesian frame (x, y, z) with x along the wall (in the direction of U_∞), z normal to it and pointing into the liquid, y out of the plane. The basis $\{\bar{e}_x, \bar{e}_y, \bar{e}_z\}$ is orthonormal and right-handed, so no change of basis is needed (det $R = 1$, $RR^T = I$ with $R = I$).
- Field form: $\bar{v} = v_x(z)\,\bar{e}_x + v_z(z)\,\bar{e}_z, T = T(z), p = p(x, z)$.
- The wall is at $z = 0$ and the liquid occupies $z > 0$.

Location	Condition	Origin
$z = 0$	$v_x = 0$	no-slip on the wall at rest
$z = 0$	$v_z = -V_w$	suction through the pores (statement)
$z = 0$	$T = T_w$	thermal equilibrium at the wall
$z \to \infty$	$v_x \to U_\infty$	free stream (statement)
$z \to \infty$	$p + \rho g z \to P_\infty$	free stream (statement)
$z \to \infty$	$T \to T_\infty$	free stream (statement)

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the normal velocity

$$\nabla \cdot \bar{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_z}{\partial z} = \underbrace{\cancel{\frac{\partial v_x}{\partial x}}}_{v_x = v_x(z)} + \frac{dv_z}{dz} = 0 \implies \frac{dv_z}{dz} = 0 \implies v_z = \text{cons} \quad (1)$$

The constant is fixed at the wall: $v_z(0) = -V_w$, therefore

$$\boxed{v_z(z) = -V_w} \quad (2)$$

3.2 Momentum equations

For $\bar{f}_m = -g\,\bar{e}_z$ the Cartesian components of (4.41) with constant μ are (Notes.pdf, Section on the Navier-Stokes equations in Cartesian coordinates). Every term that contains $\partial/\partial t$, $\partial/\partial x$ or $\partial/\partial y$ of a velocity component is zero.

z-component:

$$\rho\Big(\underbrace{\cancel{\frac{\partial v_z}{\partial t}}}_{\text{steady}} + \underbrace{\cancel{v_x \frac{\partial v_z}{\partial x}}}_{v_z = v_z(z)} + v_z \underbrace{\frac{dv_z}{dz}}_{=0 \text{ by } (1)}\Big) = -\frac{\partial p}{\partial z} + \mu \underbrace{\frac{d^2 v_z}{dz^2}}_{=0} - \rho g \implies \frac{\partial p}{\partial z} = -\rho g \quad (3)$$

Integrating (3) with respect to z at fixed x (the "constant" may depend on x):

$$p + \rho g z = f(x) \quad (4)$$

The far-field condition $p + \rho g z \to P_\infty$ holds for every x, with P_∞ a constant, so $f(x) = P_\infty$:

$$\boxed{p(x,z) = P_\infty - \rho g z}, \qquad \frac{\partial p}{\partial x} = 0 \tag{5}$$

x-component:

$$\rho\Big(\underbrace{\cancel{\frac{\partial v_x}{\partial t}}}_{\text{steady}} + \underbrace{\cancel{v_x\frac{\partial v_x}{\partial x}}}_{\partial_x = 0} + v_z\frac{dv_x}{dz}\Big) = -\underbrace{\frac{\partial p}{\partial x}}_{=0 \text{ by } (5)} + \mu\frac{d^2 v_x}{dz^2} \implies -\rho V_w \frac{dv_x}{dz} = \mu\frac{d^2 v_x}{dz^2} \tag{6}$$

Dividing by μ and using $\nu = \mu/\rho$:

$$\frac{d^2 v_x}{dz^2} + \frac{V_w}{\nu}\frac{dv_x}{dz} = 0 \tag{7}$$

This is a linear ODE with constant coefficients. Trying $v_x = e^{\lambda z}$ gives $\lambda^2 + (V_w/\nu)\lambda = 0$, so $\lambda = 0$ or $\lambda = -V_w/\nu$:

$$v_x = C_1 + C_2\, e^{-V_w z/\nu} \tag{8}$$

Boundary conditions: $v_x(\infty) = U_\infty$ gives $C_1 = U_\infty$ (the exponential vanishes because $V_w > 0$); $v_x(0) = 0$ gives $C_1 + C_2 = 0$, so $C_2 = -U_\infty$:

$$\boxed{v_x(z) = U_\infty\left(1 - e^{-V_w z/\nu}\right)} \tag{9}$$

3.3 Energy equation and the temperature field

With $e = e_o + cT$, $\bar{v} = v_x \bar{e}_x + v_z \bar{e}_z$ and $\partial/\partial x = \partial/\partial t = 0$, (4.42) reads

$$\rho c\, v_z \frac{dT}{dz} = \phi_v + k \frac{d^2T}{dz^2} \tag{10}$$

The viscous dissipation of this flow, from the Cartesian expression of ϕ_v (Notes.pdf, Section on the Navier-Stokes equations in Cartesian coordinates), has a single non-zero shear rate $\partial v_x/\partial z$:

$$\phi_v = \mu \left(\frac{dv_x}{dz} \right)^2, \qquad \frac{dv_x}{dz} = \frac{U_\infty V_w}{\nu}\, e^{-V_w z/\nu} \implies \phi_v = \frac{\mu U_\infty^2 V_w^2}{\nu^2}\, e^{-2V_w z/\nu} \tag{11}$$

Substituting $v_z = -V_w$ and (11) in (10) and dividing by k, with $\alpha = k/(\rho c)$:

$$\frac{d^2T}{dz^2} + \frac{V_w}{\alpha}\frac{dT}{dz} = -\frac{\mu U_\infty^2 V_w^2}{k\,\nu^2}\, e^{-2V_w z/\nu} \tag{12}$$

The solution is the sum of the homogeneous solution $B_1 + B_2 e^{-V_w z/\alpha}$ and a particular solution $A\, e^{-2V_w z/\nu}$. Substituting the particular form in (12), with $T_p' = -(2V_w/\nu) A\, e^{-2V_w z/\nu}$ and $T_p'' = (4V_w^2/\nu^2) A\, e^{-2V_w z/\nu}$:

$$A\left(\frac{4V_w^2}{\nu^2} - \frac{2V_w^2}{\alpha \nu} \right) = -\frac{\mu U_\infty^2 V_w^2}{k \nu^2} \implies A\,(4 - 2\,\mathrm{Pr}) = -\frac{\mu U_\infty^2}{k} = -\frac{\mathrm{Pr}\, U_\infty^2}{c} \tag{13}$$

where $\mathrm{Pr} = \nu/\alpha = \mu c/k$ was used. Hence (for $\mathrm{Pr} \neq 2$)

$$A = \frac{\mathrm{Pr}}{2(\mathrm{Pr} - 2)} \frac{U_\infty^2}{c} \tag{14}$$

so that $T = B_1 + B_2 e^{-V_w z/\alpha} + A\, e^{-2V_w z/\nu}$. The condition $T(\infty) = T_\infty$ gives $B_1 = T_\infty$ and $T(0) = T_w$ gives $B_2 = T_w - T_\infty - A$:

$$\boxed{T(z) = T_\infty + \left(T_w - T_\infty - \frac{\mathrm{Pr}\, U_\infty^2}{2(\mathrm{Pr}-2)\, c}\right) e^{-V_w z/\alpha} + \frac{\mathrm{Pr}\, U_\infty^2}{2(\mathrm{Pr}-2)\, c}\, e^{-2V_w z/\nu}} \tag{15}$$

Note that $V_w z/\alpha = \mathrm{Pr}\, V_w z/\nu$. For $\mathrm{Pr} = 2$ the two exponents coincide and the particular solution takes the form $A z\, e^{-2V_w z/\nu}$ (a resonant case, not worked out in the sheet).

3.4 Streamline through $(x_o, 0)$

The streamlines are the integral curves of $\bar{v}$: $dx/v_x = dz/v_z$. With (2) and (9):

$$\frac{dx}{dz} = \frac{v_x}{v_z} = -\frac{U_\infty}{V_w}\left(1 - e^{-V_w z/\nu}\right) \tag{16}$$

Integrating from $z' = 0$ (where $x = x_o$) to $z' = z$. The antiderivative of $1 - e^{-az'}$ with $a = V_w/\nu$ is $z' + e^{-az'}/a$, and Barrow's rule gives $\left[z' + e^{-az'}/a\right]_0^z = z + (e^{-az} - 1)/a$:

$$\boxed{x - x_o = -\frac{U_\infty}{V_w}\left[z + \frac{\nu}{V_w}\left(e^{-V_w z/\nu} - 1\right)\right]} \tag{17}$$

3.5 Viscous stress on the wall

The wall is the surface $z = 0$. Take the unit normal pointing into the liquid, $\bar{n} = \bar{e}_z$: then $\bar{\bar{\tau}}' \cdot \bar{n}$ is the viscous force per unit area that the liquid exerts on the wall. The components of $\bar{\bar{\tau}}'$ with the Navier-Poisson law (Notes.pdf, Section on stress components, Cartesian coordinates) are

$$\bar{\bar{\tau}}' \cdot \bar{e}_z = \tau'_{xz}\, \bar{e}_x + \tau'_{zz}\, \bar{e}_z, \qquad \tau'_{xz} = \mu\left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x}\right) = \mu \frac{dv_x}{dz}, \qquad \tau'_{zz} = 2\mu \frac{dv_z}{dz} + \left(\mu_B - \tfrac{2}{3}\mu\right) \nabla \cdot \bar{v} = 0 \tag{18}$$

At the wall, from (9), $dv_x/dz\big|_{z=0} = U_\infty V_w/\nu$, so

$$\boxed{\bar{\bar{\tau}}' \cdot \bar{n}\Big|_w = \mu \frac{U_\infty V_w}{\nu}\, \bar{e}_x = \rho V_w U_\infty\, \bar{e}_x} \tag{19}$$

The liquid drags the wall in the direction of the free stream.

3.6 Heat flux from the liquid to the wall

The heat flux from the liquid to the wall is $\bar{q} \cdot \bar{n}$ with $\bar{n} = -\bar{e}_z$ (outward from the liquid, into the wall), and $\bar{q} = -k\nabla T$ (Notes.pdf, Eq. 4.36):

$$\bar{q} \cdot \bar{n}\Big|_w = -k \frac{dT}{dz} \bar{e}_z \cdot (-\bar{e}_z)\Big|_{z=0} = k \frac{dT}{dz}\Big|_{z=0} \tag{20}$$

Differentiating (15) and evaluating at $z = 0$:

$$\frac{dT}{dz}\Big|_0 = -\frac{V_w}{\alpha} B_2 - \frac{2V_w}{\nu} A = -\frac{V_w}{\nu} \left(\mathrm{Pr}\, B_2 + 2A\right) \tag{21}$$

Using $k/\nu = \rho c/\mathrm{Pr}$ and $B_2 = T_w - T_\infty - A$:

$$k \frac{dT}{dz}\Big|_0 = -\rho c V_w \left(B_2 + \frac{2A}{\mathrm{Pr}}\right) = -\rho c V_w \left[T_w - T_\infty - A\,\frac{\mathrm{Pr} - 2}{\mathrm{Pr}}\right] = -\rho c V_w \left[T_w - T_\infty - \frac{U_\infty^2}{2c}\right] \tag{22}$$

$$\boxed{\bar{q} \cdot \bar{n}\Big|_w = \rho c V_w \left[T_\infty - T_w + \frac{U_\infty^2}{2c}\right]} \tag{23}$$

The result does not depend on the Prandtl number.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** Suction pins the shear layer to the wall: the velocity reaches 99% of U_∞ at $z \approx 4.6\,\nu/V_w$ and the thickness ν/V_w does not grow with x. The liquid is pulled towards the wall by the pores, and the viscous diffusion away from the wall balances this convection (equation (6)).
- **Limits.** For $V_w \to 0$ the layer thickens without bound, consistent with the absence of a steady unidirectional solution. For $z \to \infty$, $v_x \to U_\infty$ and $T \to T_\infty$. At $z = 0$, $v_x = 0$ and $T = T_w$.
- **Dimensional check.** $V_w z/\nu$ and $V_w z/\alpha$ are dimensionless. In (23), $\rho c V_w\,\Delta T$ has units $(\mathrm{kg/m^3})(\mathrm{J/(kg\,K)})(\mathrm{m/s})\,\mathrm{K} = \mathrm{W/m^2}$, and $U_\infty^2/(2c)$ has units $(\mathrm{m^2/s^2})/(\mathrm{J/(kg\,K)}) = \mathrm{K}$. In (19), $\rho V_w U_\infty$ has units $\mathrm{kg/(m^2\,s)} \cdot \mathrm{m/s} = \mathrm{N/m^2}$.
- **Energy check.** Equation (23) is the heat that must flow into the wall in steady state: per unit area the suction brings in from the free stream the energy $\rho V_w (c T_\infty + U_\infty^2/2)$ and the pores take out $\rho V_w\, c T_w$; the pressure work and the work of gravity cancel, and the shear does no work on the wall because it is at rest. The difference is $\rho c V_w [T_\infty - T_w + U_\infty^2/(2c)]$, the result (23).
- **Special case.** For $U_\infty \to 0$ there is no dissipation and (23) reduces to $\rho c V_w (T_\infty - T_w)$, the advection-conduction balance of pure suction.

PROBLEM NS2

Rotating Coaxial Cylinders

Original statement · NS2.pdf

Circular Couette flow | Torque and power | Viscous dissipation | Temperature profile

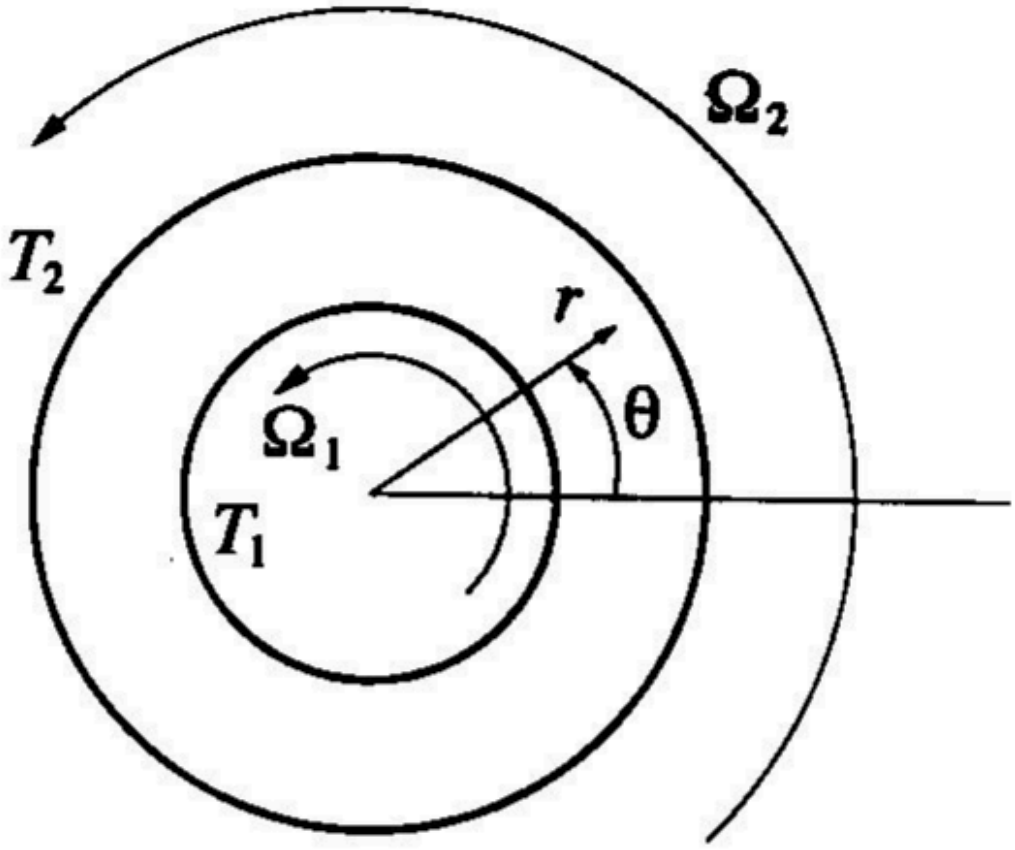
OFFICIAL PROBLEM STATEMENT

Consider the steady planar motion of a perfect liquid of density ρ, viscosity μ, thermal conductivity k and specific heat c in the gap formed between two coaxial cylinders of radii R_1 and R_2 that are rotating with angular velocities Ω_1 and Ω_2, respectively. Because of the symmetry of the problem, $v_z = 0$ and $\partial/\partial z = \partial/\partial\theta = 0$.

1. Obtain the velocity and pressure fields.

2. Calculate the torque exerted on each cylinder, as well as the power needed to keep them rotating.

3. If the temperatures on the cylinder surfaces are T_1 and T_2, determine the temperature distribution $T(r)$.



Statement figure (crop of NS2.pdf, p. 1): inner cylinder R_1 at T_1 rotating with Ω_1, outer cylinder R_2 at T_2 rotating with Ω_2, polar coordinates r, θ

`Source: NS2.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, perfect-liquid momentum Eq. 4.41 in cylindrical coordinates, energy Eq. 4.42, stress components in cylindrical coordinates.`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady planar motion ($v_z = 0, \partial/\partial z = 0$) of a perfect liquid with constant ρ, μ, k, c: equations (4.40)-(4.42) in cylindrical coordinates.
- Axisymmetric: $\partial/\partial\theta = 0$ for every variable.

- No mass force enters the problem in the plane of motion (the statement lists none; gravity, if present, only shifts the pressure by a hydrostatic term and is absorbed in the reduced pressure of Notes.pdf, Eq. 4.13). No chemical reaction and no radiation: $Q_c = Q_r = 0$.
- The cylinder surfaces are impermeable and move with their own angular velocity (no-slip).
- Data: $[\rho,\ \mu,\ k,\ c,\ R_1,\ R_2,\ \Omega_1,\ \Omega_2,\ T_1,\ T_2]$ with $R_2 > R_1$. Unknowns: $v_r(r), v_\theta(r)$, $p(r)$, the torques, the powers and $T(r)$.
- Degrees of freedom: two velocity components, the pressure and the temperature, each a function of r alone.

Symbol	Meaning	SI unit
R_1, R_2	inner and outer radii	m
Ω_1, Ω_2	angular velocities (positive counter-clockwise, towards increasing θ)	rad/s
T_1, T_2	temperatures of the inner and outer surfaces	K
ρ, μ, k	density, viscosity, thermal conductivity	kg/m^3, Pa s, W/(m K)
C_1, C_2	integration constants of v_θ	1/s, m^2/s

PHASE 2

Frames, Field Form and Boundary Conditions

- Cylindrical frame (r, θ, z) centred on the common axis, with $\bar{e}_r, \bar{e}_\theta, \bar{e}_z$ orthonormal and right-handed ($\bar{e}_r \wedge \bar{e}_\theta = \bar{e}_z$); the frame is attached to the laboratory, which is inertial.
- Field form: $\bar{v} = v_r(r)\,\bar{e}_r + v_\theta(r)\,\bar{e}_\theta, p = p(r), T = T(r)$.

Location	Condition	Origin
$r = R_1$	$v_r = 0, v_\theta = \Omega_1 R_1$	impermeable inner cylinder, no-slip
$r = R_2$	$v_r = 0, v_\theta = \Omega_2 R_2$	impermeable outer cylinder, no-slip
$r = R_1$	$T = T_1$	thermal equilibrium at the wall
$r = R_2$	$T = T_2$	thermal equilibrium at the wall

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the radial velocity

$$\frac{1}{r}\frac{d(r v_r)}{dr} + \underbrace{\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}}_{\partial_\theta = 0} + \underbrace{\frac{\partial v_z}{\partial z}}_{v_z = 0} = 0 \implies r\, v_r = \text{const} \tag{1}$$

The constant is fixed at the inner wall, where $v_r(R_1) = 0$: $r\, v_r = 0$ for every r, therefore

$$\boxed{v_r(r) = 0} \tag{2}$$

3.2 Azimuthal momentum and the velocity profile

The θ component of (4.41) in cylindrical coordinates, with $v_r = 0$, $\partial/\partial t = \partial_\theta = \partial_z = 0$ and no mass force:

$$\rho\Big(\underbrace{\frac{\partial v_\theta}{\partial t}}_{\text{steady}} + \underbrace{v_r\frac{\partial v_\theta}{\partial r}}_{v_r=0} + \underbrace{\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}}_{\partial_\theta=0} + \underbrace{\frac{v_r v_\theta}{r}}_{v_r=0} + \underbrace{v_z\frac{\partial v_\theta}{\partial z}}_{v_z=0}\Big) = -\underbrace{\frac{1}{r}\frac{\partial p}{\partial \theta}}_{\partial_\theta=0} + \mu\frac{d}{dr}\left[\frac{1}{r}\frac{d(r v_\theta)}{dr}\right] \tag{3}$$

Hence $\dfrac{d}{dr}\left[\dfrac{1}{r}\dfrac{d(r v_\theta)}{dr}\right] = 0$. Integrating once, $\dfrac{1}{r}\dfrac{d(r v_\theta)}{dr} = 2C_1$ (the factor 2 is a convenient choice of constant). Multiplying by r and integrating again, $r v_\theta = C_1 r^2 + C_2$:

$$v_\theta = C_1\, r + \frac{C_2}{r} \tag{4}$$

The boundary conditions $v_\theta(R_1) = \Omega_1 R_1$ and $v_\theta(R_2) = \Omega_2 R_2$, each multiplied by R_i, give a linear system:

$$C_1 R_1^2 + C_2 = \Omega_1 R_1^2, \qquad C_1 R_2^2 + C_2 = \Omega_2 R_2^2 \tag{5}$$

Subtracting the first equation from the second, $C_1 (R_2^2 - R_1^2) = \Omega_2 R_2^2 - \Omega_1 R_1^2$, and then $C_2 = \Omega_1 R_1^2 - C_1 R_1^2$:

$$C_1 = \frac{\Omega_2 R_2^2 - \Omega_1 R_1^2}{R_2^2 - R_1^2}, \qquad C_2 = \frac{(\Omega_1 - \Omega_2)\, R_1^2 R_2^2}{R_2^2 - R_1^2} \tag{6}$$

$$\boxed{v_\theta(r) = \frac{\Omega_2 R_2^2 - \Omega_1 R_1^2}{R_2^2 - R_1^2}\, r + \frac{(\Omega_1 - \Omega_2) R_1^2 R_2^2}{R_2^2 - R_1^2}\, \frac{1}{r}} \tag{7}$$

3.3 Radial momentum and the pressure field

The r component of (4.41) with $v_r = 0$ keeps only the centrifugal term and the pressure gradient (all other terms vanish for the same reasons as in (3)):

$$-\rho \frac{v_\theta^2}{r} = -\frac{dp}{dr} \quad \Longrightarrow \quad \frac{dp}{dr} = \rho \frac{v_\theta^2}{r} = \rho \left(C_1^2\, r + \frac{2 C_1 C_2}{r} + \frac{C_2^2}{r^3} \right) \tag{8}$$

Integrating from R_1 to r, the antiderivatives are $C_1^2 r^2/2$, $2C_1C_2 \ln r$ and $-C_2^2/(2r^2)$, and Barrow's rule gives

$$\boxed{p(r) - p(R_1) = \rho \left[\frac{C_1^2}{2} \left(r^2 - R_1^2 \right) + 2 C_1 C_2 \ln \frac{r}{R_1} - \frac{C_2^2}{2} \left(\frac{1}{r^2} - \frac{1}{R_1^2} \right) \right]} \tag{9}$$

The constant $p(R_1)$ is set by the reference pressure of the problem; the pressure grows outwards because the fluid is whirling and needs a centripetal force.

3.4 Viscous stress and torque on each cylinder

The only non-zero viscous shear stress is (Notes.pdf, stress components in cylindrical coordinates, with $v_r = 0$):

$$\tau'_{r\theta} = \mu \left[r \frac{d}{dr}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\underbrace{\frac{\partial v_r}{\partial \theta}}_{=0} \right] = \mu\, r \frac{d}{dr}\left(C_1 + \frac{C_2}{r^2}\right) = -\frac{2\mu C_2}{r^2} \tag{10}$$

The normal viscous stresses τ'_{rr} and $\tau'_{\theta\theta}$ are zero ($v_r = 0$, $\nabla \cdot \bar{v} = 0$), and the pressure force on a cylindrical surface is radial, so it has no moment about the axis. Only $\tau'_{r\theta}$ contributes to the torque.

Inner cylinder (surface $r = R_1$). The normal pointing into the liquid is $\bar{n} = +\bar{e}_r$, so $\bar{\bar{\tau}}' \cdot \bar{n} = \tau'_{r\theta}\,\bar{e}_\theta$ is the force per unit area that the liquid exerts on the cylinder. The torque per unit length about the axis, with $d\sigma = R_1\, d\theta\,(1)$ and arm $R_1 \bar{e}_r$:

$$\bar{M}_1 = \int_0^{2\pi} R_1 \bar{e}_r \wedge \tau'_{r\theta}(R_1)\bar{e}_\theta\, R_1\, d\theta = 2\pi R_1^2\, \tau'_{r\theta}(R_1)\, \bar{e}_z = -4\pi\mu C_2\, \bar{e}_z \tag{11}$$

Outer cylinder (surface $r = R_2$). The normal pointing into the liquid is $\bar{n} = -\bar{e}_r$:

$$\bar{M}_2 = \int_0^{2\pi} R_2 \bar{e}_r \wedge [-\tau'_{r\theta}(R_2)\bar{e}_\theta]\, R_2\, d\theta = -2\pi R_2^2\, \tau'_{r\theta}(R_2)\, \bar{e}_z = +4\pi\mu C_2\, \bar{e}_z \tag{12}$$

$$\boxed{\bar{M}_1 = -4\pi\mu \frac{(\Omega_1 - \Omega_2) R_1^2 R_2^2}{R_2^2 - R_1^2}\, \bar{e}_z = -\bar{M}_2} \tag{13}$$

The two torques are equal and opposite, as the angular momentum balance of the liquid requires: in steady motion the total moment on the liquid between the cylinders is zero, so the moments that the cylinders exert on it, $-\bar{M}_1$ and $-\bar{M}_2$, cancel.

3.5 Power needed to keep the cylinders rotating

The external agent must exert on each cylinder the opposite of the liquid torque, so the power it supplies is $\dot{W}_i = -\bar{M}_i \cdot \Omega_i \bar{e}_z$:

$$\dot{W}_1 = +4\pi\mu C_2\,\Omega_1, \qquad \dot{W}_2 = -4\pi\mu C_2\,\Omega_2 \tag{14}$$

$$\boxed{\dot{W} = \dot{W}_1 + \dot{W}_2 = 4\pi\mu C_2\,(\Omega_1 - \Omega_2) = 4\pi\mu \frac{(\Omega_1 - \Omega_2)^2 R_1^2 R_2^2}{R_2^2 - R_1^2}} \tag{15}$$

The total power is non-negative, as it must be: it equals the viscous dissipation in the gap (checked in 3.6).

3.6 Temperature distribution

With $v_r = 0$ the convective terms of (4.42) vanish. The energy equation for the steady, axisymmetric flow reads

$$0 = \phi_v + \frac{k}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) \tag{16}$$

and the viscous dissipation of this flow (cylindrical ϕ_v of Notes.pdf, keeping the only non-zero shear $\tau'_{r\theta}$) is

$$\phi_v = \mu\left[r\frac{d}{dr}\left(\frac{v_\theta}{r}\right)\right]^2 = \mu\left(\frac{2C_2}{r^2}\right)^2 = \frac{4\mu C_2^2}{r^4} \tag{17}$$

Substituting (17) in (16) and multiplying by r/k:

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) = -\frac{4\mu C_2^2}{k\,r^3} \tag{18}$$

Integrating once, $r\,dT/dr = \dfrac{2\mu C_2^2}{k\,r^2} + B_1$; dividing by r and integrating again:

$$T(r) = -\frac{\mu C_2^2}{k\,r^2} + B_1 \ln r + B_2 \tag{19}$$

Write $a = \mu C_2^2/k$. The conditions $T(R_1) = T_1$ and $T(R_2) = T_2$ give, after eliminating B_2,

$$T - T_1 = a\left(\frac{1}{R_1^2} - \frac{1}{r^2}\right) + B \ln\frac{r}{R_1}, \qquad B \ln\frac{R_2}{R_1} = T_2 - T_1 - a\left(\frac{1}{R_1^2} - \frac{1}{R_2^2}\right) \tag{20}$$

$$\boxed{T(r) - T_1 = \frac{\mu C_2^2}{k}\left(\frac{1}{R_1^2} - \frac{1}{r^2}\right) + \left[T_2 - T_1 - \frac{\mu C_2^2}{k}\left(\frac{1}{R_1^2} - \frac{1}{R_2^2}\right)\right]\frac{\ln(r/R_1)}{\ln(R_2/R_1)}} \tag{21}$$

Check of the power. Integrating the dissipation over the gap per unit length, with $dV = 2\pi r\,dr$ and the antiderivative of r^{-3} equal to $-1/(2r^2)$:

$$\int \phi_v\,dV = 2\pi\int_{R_1}^{R_2} \frac{4\mu C_2^2}{r^4}\,r\,dr = 8\pi\mu C_2^2\left[-\frac{1}{2r^2}\right]_{R_1}^{R_2} = 4\pi\mu C_2^2\,\frac{R_2^2 - R_1^2}{R_1^2 R_2^2} = 4\pi\mu C_2\,(\Omega_1 - \Omega_2) \tag{22}$$

where $C_2(R_2^2 - R_1^2)/(R_1^2R_2^2) = \Omega_1 - \Omega_2$ from (6) was used. It coincides with (15): the power supplied by the cylinders is dissipated in the liquid and leaves as heat through the two walls.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** $v_\theta = C_1 r + C_2/r$ is a rigid rotation (the $C_1 r$ term) plus an irrotational vortex (the C_2/r term). Only the vortex part is sheared, with $\tau'_{r\theta} \propto 1/r^2$, so the stress is larger on the inner cylinder.

- **Limits.** For $\Omega_1 = \Omega_2 = \Omega$ the liquid rotates as a rigid body, $v_\theta = \Omega r$, $C_2 = 0$, the torques and the power vanish and $p - p(R_1) = \rho\Omega^2(r^2 - R_1^2)/2$. For $\Omega_2 = 0$ and $R_2 \to \infty$, $v_\theta = \Omega_1 R_1^2/r$ and $\bar{M}_1 = -4\pi\mu\Omega_1 R_1^2\,\bar{e}_z$, the viscous torque on a spinning cylinder in an unbounded liquid.
- **Dimensional check.** $v_\theta = C_1 r + C_2/r$: C_1 has units 1/s and C_2 has units $\mathrm{m^2/s}$. The torque per unit length $4\pi\mu C_2$ has units $(\mathrm{Pa\,s})(\mathrm{m^2/s}) = \mathrm{N}$, that is N m per metre of length. The power per unit length is N m/s per metre $=$ W/m. In (21), $\mu C_2^2/(k r^2)$ has units $(\mathrm{Pa\,s})(\mathrm{m^4/s^2})/[(\mathrm{W/(m\,K)})\,\mathrm{m^2}] = \mathrm{K}$.
- **Role of the dissipation.** If the viscous heating is negligible ($a \to 0$) the temperature is the purely logarithmic conduction profile between T_1 and T_2.

PROBLEM NS3

Spherical Balloon of Variable Radius

Original statement · NS3.pdf

Spherical symmetry | Unsteady flow | Inflation power | Energy balance

OFFICIAL PROBLEM STATEMENT

A spherical balloon of variable radius $R(t)$ is immersed in an infinite liquid medium of density ρ and viscosity μ. Far from the balloon, the liquid is at rest with pressure $p = p_\infty$. Determine the radial velocity $v_r(r,t)$ and pressure distribution $p(r,t)$ for the buoyancy-free spherically-symmetrical motion induced in the surrounding liquid. Obtain also the power required to inflate the balloon.

Additional question (handwritten on the sheet): where does the work go? Consider the following control volume.

The sheet carries no printed figure; the handwritten solution sketches the balloon of radius $r = R$ with surface velocity $v_r = \dot{R}$.

Source: NS3.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 in spherical coordinates, mechanical-energy balance Eq. 4.24, viscous dissipation ϕ_v in spherical coordinates.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

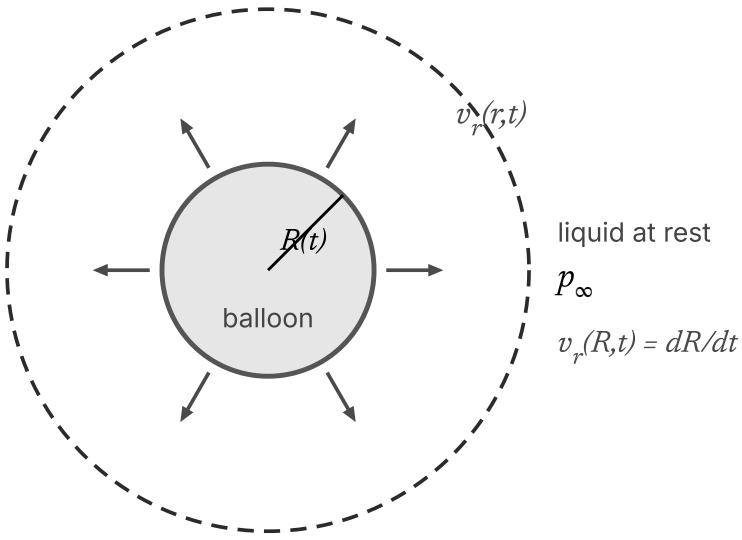
- Incompressible liquid with constant ρ and μ; equations (4.40) and (4.41) in spherical coordinates (r, θ, ϕ).
- Spherical symmetry: $\bar{v} = v_r(r,t)\,\bar{e}_r$, $v_\theta = v_\phi = 0$, and $\partial/\partial\theta = \partial/\partial\phi = 0$.
- Buoyancy-free: the mass force is neglected (or absorbed in the reduced pressure of Notes.pdf, Eq. 4.13).
- Unsteady: the balloon radius changes with time, so $\partial/\partial t \neq 0$.
- The liquid occupies $r > R(t)$; far from the balloon it is at rest with pressure p_∞.
- Data: $[\rho,\ \mu,\ p_\infty,\ R(t)]$. Unknowns: $v_r(r,t), p(r,t)$ and the inflation power $\dot{W}$.

Symbol	Meaning	SI unit
$R(t)$, $\dot{R}$, $\ddot{R}$	balloon radius, its rate of change and acceleration	m, m/s, m/s^2
ρ, μ	liquid density and viscosity	kg/m^3, Pa s
p_∞	pressure far from the balloon	Pa
$\dot{W}$	power delivered to the liquid by the balloon	W

PHASE 2

Frames, Field Form and Boundary Conditions

- Spherical frame (r, θ, ϕ) centred on the balloon centre, which stays fixed; $\bar{e}_r$ points away from the centre.
- Field form: $\bar{v} = v_r(r,t)\,\bar{e}_r, p = p(r,t)$.



control volume of the energy balance (dashed)

Balloon of radius $R(t)$ in the liquid: radial velocity $v_r(r,t) = R^2\dot{R}/r^2$, liquid at rest with pressure p_∞ far from the balloon, and the (dashed) control volume of the energy balance of Phase 3

Location	Condition	Origin
$r = R(t)$	$v_r = \dot{R}$	the liquid in contact with the balloon moves with its surface (kinematic condition)
$r \to \infty$	$v_r \to 0$	liquid at rest far from the balloon
$r \to \infty$	$p \to p_\infty$	statement

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the radial velocity

$$\nabla \cdot \bar{v} = \frac{1}{r^2}\frac{\partial (r^2 v_r)}{\partial r} + \underbrace{\frac{1}{r\sin\theta}\frac{\partial (v_\theta \sin\theta)}{\partial \theta}}_{v_\theta = 0} + \underbrace{\frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}}_{v_\phi = 0} = 0 \implies r^2\, v_r = C(t) \tag{1}$$

The kinematic condition $v_r(R,t) = \dot{R}$ gives $C(t) = R^2\dot{R}$:

$$\boxed{v_r(r,t) = \frac{R^2\dot{R}}{r^2}} \tag{2}$$

The condition at infinity is satisfied automatically. The velocity at every point is fixed by the instantaneous radius and its rate of change.

3.2 Radial momentum and the pressure field

The viscous term of the r component of (4.41) is $\mu\,\partial/\partial r\left[r^{-2}\,\partial(r^2 v_r)/\partial r\right]$ (all other viscous terms contain v_θ, v_ϕ or derivatives with respect to θ, ϕ). By (1), $r^2 v_r = C(t)$ does not depend on r, so $\partial(r^2 v_r)/\partial r = 0$ and the viscous force vanishes identically: the flow is irrotational, $\nabla \wedge \bar{v} = 0$. The momentum equation reduces to

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r}\right) = -\frac{\partial p}{\partial r} \tag{3}$$

The two accelerations are computed with the product and chain rules, using $d(R^2\dot{R})/dt = 2R\dot{R}^2 + R^2\ddot{R}$ and $\partial(r^{-2})/\partial r = -2r^{-3}$:

$$\frac{\partial v_r}{\partial t} = \frac{2R\dot{R}^2 + R^2\ddot{R}}{r^2}, \qquad v_r\frac{\partial v_r}{\partial r} = \frac{R^2\dot{R}}{r^2}\left(-\frac{2R^2\dot{R}}{r^3}\right) = -\frac{2R^4\dot{R}^2}{r^5} \tag{4}$$

$$\frac{\partial p}{\partial r} = -\rho\left[\frac{2R\dot{R}^2 + R^2\ddot{R}}{r^2} - \frac{2R^4\dot{R}^2}{r^5}\right] \tag{5}$$

Integrate (5) from r to ∞, with $p(\infty) = p_\infty$. The antiderivatives are $\int r'^{-2}dr' = -1/r'$ and $\int r'^{-5}dr' = -1/(4r'^4)$, and Barrow's rule between r and ∞ gives $\int_r^\infty r'^{-2}dr' = 1/r$ and $\int_r^\infty r'^{-5}dr' = 1/(4r^4)$:

$$p_\infty - p(r,t) = -\rho\left[\frac{2R\dot{R}^2 + R^2\ddot{R}}{r} - \frac{R^4\dot{R}^2}{2r^4}\right]$$

$$\boxed{p(r,t) = p_\infty + \rho\left[\frac{2R\dot{R}^2 + R^2\ddot{R}}{r} - \frac{R^4\dot{R}^2}{2r^4}\right]} \tag{6}$$

At the balloon surface $r = R$:

$$p(R,t) = p_\infty + \rho\left[2\dot{R}^2 + R\ddot{R} - \frac{\dot{R}^2}{2}\right] = p_\infty + \rho\left(\frac{3}{2}\dot{R}^2 + R\ddot{R}\right) \tag{7}$$

3.3 Viscous stress at the balloon surface

With the Navier-Poisson law and $\nabla \cdot \bar{v} = 0$, the radial normal stress is (Notes.pdf, stress components in spherical coordinates)

$$\tau'_{rr} = 2\mu\frac{\partial v_r}{\partial r} = 2\mu\left(-\frac{2R^2\dot{R}}{r^3}\right), \qquad \tau'_{rr}\Big|_{r=R} = -\frac{4\mu\dot{R}}{R} \tag{8}$$

The shear components $\tau'_{r\theta}$ and $\tau'_{r\phi}$ are zero by symmetry, so the viscous traction on the surface is purely radial.

3.4 Power required to inflate the balloon

The balloon surface moves with velocity $\dot{R}\,\bar{e}_r$. The liquid occupies $r > R$, so the unit normal pointing out of the liquid at the balloon surface is $-\bar{e}_r$. The traction that the balloon exerts on the liquid is $(-p\bar{\bar{I}} + \bar{\bar{\tau}}') \cdot (-\bar{e}_r) = (p - \tau'_{rr})\,\bar{e}_r$, and the power it delivers is the force times the velocity of the surface, integrated over the area $4\pi R^2$:

$$\dot{W} = 4\pi R^2\,\dot{R}\,[p(R,t) - \tau'_{rr}(R)] = 4\pi R^2 \dot{R}\,p(R,t) + 16\pi\mu R\dot{R}^2 \tag{9}$$

Substituting (7):

$$\boxed{\dot{W} = 4\pi R^2 \dot{R}\,p_\infty + 4\pi\rho R^2\dot{R}\left(\frac{3}{2}\dot{R}^2 + R\ddot{R}\right) + 16\pi\mu R\dot{R}^2} \tag{10}$$

3.5 Where does the work go? (additional question)

The three terms of (10) are identified by computing independently the rate of change of the kinetic energy of the liquid and its viscous dissipation.

Kinetic energy of the liquid. With $v_r^2 = R^4\dot{R}^2/r^4$ and $dV = 4\pi r^2\,dr$:

$$E_k = \int_R^\infty \frac{\rho}{2}\frac{R^4\dot{R}^2}{r^4}\,4\pi r^2\,dr = 2\pi\rho R^4\dot{R}^2\left[-\frac{1}{r}\right]_R^\infty = 2\pi\rho R^3\dot{R}^2 \tag{11}$$

Differentiating with the product rule,

$$\frac{dE_k}{dt} = 2\pi\rho\left(3R^2\dot{R}^3 + 2R^3\dot{R}\ddot{R}\right) = 4\pi\rho R^2\dot{R}\left(\frac{3}{2}\dot{R}^2 + R\ddot{R}\right) \tag{12}$$

which is exactly the second term of (10).

Viscous dissipation. In spherical coordinates with $v_\theta = v_\phi = 0$ the dissipation is $\phi_v = 2\mu\left[(\partial v_r/\partial r)^2 + 2(v_r/r)^2\right]$ (Notes.pdf, ϕ_v in spherical coordinates). With $\partial v_r/\partial r = -2R^2\dot{R}/r^3$ and $v_r/r = R^2\dot{R}/r^3$:

$$\phi_v = 2\mu\left[\frac{4R^4\dot{R}^2}{r^6} + \frac{2R^4\dot{R}^2}{r^6}\right] = \frac{12\mu R^4\dot{R}^2}{r^6} \tag{13}$$

$$\Phi = \int_R^\infty \frac{12\mu R^4\dot{R}^2}{r^6}\,4\pi r^2\,dr = 48\pi\mu R^4\dot{R}^2\left[-\frac{1}{3r^3}\right]_R^\infty = 16\pi\mu R\dot{R}^2 \tag{14}$$

which is the third term of (10).

Pressure term. $4\pi R^2\dot{R}\,p_\infty = p_\infty\,dV_b/dt$ with $V_b = 4\pi R^3/3$ the balloon volume: the work done against the ambient pressure to displace the liquid, that is, to set the ambient liquid in motion (the outward volume flux through any sphere is $4\pi r^2 v_r = 4\pi R^2\dot{R}$).

$$\boxed{\dot{W} = p_\infty\frac{dV_b}{dt} + \frac{dE_k}{dt} + \Phi} \tag{15}$$

The work supplied by the balloon increases the kinetic energy of the liquid, displaces the ambient pressure p_∞ and is partly converted into internal energy by viscous dissipation. This agrees with the mechanical energy balance (Notes.pdf, Eq. 4.24) applied to the liquid volume between the balloon and a distant material sphere.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The velocity decays as $1/r^2$ because the volume flux $4\pi r^2 v_r$ through every sphere equals the rate of volume growth of the balloon. The motion is irrotational, so viscosity does not enter the pressure field (6); it appears only in the boundary traction (8) and the power (10).
- **Limits.** For $r \to \infty$, $v_r \to 0$ and $p \to p_\infty$. For a balloon at constant radius ($\dot{R} = 0$) every term vanishes: no flow, $p = p_\infty$ and no power. For a constant inflation rate ($\ddot{R} = 0$) the pressure is above p_∞ near the balloon, by the amount $\rho\dot{R}^2(3/2)$ at $r = R$.

- **Dimensional check.** v_r: $\mathrm{m^2}\cdot\mathrm{m/s/m^2} = \mathrm{m/s}$. In (6), $\rho R\dot{R}^2/r$ has units $(\mathrm{kg/m^3})(\mathrm{m^2/s^2}) = \mathrm{Pa}$ and $\rho R^2\ddot{R}/r$ has units $(\mathrm{kg/m^3})\,\mathrm{m^2}\,(\mathrm{m/s^2})/\mathrm{m} = \mathrm{Pa}$. In (10), $p_\infty\, 4\pi R^2\dot{R}$ is $\mathrm{Pa}\cdot\mathrm{m^3/s} = \mathrm{W}$ and $16\pi\mu R\dot{R}^2$ is $(\mathrm{Pa\,s})\,\mathrm{m}\,(\mathrm{m^2/s^2}) = \mathrm{W}$.
- **Consistency.** Equation (15) is verified term by term: (12) equals the second term of (10) and (14) equals the third, so the power computed from the surface traction equals the power accounted for inside the liquid.

PROBLEM NS4

Couette Flow with Heat Transfer

Original statement · NS4.pdf

Plane Couette flow | Viscous dissipation | Heat flux to the walls | Energy balance

OFFICIAL PROBLEM STATEMENT

Consider the steady planar motion of a perfect liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c in the gap of thickness h that forms between two infinite horizontal walls. The upper wall, at temperature T_u, moves parallel to itself with velocity U, while the lower wall remains at rest, at temperature T_l.

1. Determine the velocity components $v_x(z)$ and $v_z(z)$, as well as the accompanying pressure field $p(x, z)$, assuming that the pressure on the upper wall has the constant value $p = p_u$.

2. Calculate the volume flux for the flow established in the gap.

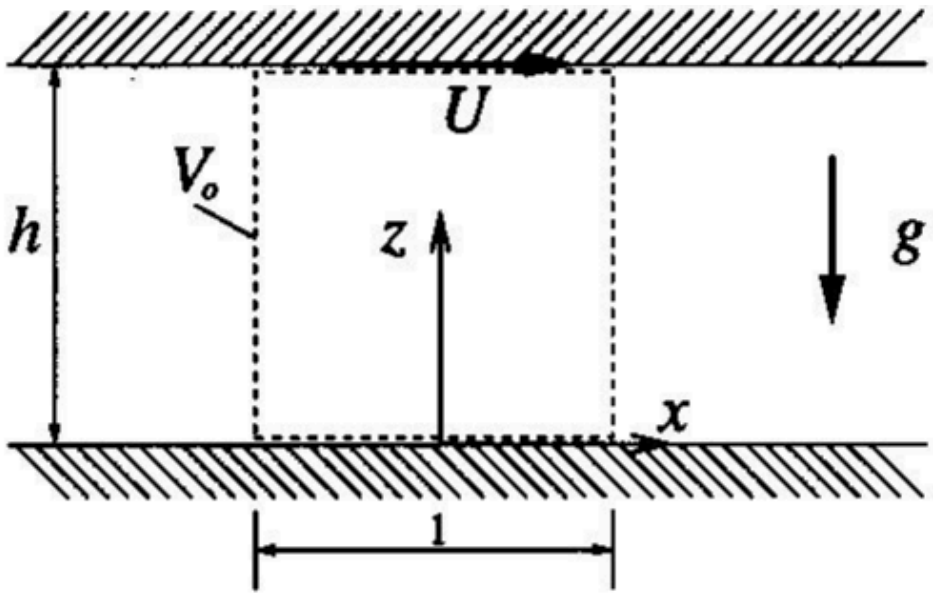
3. Obtain the pressure distribution $T(z)$.

4. Compute the force per unit surface acting on the upper wall as well as the power needed to move it.

5. Determine the conductive heat flux from the fluid to the upper and lower walls.

6. Use the fixed control volume V_o indicated in the figure to show that the power employed to move the upper wall is dissipated inside the liquid, generating an excess internal energy that is transfered by heat condution to the walls.

Transcribed as printed: in item 3 the sheet says "pressure distribution" where the temperature distribution $T(z)$ is meant, and "transfered" and "condution" are spelling errors of the sheet.



Statement figure (crop of NS4.pdf, p. 1): gap of thickness h, upper wall moving with U, fixed control volume V_o of unit length in x, axes x, z and gravity g

`Source: NS4.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41, energy Eq. 4.42, internal-energy balance Eq. 4.25, total-energy balance Eq. 4.3.`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady planar flow of a perfect liquid (constant ρ, μ, k, c): equations (4.40)-(4.42).
- Infinite horizontal walls, gravity along $-\bar{e}_z$; no chemical reaction and no radiation ($Q_c = Q_r = 0$).
- The velocity and temperature depend only on the distance z to the lower wall: $\partial/\partial x = \partial/\partial y = 0$ for them. The pressure may depend on x and z until it is determined.
- The walls are impermeable. The lower wall ($z = 0$) is at rest at T_l; the upper wall ($z = h$) moves with $U\,\bar{e}_x$ at T_u and the pressure on it is p_u.
- Data: $[\rho,\ \mu,\ k,\ c,\ h,\ U,\ T_l,\ T_u,\ p_u,\ g]$. Unknowns: $v_x(z), v_z(z), p(x,z), T(z)$, the flow rate, the wall force and power, the wall heat fluxes.

- Degrees of freedom: two velocity components, pressure and temperature, each a function of z (the pressure also of x).

Symbol	Meaning	SI unit
h	gap between the walls	m
U	velocity of the upper wall	m/s
T_l, T_u	temperatures of the lower and upper walls	K
p_u	pressure on the upper wall	Pa
ρ, μ, k, c	density, viscosity, conductivity, specific heat	kg/m^3, Pa s, W/(m K), J/(kg K)

PHASE 2

Frames, Field Form and Boundary Conditions

- Cartesian frame (x, y, z): x along the walls in the direction of U, z normal to the walls and upward from the lower wall, y out of the plane. The basis is orthonormal and right-handed (no change of basis).
- Field form: $\bar{v} = v_x(z)\,\bar{e}_x + v_z(z)\,\bar{e}_z, T = T(z), p = p(x, z)$.
- Fixed control volume V_o of the figure: the box $0 < z < h$ of unit length in x and unit width in y; its surfaces are the lower wall Σ_l ($\bar{n} = -\bar{e}_z$), the upper wall Σ_u ($\bar{n} = +\bar{e}_z$) and the two vertical faces Σ_1 ($\bar{n} = -\bar{e}_x$) and Σ_2 ($\bar{n} = +\bar{e}_x$).

Location	Condition	Origin
$z = 0$	$v_x = 0, v_z = 0, T = T_l$	lower wall at rest, impermeable
$z = h$	$v_x = U, v_z = 0, T = T_u$	upper wall moving, impermeable
$z = h$	$p = p_u$	statement

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the normal velocity

$$\frac{\partial v_x}{\partial x} + \frac{dv_z}{dz} = \underbrace{\frac{\partial v_x}{\partial x}}_{v_x = v_x(z)} + \frac{dv_z}{dz} = 0 \implies v_z = \text{const} \qquad (1)$$

The wall is impermeable, $v_z(0) = 0$, so

$$\boxed{v_z(z) = 0} \tag{2}$$

3.2 Momentum equations

z-component: with $v_z = 0$ every inertial and viscous term vanishes,

$$0 = -\frac{\partial p}{\partial z} - \rho g \implies p + \rho g z = f(x) \tag{3}$$

x-component: the convective terms vanish ($\partial_x = 0$, $v_z = 0$),

$$\rho\Big(\underbrace{\cancel{\frac{\partial v_x}{\partial t}}}_{\text{steady}} + \underbrace{v_x\cancel{\frac{\partial v_x}{\partial x}}}_{\partial_x = 0} + \underbrace{v_z\cancel{\frac{\partial v_x}{\partial z}}}_{v_z = 0}\Big) = -\frac{\partial p}{\partial x} + \mu\frac{d^2 v_x}{dz^2} \implies \mu\frac{d^2 v_x}{dz^2} = \frac{\partial p}{\partial x} = f'(x) \tag{4}$$

The left side depends only on z and the right side only on x, so both are equal to a constant. The pressure on the upper wall is $p(x, h) = p_u$ for every x, and by (3) $f(x) = p_u + \rho g h$ is a constant, so $f'(x) = 0$:

$$\boxed{p(x, z) = p_u + \rho g\,(h - z)}, \qquad \frac{\partial p}{\partial x} = 0 \tag{5}$$

Equation (4) becomes $d^2 v_x / dz^2 = 0$; integrating twice, $v_x = C_1 + C_2 z$. The conditions $v_x(0) = 0$ and $v_x(h) = U$ give $C_1 = 0$ and $C_2 = U/h$:

$$\boxed{v_x(z) = U\,\frac{z}{h}} \tag{6}$$

3.3 Volume flux

The flux per unit width in y through a section $x = \text{const}$ is $Q = \int_0^h v_x \, dz$. The antiderivative of Uz/h is $Uz^2/(2h)$ and Barrow's rule gives:

$$\boxed{Q = \left[\frac{U}{h} \frac{z^2}{2} \right]_0^h = \frac{Uh}{2}} \tag{7}$$

3.4 Temperature distribution

The energy equation (4.42) with $v_z = 0$, $\partial_x = 0$ and $\phi_v = \mu (dv_x/dz)^2 = \mu U^2/h^2$:

$$\underbrace{\rho c v_x \frac{\partial T}{\partial x}}_{\partial_x = 0} = \phi_v + k \frac{d^2 T}{dz^2} \quad \Longrightarrow \quad \frac{d^2 T}{dz^2} = -\frac{\mu U^2}{k \, h^2} \tag{8}$$

Integrating twice, $T = B_1 + B_2 z - \dfrac{\mu U^2}{2 k h^2} z^2$. The conditions $T(0) = T_l$ and $T(h) = T_u$ give $B_1 = T_l$ and $B_2 = \dfrac{T_u - T_l}{h} + \dfrac{\mu U^2}{2kh}$:

$$\boxed{T(z) = T_l + (T_u - T_l) \, \frac{z}{h} + \frac{\mu U^2}{2k} \, \frac{z}{h} \left(1 - \frac{z}{h} \right)} \tag{9}$$

3.5 Force on the upper wall and power needed to move it

On the upper wall, the normal pointing into the liquid is $\bar{n} = -\bar{e}_z$, so $\left(-p \bar{\bar{I}} + \bar{\bar{\tau}}' \right) \cdot \bar{n}$ is the force per unit area that the liquid exerts on the wall. The viscous components are $\tau'_{xz} = \mu \, dv_x/dz = \mu U/h$ and $\tau'_{zz} = 0$ (Notes.pdf, stress components, Cartesian coordinates):

$$\bar{f}_u = -p_u (-\bar{e}_z) + \bar{\bar{\tau}}' \cdot (-\bar{e}_z) = p_u \, \bar{e}_z - \mu \frac{U}{h} \, \bar{e}_x \tag{10}$$

The viscous part opposes the motion of the wall. The external agent must supply $+\mu U/h \, \bar{e}_x$ per unit area, with the wall velocity $U \bar{e}_x$:

$$\boxed{\bar{f}_u = p_u\,\bar{e}_z - \mu\frac{U}{h}\,\bar{e}_x, \qquad \dot{W} = \mu\frac{U^2}{h}\ \text{per unit area}} \tag{11}$$

The pressure force $p_u\bar{e}_z$ is normal to the wall and does no work on it.

3.6 Heat fluxes to the walls

The heat flux from the liquid to a wall is $\bar{q}\cdot\bar{n}$, with $\bar{n}$ pointing out of the liquid and $\bar{q} = -k\nabla T$. From (9),

$$\frac{dT}{dz} = \frac{T_u - T_l}{h} + \frac{\mu U^2}{2kh} - \frac{\mu U^2}{kh^2}z \tag{12}$$

Upper wall ($\bar{n} = +\bar{e}_z$, $z = h$):

$$q_u = -k\frac{dT}{dz}\Big|_{z=h} = k\frac{T_l - T_u}{h} + \frac{\mu U^2}{2h} \tag{13}$$

Lower wall ($\bar{n} = -\bar{e}_z$, $z = 0$):

$$q_l = +k\frac{dT}{dz}\Big|_{z=0} = k\frac{T_u - T_l}{h} + \frac{\mu U^2}{2h} \tag{14}$$

$$\boxed{q_u = k\frac{T_l - T_u}{h} + \frac{\mu U^2}{2h}, \qquad q_l = k\frac{T_u - T_l}{h} + \frac{\mu U^2}{2h}} \tag{15}$$

The conductive contributions are equal and opposite and the viscous heating splits equally between the walls. Their sum is $q_u + q_l = \mu U^2/h$.

3.7 Integral balance in the fixed control volume V_o

Total energy (Notes.pdf, Eq. 4.3 in integral form for the fixed volume, steady):

$$\int_{\Sigma_o} \rho \left(e + \frac{|\bar{v}|^2}{2} \right) \bar{v} \cdot \bar{n} \, d\sigma = - \int_{\Sigma_o} p \, \bar{v} \cdot \bar{n} \, d\sigma + \int_{\Sigma_o} \bar{v} \cdot \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma + \underbrace{\int_{V_o} \rho \bar{g} \cdot \bar{v} \, dV}_{=0} - \int_{\Sigma_o} \bar{q} \cdot \bar{n} \, d\sigma \qquad (16)$$

Each term, surface by surface (areas of unit length times unit width):

Surface	**Convective** $\int \rho(e + v^2/2)\, \bar{v} \cdot \bar{n}$	**Pressure work** $-\int p\, \bar{v} \cdot \bar{n}$	**Viscous work** $\int \bar{v} \cdot \bar{\bar{\tau}}' \cdot \bar{n}$	**Heat** $-\int \bar{q} \cdot \bar{n}$
Σ_l ($z = 0$)	0 ($\bar{v} \cdot \bar{n} = 0$)	0	0 ($\bar{v} = 0$)	$-q_l$
Σ_u ($z = h$)	0 ($\bar{v} \cdot \bar{n} = 0$)	0	$U\, \tau'_{xz} = \mu U^2/h$	$-q_u$
$\Sigma_1 + \Sigma_2$	the two faces cancel ($\bar{v}$, e, p do not depend on x; opposite normals)	cancel	cancel	0 (T independent of x)

The mass-force work vanishes because $\bar{g} \cdot \bar{v} = -g\, v_z = 0$. Equation (16) gives

$$0 = \mu \frac{U^2}{h} - q_u - q_l \implies \boxed{\mu \frac{U^2}{h} = q_u + q_l} \qquad (17)$$

Internal energy (Notes.pdf, Eq. 4.25, steady, $\nabla \cdot \bar{v} = 0$): $0 = -\int_{\Sigma_o} \bar{q} \cdot \bar{n} \, d\sigma + \int_{V_o} \phi_v \, dV$, with $\int_{V_o} \phi_v \, dV = \dfrac{\mu U^2}{h^2}\, h = \dfrac{\mu U^2}{h}$ (area 1, height h), which again gives $q_u + q_l = \mu U^2/h$.

Equations (11), (15) and (17) show the whole chain: the power spent moving the upper wall, $\mu U^2/h$ per unit area, equals the viscous dissipation in the liquid and leaves the volume as the heat conducted to the two walls.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The pressure is hydrostatic and the velocity is linear: viscous forces alone transmit the motion. The dissipation $\phi_v = \mu U^2/h^2$ is uniform, and it heats the liquid; the parabolic term of (9) is the temperature rise it produces, with its maximum in

the middle of the gap.

- **Limits.** For $U \to 0$ there is no flow and no dissipation: T is the linear conduction profile between T_l and T_u and $q_u = -q_l = k(T_l - T_u)/h$. For $T_u = T_l$, the temperature is symmetric about $z = h/2$ and each wall receives $\mu U^2/(2h)$.
- **Dimensional check.** $\mu U/h$: $(\mathrm{Pa\,s})(\mathrm{m/s})/\mathrm{m} = \mathrm{Pa}$. $\mu U^2/h$: $\mathrm{Pa \cdot m/s} = \mathrm{W/m^2}$. In (9), $\mu U^2/(2k)$ has units $(\mathrm{Pa\,s})(\mathrm{m^2/s^2})/[\mathrm{W/(m\,K)}] = \mathrm{K}$.
- **Mean velocity.** From (7), $\bar{v} = Q/h = U/2$, half the wall velocity.

PROBLEM NS5

Cylinder Dragging a Liquid Layer

Original statement · NS5.pdf

Annular film | Gravity and drag | Volume flux | Adiabatic outer surface

OFFICIAL PROBLEM STATEMENT

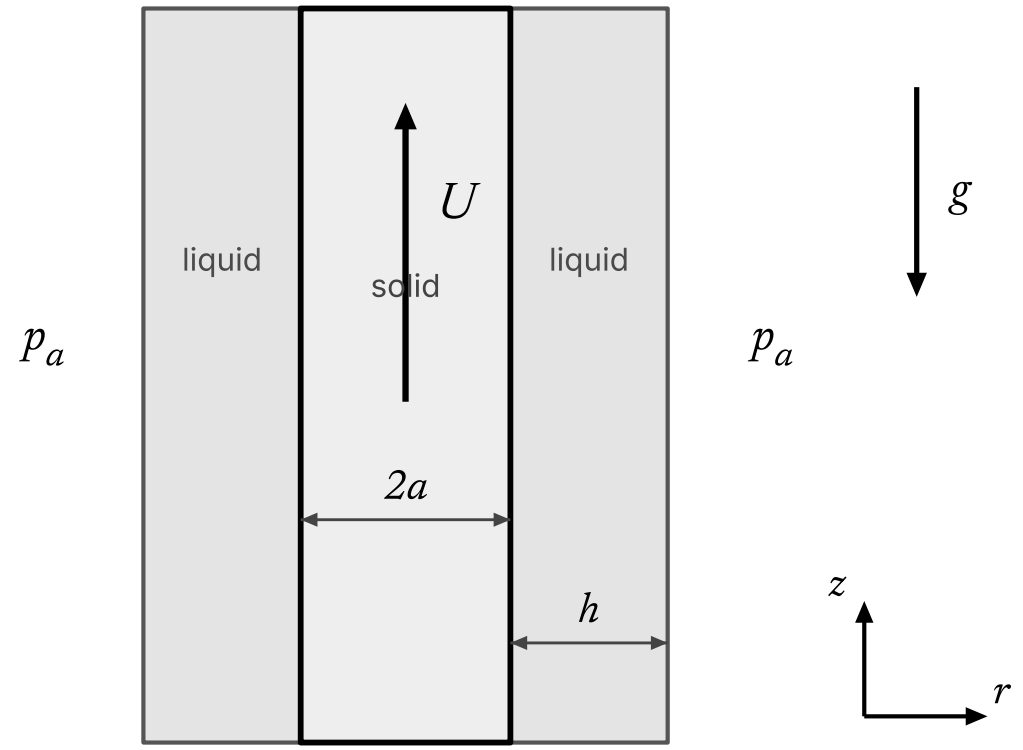
The solid cylinder of radius a shown in the figure moves vertically with velocity U, dragging in its motion a coaxial cylindrical layer of liquid of constant uniform thickness h. The liquid has constant values of its density ρ, viscosity μ, thermal conductivity k, and specific heat c. The air surrounding the liquid has a density $\rho_a \ll \rho$ and viscosity $\mu_a \ll \mu$, so that in the first approximation it moves without significant pressure differences, exerting on the liquid a negligible viscous force. Because of the symmetry of the problem, $\partial/\partial\theta = 0$ and the only nonzero velocity component is v_z.

1. Obtain the velocity $v_z(r, z)$ and pressure $p(r, z)$ in the liquid.

2. Compute the force on the cylinder (per unit length).

3. Calculate the volume flux Q dragged by the cylinder, as well as the value of U for which $Q = 0$.

4. Assuming that the thermal conductivity of air is very small compared with that of the liquid and that the cylinder is at constant uniform temperature T_c, obtain the temperature distribution in the liquid $T(r)$ as well as the conductive heat flux to the wall.



Statement figure (redrawn from NS5.pdf, p. 1): solid cylinder of radius a moving upward with U, coaxial liquid layer of thickness h, surrounding air at pressure p_a, gravity g and axes r, z

`Source: NS5.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 in cylindrical coordinates, energy Eq. 4.42, stress components in cylindrical coordinates.`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady motion of a perfect liquid (constant ρ, μ, k, c): equations (4.40)-(4.42) in cylindrical coordinates.
- Axisymmetric ($\partial/\partial\theta = 0$) with $v_r = v_\theta = 0$ (given). Gravity acts along $-\bar{e}_z$, opposite to the motion of the cylinder.
- The layer has uniform thickness h and is far from its ends: the thickness does not change along z (fully developed). No chemical reaction and no radiation.

- The air exerts no pressure variation on the liquid and a negligible viscous force: at the free surface $r = a + h, p = p_a$ and the shear stress of the liquid is zero.
- Data: $[\rho,\ \mu,\ k,\ c,\ a,\ h,\ U,\ g,\ p_a,\ T_c]$. Notation: $b = a + h, \nu = \mu/\rho$. Unknowns: $v_z(r), p$, the force, Q, U_0 (the U with $Q = 0$), $T(r)$ and the wall heat flux.
- Degrees of freedom: one velocity component, the pressure and the temperature, each function of r (and z for the pressure until it is determined).

Symbol	Meaning	SI unit
$a, h, b = a + h$	cylinder radius, layer thickness, outer radius of the layer	m
U	velocity of the cylinder (upward)	m/s
p_a	air pressure	Pa
T_c	cylinder temperature	K
ρ, μ, k, c	liquid properties	kg/m^3, Pa s, W/(m K), J/(kg K)

PHASE 2

Frames, Field Form and Boundary Conditions

- Cylindrical frame (r, θ, z) with z along the axis, upward (direction of U). The basis $\{\bar{e}_r, \bar{e}_\theta, \bar{e}_z\}$ is orthonormal and right-handed.
- Field form: $\bar{v} = v_z\, \bar{e}_z, p = p(r, z), T = T(r)$. The liquid occupies $a < r < b$.

Location	Condition	Origin
$r = a$	$v_z = U$	no-slip on the moving cylinder
$r = b$	$\tau'_{rz} = \mu\, \partial v_z/\partial r = 0$	negligible viscous force of the air
$r = b$	$p = p_a$	no significant pressure differences in the air
$r = a$	$T = T_c$	uniform temperature of the cylinder
$r = b$	$\partial T/\partial r = 0$	conductivity of the air much smaller than k: adiabatic free surface

PHASE 3

Step-by-Step Derivation

3.1 Continuity

$$\frac{1}{r}\frac{\partial (r v_r)}{\partial r} + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = \underbrace{\frac{1}{r}\frac{\partial (r v_r)}{\partial r}}_{v_r = 0} + \underbrace{\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}}_{v_\theta = 0} + \frac{\partial v_z}{\partial z} = 0 \implies \frac{\partial v_z}{\partial z} = 0 \tag{1}$$

so $v_z = v_z(r)$ (the field does not change along the axis).

3.2 Momentum equations and the pressure

r-component: with $v_r = v_\theta = 0$ and $\partial/\partial\theta = 0$ every term vanishes except the pressure gradient,

$$0 = -\frac{\partial p}{\partial r} \implies p = p(z) \tag{2}$$

z-component:

$$\rho\Big(\underbrace{\frac{\partial v_z}{\partial t}}_{\text{steady}} + \underbrace{v_r\frac{\partial v_z}{\partial r}}_{v_r = 0} + \underbrace{\frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta}}_{v_\theta = 0} + \underbrace{v_z\frac{\partial v_z}{\partial z}}_{\text{by (1)}}\Big) = -\frac{dp}{dz} + \mu\,\frac{1}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) - \rho g \tag{3}$$

By (2), p depends only on z; at $r = b$ it equals p_a for every z, therefore

$$\boxed{p(r,z) = p_a}, \qquad \frac{dp}{dz} = 0 \tag{4}$$

The liquid is at the pressure of the surrounding air everywhere: the weight of the layer is carried by the viscous stress, not by a pressure gradient. Equation (3) becomes

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) = \frac{\rho g}{\mu} = \frac{g}{\nu} \tag{5}$$

3.3 Velocity profile

Multiply (5) by r and integrate: $r\,dv_z/dr = \dfrac{g}{2\nu}r^2 + C_1$. Divide by r and integrate again, with $\int dr/r = \ln r$:

$$v_z = \frac{g}{4\nu}r^2 + C_1 \ln r + C_2 \tag{6}$$

The shear-free condition at $r = b$, $dv_z/dr\big|_b = \dfrac{g}{2\nu}b + \dfrac{C_1}{b} = 0$, gives $C_1 = -\dfrac{g}{2\nu}b^2$. The no-slip condition $v_z(a) = U$ gives $C_2 = U - \dfrac{g}{4\nu}a^2 + \dfrac{g}{2\nu}b^2 \ln a$. Substituting,

$$\boxed{v_z(r) = U + \frac{g}{2\nu}\left[\frac{r^2 - a^2}{2} - (a+h)^2 \ln\frac{r}{a}\right]} \tag{7}$$

3.4 Force on the cylinder

The surface of the cylinder is $r = a$; the normal pointing into the liquid is $\bar{n} = +\bar{e}_r$, so $(-p\bar{\bar{I}} + \bar{\bar{\tau}}') \cdot \bar{n}$ is the force per unit area that the liquid exerts on the cylinder. The pressure is uniform ($p = p_a$) and radial, so it integrates to zero around the circumference. The viscous stress is $\tau'_{rz} = \mu\, dv_z/dr$ (Notes.pdf, stress components in cylindrical coordinates). From (7), $dv_z/dr = \dfrac{g}{2\nu}\left(r - \dfrac{b^2}{r}\right)$, so at the wall

$$\tau'_{rz}\Big|_a = \mu\frac{g}{2\nu}\left(a - \frac{b^2}{a}\right) = -\frac{\rho g\,(b^2 - a^2)}{2a} = -\frac{\rho g\, h\,(2a + h)}{2a} \tag{8}$$

The force per unit length of cylinder (area $2\pi a$):

$$\boxed{\bar{F} = 2\pi a\,\tau'_{rz}\Big|_a \bar{e}_z = -\rho g\,\pi\left[(a+h)^2 - a^2\right]\bar{e}_z = -2\pi a\,\rho g\,h\left(1 + \frac{h}{2a}\right)\bar{e}_z} \tag{9}$$

The liquid pulls the cylinder downward with a force equal to the weight of the layer per unit length, as the equilibrium of the whole layer requires (the pressure forces on it cancel).

3.5 Volume flux and the velocity for zero flux

$$Q = \int_a^b v_z\, 2\pi r\, dr \tag{10}$$

With $\kappa = g/(2\nu)$, three integrals are needed. The first is elementary, $\int_a^b r\,dr = (b^2-a^2)/2$. For the second, the antiderivative of $r(r^2-a^2)/2$ is $r^4/8 - a^2 r^2/4$, and Barrow's rule gives $\dfrac{(b^2-a^2)^2}{8}$. For the third, integration by parts with $u = \ln(r/a)$, $dv = r\,dr$ gives the antiderivative $\dfrac{r^2}{2}\ln\dfrac{r}{a} - \dfrac{r^2}{4}$, so $\displaystyle\int_a^b r\ln\frac{r}{a}\,dr = \frac{b^2}{2}\ln\frac{b}{a} - \frac{b^2-a^2}{4}$. Then

$$\frac{Q}{2\pi} = U\frac{b^2-a^2}{2} + \kappa\left[\frac{(b^2-a^2)^2}{8} - b^2\left(\frac{b^2}{2}\ln\frac{b}{a} - \frac{b^2-a^2}{4}\right)\right]$$

and, simplifying $\frac{(b^2-a^2)^2}{8} + \frac{b^2(b^2-a^2)}{4} = \frac{(b^2-a^2)(3b^2-a^2)}{8}$,

$$\boxed{Q = \pi U\left(b^2-a^2\right) + \frac{\pi g}{\nu}\left[\frac{(b^2-a^2)(3b^2-a^2)}{8} - \frac{b^4}{2}\ln\frac{b}{a}\right]} \tag{11}$$

Setting $Q = 0$ and solving for U:

$$\boxed{U_0 = \frac{g}{\nu}\left[\frac{b^4}{2\left(b^2-a^2\right)}\ln\frac{b}{a} - \frac{3b^2-a^2}{8}\right]} \tag{12}$$

For $U > U_0$ the cylinder carries a net upward flux; for $U < U_0$ the layer drains downward.

3.6 Temperature distribution and heat flux to the wall

With $v_r = 0$, $\partial T/\partial z = 0$ and the dissipation $\phi_v = \mu\,(dv_z/dr)^2$ (the only non-zero shear is τ'_{rz}), the energy equation (4.42) reduces to

$$0 = \phi_v + \frac{k}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right), \qquad \phi_v = \mu\left[\frac{g}{2\nu}\left(r - \frac{b^2}{r}\right)\right]^2 = \frac{\rho^2 g^2}{4\mu}\frac{(r^2-b^2)^2}{r^2} \tag{13}$$

Let $\Lambda = \rho^2 g^2/(4\mu k)$. Multiplying by r/k and expanding $(r^2-b^2)^2 = r^4 - 2b^2r^2 + b^4$:

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) = -\Lambda\left(r^3 - 2b^2 r + \frac{b^4}{r}\right) \tag{14}$$

Integrating once, $r\,dT/dr = -\Lambda\left(\dfrac{r^4}{4} - b^2r^2 + b^4\ln r\right) + D_1$. The adiabatic condition $dT/dr\big|_b = 0$ fixes D_1 and gives

$$r\frac{dT}{dr} = -\Lambda\left[\frac{r^4}{4} - b^2r^2 + \frac{3b^4}{4} + b^4\ln\frac{r}{b}\right] \tag{15}$$

Dividing by r and integrating again (the antiderivative of $\ln(r/b)/r$ is $\tfrac12\ln^2(r/b)$) with $T(a) = T_c$:

$$\boxed{T(r) = T_c - \frac{\rho^2 g^2}{4\mu k}\left[\frac{r^4 - a^4}{16} - \frac{b^2(r^2-a^2)}{2} + \frac{3b^4}{4}\ln\frac{r}{a} + \frac{b^4}{2}\left(\ln^2\frac{r}{b} - \ln^2\frac{a}{b}\right)\right]} \tag{16}$$

Heat flux to the wall. The heat flux from the liquid to the cylinder is $\bar q\cdot\bar n$ with $\bar n = -\bar e_r$ (out of the liquid): $q_w = k\,dT/dr\big|_a$, and from (15),

$$\boxed{q_w = -\frac{\rho^2 g^2}{4\mu}\left[\frac{a^3}{4} - a b^2 + \frac{3b^4}{4a} - \frac{b^4}{a}\ln\frac{b}{a}\right]}, \qquad \dot{Q}_w = 2\pi a\, q_w \text{ per unit length} \tag{17}$$

Check. The outer surface is adiabatic, so all the viscous heating generated in the layer must leave through the wall: $\dot{Q}_w = \int_a^b \phi_v \, 2\pi r \, dr$. Evaluating both sides numerically for $\rho = 1000\ \mathrm{kg/m^3}$, $\mu = 0.1\ \mathrm{Pa\,s}$, $k = 0.6\ \mathrm{W/(m\,K)}$, $a = 20\ \mathrm{mm}$ and $h = 5\ \mathrm{mm}$ gives the same value, $6.344\ \mathrm{W/m}$, to the precision of the integration.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The velocity decreases from U at the wall; the shear stress is largest at the wall and zero at the free surface. The cylinder must support the weight of the layer, which is what (9) says. The pressure plays no role because the free surface is at p_a and the layer is fully developed.
- **Limits.** For $h \ll a$ the layer behaves as a flat film: (11) tends to $Q \simeq 2\pi a\left(U h - g h^3/(3\nu)\right)$ and (12) to $U_0 \simeq g h^2/(3\nu)$. For $h \to 0$ the force and the flux vanish. For $g \to 0$ the profile is uniform, $v_z = U$.
- **Dimensional check.** g/ν has units $(\mathrm{m/s^2})/(\mathrm{m^2/s}) = 1/(\mathrm{m\,s})$; in (7) $(g/\nu)\,r^2$ has units m/s. The force per unit length $\rho g \pi b^2$ has units $(\mathrm{kg/m^3})(\mathrm{m/s^2})\,\mathrm{m^2} = \mathrm{N/m}$. In (16), $\Lambda r^4 = \rho^2 g^2 r^4/(\mu k)$ has units $(\mathrm{kg^2\,s^{-4}})/(\mathrm{kg^2\,s^{-4}\,K^{-1}}) = \mathrm{K}$, with μk in $(\mathrm{kg\,m^{-1}s^{-1}})(\mathrm{kg\,m\,s^{-3}K^{-1}})$.
- **Heat balance.** The wall heat flux (17) equals the viscous dissipation of the layer (per unit area of the wall), because no heat leaves through the free surface.

PROBLEM NS6

Sphere with Radial Suction

Original statement · NS6.pdf

Spherical symmetry | Suction | Trajectory and time | Convection-conduction

OFFICIAL PROBLEM STATEMENT

A sphere of radius a is immersed in a liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c. The liquid is sucked with radial velocity $v_r = -V_o$ through the porous surface of the sphere. For the buoyancy-free, steady sphericosymmetrical motion induced in the surrounding liquid,

1. Obtain the velocity $v_r(r)$

2. Compute the trajectory of a fluid particle initially located at $r = r_i > a$, as well as the time t_i at which the fluid particle is sucked by the sphere.

3. Determine the pressure distribution $p - p_\infty$, where p_∞ is the pressure value far from the sphere.

4. Calculate the temperature distribution in the liquid $T(r)$ assuming that the temperature of the sphere is T_s, smaller then the value T_∞ found in the surrounding ambient atmosphere.

5. Obtain the amount of heat transfered to the sphere per unit time.

Transcribed as printed: "sphericosymmetrical", "then" (for "than") and "transfered" are spelling errors of the sheet. The sheet carries no printed figure.

Source: NS6.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 and energy Eq. 4.42 in spherical coordinates, viscous dissipation ϕ_v in spherical coordinates.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, spherically symmetric flow of a perfect liquid (constant ρ, μ, k, c): $\bar{v} = v_r(r)\,\bar{e}_r$, $v_\theta = v_\phi = 0, \partial/\partial\theta = \partial/\partial\phi = \partial/\partial t = 0$.
- Buoyancy-free: the mass force is neglected. No chemical reaction and no radiation. The temperature depends only on r.

- Suction: the porous sphere takes liquid in at the rate $4\pi a^2 V_o$, with $V_o > 0$. Far from the sphere the liquid is at rest at p_∞ and T_∞.
- Data: $[\rho,\ \mu,\ k,\ c,\ a,\ V_o,\ T_s,\ T_\infty,\ p_\infty,\ r_i]$ with $T_s < T_\infty$. Unknowns: $v_r(r), r(t), t_i$, $p(r), T(r)$ and the heat rate to the sphere.
- Dimensionless numbers used: the suction Péclet number $\mathrm{Pe} = \rho c a V_o / k$ and the length $\beta = \rho c a^2 V_o / k = a\,\mathrm{Pe}$.

Symbol	Meaning	SI unit
a	sphere radius	m
V_o	suction velocity at the sphere surface	m/s
r_i, t_i	initial position of the particle, time to reach the sphere	m, s
T_s, T_∞	sphere and ambient temperatures	K
Pe, β	suction Péclet number, length $\beta = a\,\mathrm{Pe}$	1, m
$\Delta T_d = 12\mu V_o/(\rho c a)$	temperature scale of the viscous heating	K

PHASE 2

Frames, Field Form and Boundary Conditions

- Spherical frame (r, θ, ϕ) centred on the sphere; $\bar{e}_r$ points away from the centre; the liquid occupies $r > a$.
- Field form: $\bar{v} = v_r(r)\,\bar{e}_r, p = p(r), T = T(r)$.

Location	Condition	Origin
$r = a$	$v_r = -V_o$	suction through the porous surface
$r \to \infty$	$p \to p_\infty$	statement
$r = a$	$T = T_s$	temperature of the sphere
$r \to \infty$	$T \to T_\infty$	ambient temperature

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the radial velocity

$$\frac{1}{r^2}\frac{d(r^2 v_r)}{dr} = 0 \implies r^2 v_r = \text{const} = -a^2 V_o \tag{1}$$

where the constant is fixed with $v_r(a) = -V_o$:

$$\boxed{v_r(r) = -\frac{a^2 V_o}{r^2}} \tag{2}$$

3.2 Trajectory and suction time

A fluid particle moves with the local velocity: $dr/dt = v_r = -a^2 V_o / r^2$. Separating variables, $r^2\, dr = -a^2 V_o\, dt$. Integrating from r_i at $t = 0$:

$$\frac{r^3 - r_i^3}{3} = -a^2 V_o\, t \implies \boxed{r(t) = \left(r_i^3 - 3a^2 V_o\, t\right)^{1/3}} \tag{3}$$

The particle is sucked when $r = a$: $a^3 = r_i^3 - 3a^2 V_o\, t_i$, so

$$\boxed{t_i = \frac{r_i^3 - a^3}{3\, a^2 V_o}} \tag{4}$$

3.3 Pressure distribution

In the r component of (4.41) the viscous term $\mu\,\partial\left[r^{-2}\partial(r^2 v_r)/\partial r\right]/\partial r$ vanishes because $r^2 v_r$ is constant (1): the flow is irrotational. The rest is the steady Euler equation:

$$\rho\, v_r \frac{dv_r}{dr} = -\frac{dp}{dr} \;\Longrightarrow\; p + \frac{\rho v_r^2}{2} = \text{const} = p_\infty \tag{5}$$

(the constant is evaluated at infinity, where $v_r \to 0$). Substituting (2):

$$\boxed{p - p_\infty = -\frac{\rho\, a^4 V_o^2}{2\, r^4}} \tag{6}$$

The pressure is below the ambient value, the minimum being at the sphere surface, where the liquid has the highest speed.

3.4 Temperature distribution

With $v_\theta = v_\phi = 0$ the energy equation (4.42) in spherical coordinates is

$$\rho c\, v_r \frac{dT}{dr} = \phi_v + \frac{k}{r^2}\frac{d}{dr}\left(r^2 \frac{dT}{dr}\right) \tag{7}$$

The viscous dissipation is $\phi_v = 2\mu\left[(dv_r/dr)^2 + 2(v_r/r)^2\right]$ with $dv_r/dr = 2a^2 V_o / r^3$ and $v_r/r = -a^2 V_o / r^3$:

$$\phi_v = 2\mu\left[\frac{4a^4 V_o^2}{r^6} + \frac{2a^4 V_o^2}{r^6}\right] = \frac{12\mu\, a^4 V_o^2}{r^6} \tag{8}$$

Substituting v_r and ϕ_v in (7), dividing by k and multiplying by r^2:

$$\frac{d}{dr}\left(r^2\frac{dT}{dr}\right) + \beta\frac{dT}{dr} = -\frac{\Gamma}{r^4}, \qquad \beta = \frac{\rho c a^2 V_o}{k}, \quad \Gamma = \frac{12\mu a^4 V_o^2}{k} \tag{9}$$

Homogeneous solution. With $y = T'$: $r^2 y' + (2r + \beta) y = 0$; the integrating factor is $\exp \int (2/r + \beta/r^2)\, dr = r^2 e^{-\beta/r}$, so $y = D\, e^{\beta/r}/r^2$, and since $\int e^{\beta/r} r^{-2} dr = -e^{\beta/r}/\beta$:

$$T_h = B_2 + B_1\, e^{\beta/r} \tag{10}$$

Particular solution. Work with $y = T_p'$ in the first-order form $r^2 y' + (2r + \beta) y = -\Gamma/r^4$. The right-hand side is a power of $1/r$, so try a sum of powers, $y = -\Gamma \left(\frac{c_4}{r^4} + \frac{c_3}{r^3} + \frac{c_2}{r^2} \right)$, and match powers one at a time. The three terms of the equation are $r^2 y' = \Gamma \left(\frac{4c_4}{r^3} + \frac{3c_3}{r^2} + \frac{2c_2}{r} \right)$, $2ry = -\Gamma \left(\frac{2c_4}{r^3} + \frac{2c_3}{r^2} + \frac{2c_2}{r} \right)$ and $\beta y = -\Gamma \left(\frac{\beta c_4}{r^4} + \frac{\beta c_3}{r^3} + \frac{\beta c_2}{r^2} \right)$. Their sum is $\Gamma \left[-\frac{\beta c_4}{r^4} + \frac{2c_4 - \beta c_3}{r^3} + \frac{c_3 - \beta c_2}{r^2} \right]$, which must equal $-\Gamma/r^4$: $c_4 = 1/\beta$, $c_3 = 2c_4/\beta = 2/\beta^2$ and $c_2 = c_3/\beta = 2/\beta^3$. Hence

$$y = T_p' = -\Gamma \left(\frac{1}{\beta r^4} + \frac{2}{\beta^2 r^3} + \frac{2}{\beta^3 r^2} \right)$$

and integrating y with respect to r (the antiderivative of r^{-n} is $-r^{1-n}/(n-1)$):

$$T_p(r) = \Gamma \left(\frac{1}{3\beta r^3} + \frac{1}{\beta^2 r^2} + \frac{2}{\beta^3 r} \right) = \Delta T_d \left[\frac{1}{3} \left(\frac{a}{r} \right)^3 + \frac{1}{\mathrm{Pe}} \left(\frac{a}{r} \right)^2 + \frac{2}{\mathrm{Pe}^2} \frac{a}{r} \right] \tag{11}$$

with $\Delta T_d = 12 \mu V_o / (\rho c a)$ (use $\Gamma/\beta = 12 \mu a^2 V_o/(\rho c)$ and $\beta = a\,\mathrm{Pe}$). The general solution is $T = B_2 + B_1 e^{\beta/r} + T_p(r)$.

Boundary conditions. $T(\infty) = T_\infty$: $e^{\beta/r} \to 1$ and $T_p \to 0$, so $B_2 + B_1 = T_\infty$. $T(a) = T_s$: $B_2 + B_1 e^{\mathrm{Pe}} + T_p(a) = T_s$, because $\beta/a = \mathrm{Pe}$. Subtracting,

$$B_1 = -\frac{T_\infty - T_s + T_p(a)}{e^{\mathrm{Pe}} - 1}, \qquad T_p(a) = \Delta T_d \left(\frac{1}{3} + \frac{1}{\mathrm{Pe}} + \frac{2}{\mathrm{Pe}^2} \right)$$

$$\boxed{T(r) = T_\infty + T_p(r) - [T_\infty - T_s + T_p(a)]\,\frac{e^{\mathrm{Pe}\,a/r} - 1}{e^{\mathrm{Pe}} - 1}} \qquad (12)$$

Discrepancy with the official solution

The handwritten solution gets the particular solution $12\mu V_o/(\rho c\, r)$, as read from the handwriting. Its ODE has the dissipation term as $12\mu V_o^2 a^2/(k\, r^2)$, but multiplying ϕ_v/k of (8) by r^2 gives $12\mu a^4 V_o^2/(k\, r^4)$, as in (9). The term A/r cannot satisfy (9), whose right-hand side decays as r^{-4}: the check above shows that (11) does. The computed result (12) is kept; with $\Delta T_d \to 0$ (negligible dissipation) it reduces to the sheet's result $T = T_\infty + (T_s - T_\infty)\dfrac{e^{\mathrm{Pe}\,a/r} - 1}{e^{\mathrm{Pe}} - 1}$.

3.5 Heat transferred to the sphere

The heat that flows from the liquid into the sphere per unit time is $4\pi a^2\, \bar{q}\cdot\bar{n}$ with $\bar{n} = -\bar{e}_r$ (outward from the liquid) and $\bar{q} = -k\nabla T$, that is $\dot{Q}_s = 4\pi a^2 k\, dT/dr\big|_a$. From (12),

$$\left.\frac{dT}{dr}\right|_a = T_p'(a) + \frac{\mathrm{Pe}}{a}\,[T_\infty - T_s + T_p(a)]\,\frac{e^{\mathrm{Pe}}}{e^{\mathrm{Pe}} - 1}, \qquad T_p'(a) = -\frac{\Delta T_d}{a}\left(1 + \frac{2}{\mathrm{Pe}} + \frac{2}{\mathrm{Pe}^2}\right) \qquad (13)$$

Using $4\pi a\, k\,\mathrm{Pe} = 4\pi\rho c\, a^2 V_o$ and $e^{\mathrm{Pe}}/(e^{\mathrm{Pe}} - 1) = 1/(1 - e^{-\mathrm{Pe}})$:

$$\boxed{\dot{Q}_s = \frac{4\pi\rho c\, a^2 V_o\,[T_\infty - T_s + T_p(a)]}{1 - e^{-\mathrm{Pe}}} - 4\pi a\, k\,\Delta T_d\left(1 + \frac{2}{\mathrm{Pe}} + \frac{2}{\mathrm{Pe}^2}\right)} \qquad (14)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The suction brings ambient liquid at T_∞ towards the cold sphere, which compresses the thermal layer against the surface: $\beta = a\,\mathrm{Pe}$ is the length over which convection and conduction compete. The temperature field (12) has no solution that is regular at infinity for blowing ($V_o < 0$), where the exponent would change sign.

- **Limits.** For negligible dissipation ($\Delta T_d \to 0$): $\dot{Q}_s = 4\pi\rho c a^2 V_o\,(T_\infty - T_s)/(1 - e^{-\mathrm{Pe}})$. For $\mathrm{Pe} \gg 1$ this is $\dot{m}\,c\,(T_\infty - T_s)$ with $\dot{m} = 4\pi\rho a^2 V_o$ the sucked mass flow: the sphere only has to cool the liquid it takes in. For $\mathrm{Pe} \to 0$ it tends to $4\pi a k\,(T_\infty - T_s)$, pure conduction to a sphere in an unbounded medium.
- **Dimensional check.** $a^2 V_o / r^2$ is a velocity. ρv_r^2 in (6) is a pressure. $\beta = \rho c a^2 V_o / k$ has units $(\mathrm{J}/(\mathrm{m^3 K}))\,\mathrm{m^3/s}/(\mathrm{W}/(\mathrm{m\,K})) = \mathrm{m}$. $\Delta T_d = 12\mu V_o/(\rho c a)$ has units $\mathrm{Pa\,s}\cdot\mathrm{m/s}/[(\mathrm{Pa/K})\,\mathrm{m}] = \mathrm{K}$. $\dot{Q}_s$ in (14): $\rho c a^2 V_o\,\Delta T$ has units $(\mathrm{Pa/K})\,\mathrm{m^3/s}\cdot\mathrm{K} = \mathrm{W}$.
- **Check of (11).** Differentiating the three terms of (11) and substituting in (9) reproduces $-\Gamma/r^4$ term by term, as written in 3.4.

PROBLEM NS7

Liquid Film on an Inclined Plane

Original statement · NS7.pdf

Free-surface film | Gravity-driven flow | Viscous dissipation | Energy balance

OFFICIAL PROBLEM STATEMENT

Consider the steady planar motion of a liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c that flows down over an inclined plane of angle α due to the action of gravity, forming a layer of constant thickness h. To solve the problem, use the coordinate system (x, y) indicated in the figure, in which gravity can be expressed as $\bar{g} = g\,\mathrm{sin}(\alpha)\bar{e}_x - g\,\mathrm{cos}(\alpha)\bar{e}_y$. To analyze the resulting liquid flow, assume that the velocity components $v_x(y)$ and $v_y(y)$ are a function of the distance to the wall, and that the air surrounding the liquid has a density $\rho_a \ll \rho$ and a viscosity $\mu_a \ll \mu$, such that in the first approximation it moves without significant pressure differences, exerting on the liquid a negligible viscous force.

▪ Using continuity, determine the velocity component $v_y(y)$.

Obtain the pressure field $p(x, y)$ by integration of the y component of the momentum equation, demonstrating in particular that $\partial p/\partial x = 0$.

Compute $v_x(y)$.

Calculate the force per unit surface that the liquid exerts on the wall.

Determine the temperature in the liquid $T(y)$. In the calculation, assume that the thermal conductivity of air is very small compared with that of the liquid ($k_a \ll k$) and that the wall is at constant uniform temperature T_w.

Obtain the conductive heat flux from the liquid to the wall $(\bar{q} \cdot \bar{n})_w$, where $\bar{n}$ is appropriately oriented.

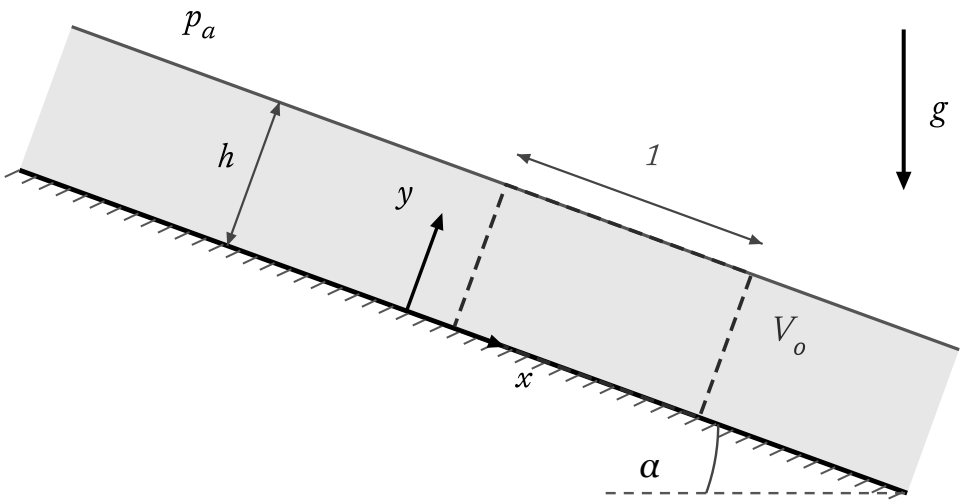
To analyze the energy balances emerging in the problem, consider the fixed control volume V_o of unity length indicated in the figure.

▪ Compute the total energy dissipated per unit time inside the control volume $\int_{V_o} \phi_v \, dV$.

▪ Calculate the work per unit time done by gravity $\int_{V_o} \rho \bar{v} \cdot \bar{g} \, dV$.

▪ Use convervation of energy in the control volume to explain why $(\bar{q} \cdot \bar{n})_w = \int_{V_o} \phi_v \, dV = \int_{V_o} \rho \bar{v} \cdot \bar{g} \, dV$.

Transcribed as printed, including the markers of the sheet and the spelling "convervation".



Statement figure (redrawn from NS7.pdf, p. 1): layer of thickness h on a plane inclined at α, axes x (down the plane) and y (normal to it), gravity $\bar{g} = g\sin\alpha\,\bar{e}_x - g\cos\alpha\,\bar{e}_y$, air at p_a and the fixed control volume V_o of unit length

Source: NS7.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41, energy Eq. 4.42, internal-energy balance Eq. 4.25, total-energy balance Eq. 4.3.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady planar flow of a perfect liquid (constant ρ, μ, k, c): equations (4.40)-(4.42).
- The fields depend only on the distance y to the plane: $\partial/\partial x = 0$ for v_x, v_y and T. The pressure is determined below.
- The plane is impermeable and fixed; the liquid layer has uniform thickness h; no chemical reaction and no radiation.
- The air exerts no pressure variation and a negligible viscous force on the liquid, and its thermal conductivity is negligible: at the free surface $y = h, p = p_a, \tau'_{xy} = 0$ and $\partial T/\partial y = 0$.

- Data: $[\rho,\ \mu,\ k,\ c,\ h,\ \alpha,\ g,\ p_a,\ T_w]$. Unknowns: $v_y, p(x,y), v_x(y)$, the wall force, $T(y)$, the wall heat flux and the three energy integrals.

Symbol	Meaning	SI unit
h	layer thickness	m
α	inclination of the plane	rad
$g\sin\alpha,\ g\cos\alpha$	components of gravity along and normal to the plane	m/s^2
p_a, T_w	air pressure, wall temperature	Pa, K
ρ, μ, k, c	liquid properties	kg/m^3, Pa s, W/(m K), J/(kg K)

PHASE 2

Frames, Field Form and Boundary Conditions

- Cartesian frame (x, y, z) attached to the plane: x down the slope, y normal to the plane and into the liquid, z out of the plane; the basis is orthonormal and right-handed. Gravity has the components $g_x = g\sin\alpha$, $g_y = -g\cos\alpha$. The frame is rotated by α with respect to the horizontal-vertical frame; the rotation matrix between them is orthogonal with determinant 1, since $\bar{e}_x = (\cos\alpha, -\sin\alpha)$ and $\bar{e}_y = (\sin\alpha, \cos\alpha)$ in horizontal-vertical components.
- Field form: $\bar{v} = v_x(y)\,\bar{e}_x + v_y(y)\,\bar{e}_y, T = T(y), p = p(x,y)$.
- Fixed control volume V_o of the figure: $0 < y < h$, unit length in x and unit width; surfaces: the wall Σ_w ($\bar{n} = -\bar{e}_y$), the free surface Σ_s ($\bar{n} = +\bar{e}_y$) and the two faces normal to x.

Location	Condition	Origin
$y = 0$	$v_x = 0, v_y = 0, T = T_w$	fixed impermeable wall at T_w
$y = h$	$p = p_a$	no pressure variations in the air
$y = h$	$\tau'_{xy} = \mu\, dv_x/dy = 0$	negligible viscous force of the air
$y = h$	$dT/dy = 0$	$k_a \ll k$: adiabatic free surface

PHASE 3

Step-by-Step Derivation

3.1 Continuity

$$\frac{\partial v_x}{\partial x} + \frac{dv_y}{dy} = \underbrace{\cancel{\frac{\partial v_x}{\partial x}}}_{v_x = v_x(y)} + \frac{dv_y}{dy} = 0 \implies v_y = \text{const} \tag{1}$$

The wall is impermeable, $v_y(0) = 0$, so

$$\boxed{v_y(y) = 0} \tag{2}$$

3.2 y-momentum and the pressure field

With $v_y = 0$ and $\partial_x = 0$ for the velocity, the inertial and viscous terms of the y component vanish:

$$0 = -\frac{\partial p}{\partial y} - \rho g \cos\alpha \implies p = -\rho g \cos\alpha \, y + f(x) \tag{3}$$

The free-surface condition $p(x, h) = p_a$ for every x gives $f(x) = p_a + \rho g h \cos\alpha$, a constant:

$$\boxed{p(x, y) = p_a + \rho g \cos\alpha \, (h - y)}, \qquad \frac{\partial p}{\partial x} = 0 \tag{4}$$

3.3 x-momentum and the velocity profile

$$\rho\Big(\underbrace{v_x\frac{\partial v_x}{\partial x}}_{\partial_x=0}+\underbrace{v_y\frac{\partial v_x}{\partial y}}_{v_y=0}\Big)=-\underbrace{\frac{\partial p}{\partial x}}_{=0}+\mu\frac{d^2v_x}{dy^2}+\rho g\sin\alpha\;\Longrightarrow\;\frac{d^2v_x}{dy^2}=-\frac{\rho g\sin\alpha}{\mu} \tag{5}$$

Integrating twice, $v_x=-\dfrac{\rho g\sin\alpha}{2\mu}y^2+C_1y+C_2$. The wall condition $v_x(0)=0$ gives $C_2=0$. The free-surface condition $dv_x/dy\big|_h=-\dfrac{\rho g\sin\alpha}{\mu}h+C_1=0$ gives $C_1=\dfrac{\rho g\sin\alpha}{\mu}h$:

$$\boxed{v_x(y)=\frac{\rho g\sin\alpha}{\mu}\left(hy-\frac{y^2}{2}\right)=\frac{gh^2\sin\alpha}{\nu}\left[\frac{y}{h}-\frac{1}{2}\left(\frac{y}{h}\right)^2\right]} \tag{6}$$

3.4 Force per unit surface on the wall

The normal pointing into the liquid is $\bar n=+\bar e_y$, so $(-p\bar{\bar I}+\bar{\bar\tau}')\cdot\bar n$ is the force per unit area that the liquid exerts on the wall. At $y=0$, $p=p_a+\rho gh\cos\alpha$ from (4), $\tau'_{xy}=\mu\,dv_x/dy\big|_0=\rho gh\sin\alpha$ and $\tau'_{yy}=0$:

$$\boxed{\bar f_w=-\left(p_a+\rho gh\cos\alpha\right)\bar e_y+\rho gh\sin\alpha\,\bar e_x} \tag{7}$$

The normal component pushes on the wall with the air pressure plus the hydrostatic pressure of the layer; the tangential component is the shear that drags the wall down the slope.

3.5 Temperature distribution

With $v_y=0$ and $\partial_x=0$ the energy equation (4.42) reduces to $0=\phi_v+k\,d^2T/dy^2$, with $\phi_v=\mu\,(dv_x/dy)^2$ and, from (6), $dv_x/dy=(\rho g\sin\alpha/\mu)(h-y)$:

$$\phi_v=\frac{(\rho g\sin\alpha)^2}{\mu}(h-y)^2,\qquad \frac{d^2T}{dy^2}=-\frac{(\rho g\sin\alpha)^2}{\mu k}(h-y)^2 \tag{8}$$

Integrate once: the antiderivative of $(h-y)^2$ is $-(h-y)^3/3$, so $dT/dy = \frac{(\rho g \sin\alpha)^2}{3\mu k}(h-y)^3 + B_1$. The adiabatic condition $dT/dy\big|_h = 0$ gives $B_1 = 0$. Integrating again, $T = -\frac{(\rho g \sin\alpha)^2}{12\mu k}(h-y)^4 + B_2$, and $T(0) = T_w$ gives $B_2 = T_w + \frac{(\rho g \sin\alpha)^2 h^4}{12\mu k}$:

$$\boxed{T(y) = T_w + \frac{(\rho g \sin\alpha)^2}{12\,\mu k}\left[h^4 - (h-y)^4\right]} \tag{9}$$

Expanding, $h^4 - (h-y)^4 = y\left(4h^3 - 6h^2 y + 4hy^2 - y^3\right)$.

3.6 Heat flux to the wall

The heat flux from the liquid to the wall is $\bar{q}\cdot\bar{n}$ with $\bar{n} = -\bar{e}_y$ (out of the liquid) and $\bar{q} = -k\nabla T$:

$$(\bar{q}\cdot\bar{n})_w = k\frac{dT}{dy}\Big|_{y=0} = k\,\frac{(\rho g\sin\alpha)^2 h^3}{3\mu k} = \boxed{\frac{(\rho g\sin\alpha)^2 h^3}{3\mu}} \tag{10}$$

3.7 Energy balances in the fixed control volume V_o

Total viscous dissipation. The volume V_o has unit length and width and height h, so $dV = dy$:

$$\int_{V_o}\phi_v\,dV = \int_0^h \frac{(\rho g\sin\alpha)^2}{\mu}(h-y)^2\,dy = \frac{(\rho g\sin\alpha)^2}{\mu}\left[-\frac{(h-y)^3}{3}\right]_0^h = \frac{(\rho g\sin\alpha)^2 h^3}{3\mu} \tag{11}$$

Work of gravity. $\bar{v}\cdot\bar{g} = v_x g\sin\alpha$ since $v_y = 0$. The antiderivative of $hy - y^2/2$ is $hy^2/2 - y^3/6$ and Barrow's rule gives $h^3/2 - h^3/6 = h^3/3$:

$$\int_{V_o}\rho\bar{v}\cdot\bar{g}\,dV = \rho g\sin\alpha\int_0^h v_x\,dy = \frac{\rho^2 g^2\sin^2\alpha}{\mu}\left[\frac{hy^2}{2} - \frac{y^3}{6}\right]_0^h = \frac{(\rho g\sin\alpha)^2 h^3}{3\mu} \tag{12}$$

Conservation of energy. In the steady total-energy balance of V_o (Notes.pdf, Eq. 4.3 for a fixed volume), each surface contributes as follows:

Surface	Convective flux	Pressure work	Viscous work	Heat
Wall Σ_w	0 ($\bar{v}\cdot\bar{n}=0$)	0	0 ($\bar{v}=0$)	$-(\bar{q}\cdot\bar{n})_w$
Free surface Σ_s	0 ($v_y=0$)	0 ($v_y=0$)	0 ($\bar{\bar{\tau}}'\cdot\bar{n}=0$)	0 (adiabatic)
Faces normal to x	cancel (profiles independent of x)	cancel	cancel	0

The only remaining term besides the heat flux is the work of the mass force, so $0=\int_{V_o}\rho\bar{g}\cdot\bar{v}\,dV-(\bar{q}\cdot\bar{n})_w$. The internal-energy balance (Notes.pdf, Eq. 4.25), $0=\int_{V_o}\phi_v\,dV-(\bar{q}\cdot\bar{n})_w$ with no compression work, gives the second equality:

$$\boxed{(\bar{q}\cdot\bar{n})_w=\int_{V_o}\phi_v\,dV=\int_{V_o}\rho\bar{v}\cdot\bar{g}\,dV=\frac{(\rho g\sin\alpha)^2h^3}{3\mu}}\qquad(13)$$

Equations (10), (11) and (12) confirm it. The work of gravity on the liquid is entirely dissipated by viscosity and leaves through the wall as heat; the kinetic energy of the liquid in V_o does not change.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The profile (6) is half a parabola: the weight component along the slope is balanced by the shear on the wall, $\tau_w=\rho gh\sin\alpha$, and the pressure is hydrostatic with the normal component of gravity. The wall carries the weight component normal to the plane as pressure.
- **Limits.** For $\alpha=0$ the plane is horizontal and nothing drives the flow: $v_x=0$, no dissipation, no heat flux. For $\alpha=90^\circ$ (vertical wall) the pressure is uniform and equal to p_a and the shear equals the weight of the layer per unit area. The mean velocity is $\bar{v}=gh^2\sin\alpha/(3\nu)$ (flux $Q=\int_0^h v_x\,dy=gh^3\sin\alpha/(3\nu)$).
- **Dimensional check.** gh^2/ν has units $(\mathrm{m/s^2})\,\mathrm{m^2/(m^2/s)}=\mathrm{m/s}$. $\rho gh\sin\alpha$ is a pressure. $(\rho g)^2h^3/\mu$ has units $(\mathrm{N/m^3})^2\,\mathrm{m^3/(Pa\,s)}=\mathrm{N^2\,m^{-3}/(N\,m^{-2}\,s)}=\mathrm{N/(m\,s)}=\mathrm{W/m^2}$. In (9), $(\rho g)^2h^4/(\mu k)$ has units K.

- **Frame check.** With $\bar{e}_X$ horizontal (towards the descending side) and $\bar{e}_Z$ vertical upward, $\bar{e}_x = \cos\alpha\,\bar{e}_X - \sin\alpha\,\bar{e}_Z$ and $\bar{e}_y = \sin\alpha\,\bar{e}_X + \cos\alpha\,\bar{e}_Z$. The matrix of components has $\det R = \cos^2\alpha + \sin^2\alpha = 1$ and $R\,R^T = I$. Then $\bar{g}\cdot\bar{e}_x = (-g\,\bar{e}_Z)\cdot\bar{e}_x = g\sin\alpha$ and $\bar{g}\cdot\bar{e}_y = -g\cos\alpha$, the components given in the statement.

PROBLEM NS8

Porous Cylinder Moving Axially with Suction

Original statement · NS8.pdf

Axisymmetric flow | Power-law profile | Suction Reynolds number | Prandtl number

OFFICIAL PROBLEM STATEMENT

An infinitely long porous cylinder of radius R is surrounded by a liquid of density ρ, viscosity μ, specific heat c, and thermal conductivity k. The cylinder moves axially with velocity $v_z = U$ and entrains liquid through its surface pores with a suction velocity $v_r = -V$, inducing (for $r > R$) an axisymmetric velocity field with components $v_r(r)$, $v_z(r)$ and $v_\theta = 0$. At large radial distances $r \gg R$ from the cylinder the liquid is at rest with uniform pressure p_∞. Assume in the following analysis that the effect of gravity on the fluid motion can be neglected.

1. Use the continuity equation to determine the radial velocity component $v_r(r)$ for $R \le r < \infty$.

2. From the radial component of the momentum equation compute the pressure distribution, demonstrating in particular that $\partial p/\partial z = 0$.

3. Determine the axial velocity distribution $v_z(r)$ as a function of the suction Reynolds number $\mathrm{Re} = \rho V R/\mu$ (to solve the problem, assume that the two independent solutions of the resulting linear ODE are of the form $v_z \propto r^\alpha$, obtaining the two values of α that are compatible with the equation).

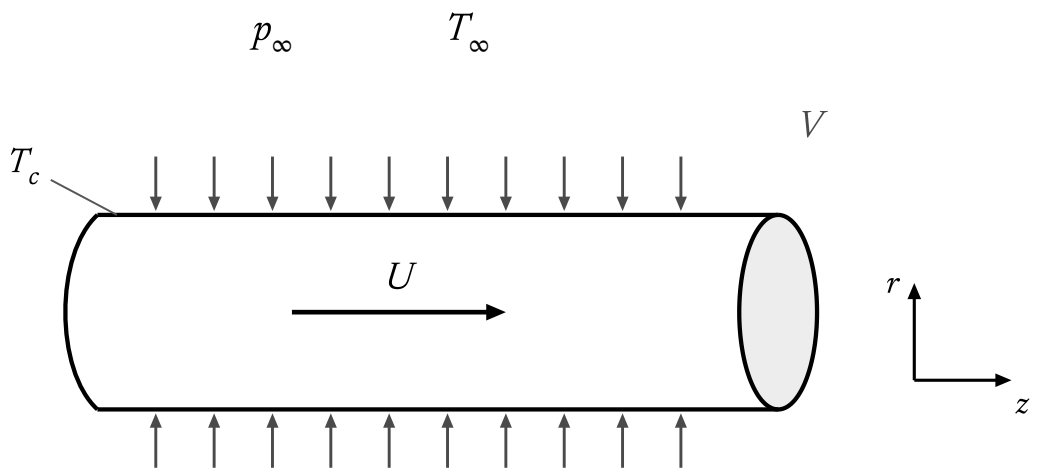
4. Obtain the force acting on the cylinder (per unit length).

5. Calculate the trajectory of the fluid particle located initially at $r = r_o$ and $z = 0$.

6. Compute the temperature field $T(r)$ when the cylinder temperature T_c is different from that of the surrounding atmosphere $T_\infty > T_c$. Neglect viscous dissipation and express the result with use made of the Prandtl number $\mathrm{Pr} = \mu c/k$.

7. Determine the heat transfered to the cylinder per unit time (per unit length).

Transcribed as printed (exam of Dec 16, 2011). On the sheet, the word "the" after "entrains" is struck out, "for $r > R$" is added by hand before "an axisymmetric", and in item 6 the words "$T(r)$" and the sentence "Neglect viscous dissipation and express the result with use made of the Prandtl number $\mathrm{Pr} = \mu c/k$" are handwritten additions. "transfered" is a spelling error of the sheet.



Statement figure (redrawn from NS8.pdf, p. 1): porous cylinder of radius R at temperature T_c moving with U along z, suction velocity V through its surface, ambient p_∞, T_∞ and axes r, z

`Source: NS8.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 and energy Eq. 4.42 in cylindrical coordinates, stress components in cylindrical coordinates.`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady, axisymmetric flow of a perfect liquid (constant ρ, μ, k, c) around an infinitely long cylinder: the fields depend only on r, $\partial/\partial\theta = \partial/\partial z = \partial/\partial t = 0$, with $v_\theta = 0$. Gravity is neglected; no chemical reaction and no radiation; viscous dissipation is neglected in the temperature equation (item 6).
- The cylinder surface is porous: it entrains liquid with $v_r(R) = -V, V > 0$, and moves axially with velocity U.
- Far from the cylinder the liquid is at rest at p_∞ and T_∞.
- Data: $[\rho,\ \mu,\ c,\ k,\ R,\ U,\ V,\ p_\infty,\ T_c,\ T_\infty,\ r_o]$ with $T_\infty > T_c$. Unknowns: $v_r(r), p(r)$, $v_z(r)$, the force, the particle trajectory, $T(r)$ and the heat rate.

- Dimensionless numbers: the suction Reynolds number $\mathrm{Re} = \rho V R/\mu$ and the Prandtl number $\mathrm{Pr} = \mu c/k$.

Symbol	Meaning	SI unit
R	cylinder radius	m
U	axial velocity of the cylinder	m/s
V	suction velocity	m/s
Re, Pr	suction Reynolds number, Prandtl number	1
T_c, T_∞	cylinder and ambient temperatures	K
r_o	initial radius of the fluid particle	m

PHASE 2

Frames, Field Form and Boundary Conditions

- Cylindrical frame (r, θ, z) with z along the axis in the direction of U; the basis $\{\bar{e}_r, \bar{e}_\theta, \bar{e}_z\}$ is orthonormal and right-handed. The liquid occupies $r > R$.
- Field form: $\bar{v} = v_r(r)\,\bar{e}_r + v_z(r)\,\bar{e}_z, p = p(r), T = T(r)$.

Location	Condition	Origin
$r = R$	$v_r = -V$	suction through the porous surface
$r = R$	$v_z = U$	liquid in contact with the surface moves with the cylinder (no-slip)
$r \to \infty$	$v_z \to 0, p \to p_\infty, T \to T_\infty$	liquid at rest far away
$r = R$	$T = T_c$	uniform temperature of the cylinder

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the radial velocity

$$\frac{1}{r}\frac{d(r v_r)}{dr} + \underbrace{\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}}_{v_\theta = 0} + \underbrace{\frac{\partial v_z}{\partial z}}_{\partial_z = 0} = 0 \implies r\,v_r = \text{const} = -RV \qquad (1)$$

where the constant is fixed with $v_r(R) = -V$:

$$\boxed{v_r(r) = -\frac{RV}{r}} \tag{2}$$

3.2 Radial momentum and the pressure

The viscous term of the r component, $\mu\,\partial\left[r^{-1}\partial(r v_r)/\partial r\right]/\partial r$, is zero because $r v_r$ is constant (1); the terms in v_θ vanish. What is left is

$$\rho\, v_r \frac{dv_r}{dr} = -\frac{dp}{dr} \;\Longrightarrow\; p + \frac{\rho v_r^2}{2} = \text{const} = p_\infty \tag{3}$$

(the constant is evaluated at infinity, where $v_r \to 0$):

$$\boxed{p(r) = p_\infty - \frac{\rho R^2 V^2}{2\,r^2}}, \qquad \frac{\partial p}{\partial z} = 0 \tag{4}$$

The pressure depends only on r, hence $\partial p/\partial z = 0$.

3.3 Axial momentum and the velocity profile

The z component of (4.41) with $\partial/\partial z = 0$, $v_\theta = 0$, no gravity and $\partial p/\partial z = 0$:

$$\rho\, v_r \frac{dv_z}{dr} = \mu\,\frac{1}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) \tag{5}$$

Substituting $v_r = -RV/r$ and multiplying by r/μ:

$$-\frac{\rho R V}{\mu}\frac{dv_z}{dr} = \frac{d}{dr}\left(r\frac{dv_z}{dr}\right) \;\Longrightarrow\; \frac{d}{dr}\left(r\frac{dv_z}{dr}\right) + \mathrm{Re}\,\frac{dv_z}{dr} = 0 \tag{6}$$

Following the hint of the statement, try $v_z = r^\alpha$: then $r\,dv_z/dr = \alpha r^\alpha$, $\dfrac{d}{dr}(r\,dv_z/dr) = \alpha^2 r^{\alpha-1}$ and $\mathrm{Re}\,dv_z/dr = \mathrm{Re}\,\alpha\, r^{\alpha-1}$. Substituting in (6) and dividing by $r^{\alpha-1}$:

$$\alpha^2 + \mathrm{Re}\,\alpha = 0 \implies \alpha = 0 \quad \text{or} \quad \alpha = -\mathrm{Re} \tag{7}$$

The two independent solutions are $r^0 = 1$ and $r^{-\mathrm{Re}}$, so $v_z = C_1 + C_2\, r^{-\mathrm{Re}}$. The condition $v_z \to 0$ for $r \to \infty$ requires $C_1 = 0$ (since $r^{-\mathrm{Re}} \to 0$ for $\mathrm{Re} > 0$), and $v_z(R) = U$ gives $C_2 = U R^{\mathrm{Re}}$:

$$\boxed{v_z(r) = U \left(\frac{R}{r} \right)^{\mathrm{Re}}} \tag{8}$$

3.4 Force on the cylinder

The normal pointing into the liquid is $\bar{n} = +\bar{e}_r$, so $(-p\bar{\bar{I}} + \bar{\bar{\tau}}') \cdot \bar{n}$ is the force per unit area that the liquid exerts on the cylinder. The pressure and τ'_{rr} act radially and integrate to zero around the circumference. The axial component is $\tau'_{rz} = \mu\,(\partial v_z/\partial r + \partial v_r/\partial z) = \mu\, dv_z/dr$. From (8), $dv_z/dr = -\mathrm{Re}\, U R^{\mathrm{Re}} r^{-\mathrm{Re}-1}$, so at $r = R$:

$$\tau'_{rz}\Big|_R = -\mu\,\mathrm{Re}\,\frac{U}{R} = -\rho\, V\, U \tag{9}$$

The force per unit length (area $2\pi R$):

$$\boxed{\bar{F} = 2\pi R\, \tau'_{rz}\Big|_R\, \bar{e}_z = -2\pi R\, \rho\, V\, U\, \bar{e}_z} \tag{10}$$

The liquid pulls the cylinder back, with the magnitude of the mass flow rate sucked in per unit length, $\dot{m} = 2\pi R \rho V$, times U: the cylinder must supply to the entrained liquid the axial momentum U per unit mass, which is zero far away.

3.5 Trajectory of a fluid particle

The particle moves with the local velocity: $dr/dt = v_r = -RV/r$ and $dz/dt = v_z$. For the radial motion, separating variables, $r\,dr = -RV\,dt$ and integrating from r_o at $t = 0$:

$$\frac{r^2 - r_o^2}{2} = -RV\,t \;\Longrightarrow\; r(t) = \sqrt{r_o^2 - 2RV\,t}, \qquad t_s = \frac{r_o^2 - R^2}{2RV} \tag{11}$$

where t_s is the time at which the particle reaches the surface ($r = R$). The axial coordinate follows from $dz/dr = v_z/v_r$:

$$\frac{dz}{dr} = -\frac{U}{RV}\left(\frac{R}{r}\right)^{\mathrm{Re}} r = -\frac{U R^{\mathrm{Re}-1}}{V}\, r^{1-\mathrm{Re}} \tag{12}$$

Integrating from r_o (where $z = 0$) to r with $\int r^{1-\mathrm{Re}} dr = r^{2-\mathrm{Re}}/(2-\mathrm{Re})$ for $\mathrm{Re} \neq 2$ and Barrow's rule:

$$\boxed{z = \frac{UR}{V\,(2-\mathrm{Re})}\left[\left(\frac{r_o}{R}\right)^{2-\mathrm{Re}} - \left(\frac{r}{R}\right)^{2-\mathrm{Re}}\right]} \qquad (\mathrm{Re} \neq 2) \tag{13}$$

and, for $\mathrm{Re} = 2$, the integral of $1/r$ gives $z = (UR/V)\ln(r_o/r)$. The particle moves towards the cylinder and is carried along z by the entrained flow; the total axial distance travelled until it reaches the surface is (13) with $r = R$.

3.6 Temperature field

Neglecting the viscous dissipation, the energy equation (4.42) reads

$$\rho c\, v_r \frac{dT}{dr} = k\,\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) \;\Longrightarrow\; -\rho c \frac{RV}{r}\frac{dT}{dr} = \frac{k}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) \tag{14}$$

Multiplying by r/k and using $\rho c RV/k = \dfrac{\rho V R}{\mu}\dfrac{\mu c}{k} = \mathrm{Pr}\,\mathrm{Re}$:

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) + \mathrm{Pr}\,\mathrm{Re}\,\frac{dT}{dr} = 0 \tag{15}$$

This is the same ODE as (6) with Re replaced by $\mathrm{Pr}\,\mathrm{Re}$, so $T = B_1 + B_2\, r^{-\mathrm{Pr}\,\mathrm{Re}}$. The conditions $T(\infty) = T_\infty$ and $T(R) = T_c$ give $B_1 = T_\infty$ and $B_2 = (T_c - T_\infty) R^{\mathrm{Pr}\,\mathrm{Re}}$:

$$\boxed{\frac{T - T_\infty}{T_c - T_\infty} = \left(\frac{R}{r}\right)^{\mathrm{Pr}\,\mathrm{Re}}} \tag{16}$$

3.7 Heat transferred to the cylinder

The heat flux from the liquid to the cylinder is $\bar{q}\cdot\bar{n}$ with $\bar{n} = -\bar{e}_r$ (out of the liquid): $q_c = k\,dT/dr\big|_R$. From (16), $dT/dr\big|_R = -\mathrm{Pr}\,\mathrm{Re}\,(T_c - T_\infty)/R = \mathrm{Pr}\,\mathrm{Re}\,(T_\infty - T_c)/R$. With $k\,\mathrm{Pr}\,\mathrm{Re}/R = \rho c V$ and the area $2\pi R$ per unit length:

$$\boxed{\dot{Q} = 2\pi R\,\rho\,c\,V\,(T_\infty - T_c)} \tag{17}$$

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The suction squeezes the momentum and thermal layers against the cylinder: the velocity decays as $r^{-\mathrm{Re}}$ and the temperature excess as $r^{-\mathrm{Pr}\,\mathrm{Re}}$, with the scales R/Re and $R/(\mathrm{Pr}\,\mathrm{Re})$ close to the wall for $\mathrm{Re} \gg 1$. The thermal layer is thinner than the viscous one when $\mathrm{Pr} > 1$.
- **Limits.** For $V \to 0$ ($\mathrm{Re} \to 0$) the exponent vanishes and (8) tends to the uniform velocity U, which does not decay at infinity: a steady solution that vanishes far away needs the suction. For $U = 0$ there is no axial motion, the force vanishes and (17) remains: the heat is carried by the entrained liquid.
- **Dimensional check.** $\mathrm{Re} = \rho V R/\mu$ is dimensionless: $(\mathrm{kg/m^3})(\mathrm{m/s})\,\mathrm{m}/(\mathrm{Pa\,s}) = 1$. $\rho V U$ in (9) has units $\mathrm{kg/(m^2\,s)}\cdot\mathrm{m/s} = \mathrm{N/m^2}$, and the force per unit length (10) has units N/m. In (17), $\rho c V\,\Delta T$ is $\mathrm{W/m^2}$, times $2\pi R$: W/m.

- **Momentum check.** On a cylindrical surface of any radius r the axial momentum flux is $2\pi r\,\rho v_r v_z = -2\pi\rho R V\, v_z(r)$ and the viscous force is $2\pi r\,\mu\, dv_z/dr = -2\pi\mu\,\mathrm{Re}\, v_z(r) = -2\pi\rho R V\, v_z(r)$, using $r\,dv_z/dr = -\mathrm{Re}\, v_z$ from (8): the two are equal at every r, and at $r = R$ the viscous force is (10).

PROBLEM NS9

Porous Cylinder Quiz with Viscous Heating

Original statement · NS9.pdf

Axisymmetric flow | Suction Reynolds number | Viscous heating | Blowing: no steady solution

OFFICIAL PROBLEM STATEMENT

Consider the steady buoyancy-free motion of a perfect liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c, induced by an infinitely long porous cylinder of radius R that moves with a constant velocity U_w parallel to its axis, while a constant flow rate $q = 2\pi R V_w$ per unit length is uniformly sucked in the radial direction through the porous wall of the cylinder. Far from the cylinder, $r \to \infty$, the fluid is at rest, at pressure p_∞ and temperature T_∞.

To solve the problem, use a cylindrical coordinate system (r, θ, z), where z is the axis of the cylinder, and assume that the velocity field is axisymmetric, $\partial \bar{v}/\partial \theta = 0$, as well as fully developed, $\partial \bar{v}/\partial z = 0$, thus $\bar{v} = v_r(r)\bar{e}_r + v_z(r)\bar{e}_z$.

1. Using continuity, determine the velocity component $v_r(r)$. (1 point)

2. Obtain the pressure field $p(r, z)$ by integration of the r-momentum equation, demonstrating in particular that $\partial p/\partial z = 0$. (2 points)

3. Using the z-momentum equation, obtain the axial velocity distribution, $v_z(r)$. (2 points)

4. Compute the trajectory of a particle initially placed at $r = R_i > R$ and $z = z_i$, as well as the corresponding sucking time, t_s. (1 point)

5. Calculate the force per unit length that the liquid exerts on the cylinder. (1 point)

6. Write the differential equation that determines the temperature in the liquid $T(r)$. Obtain the function $T(r)$, assuming that the wall is at constant uniform temperature T_w. (2 points)

7. Obtain the conductive heat flux per unit length from the liquid to the wall q_w. (1 point)

8. Show that there is no steady solution if fluid is blown through the wall, instead of being sucked. (1 extra point)

The sheet is headed "INTRODUCTION TO FLUID MECHANICS, QUIZ # 2 (Nov 26, 2012)" and carries no figure.

Source: NS9.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 and energy Eq. 4.42 in cylindrical coordinates, viscous dissipation ϕ_v and stress components in cylindrical coordinates.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

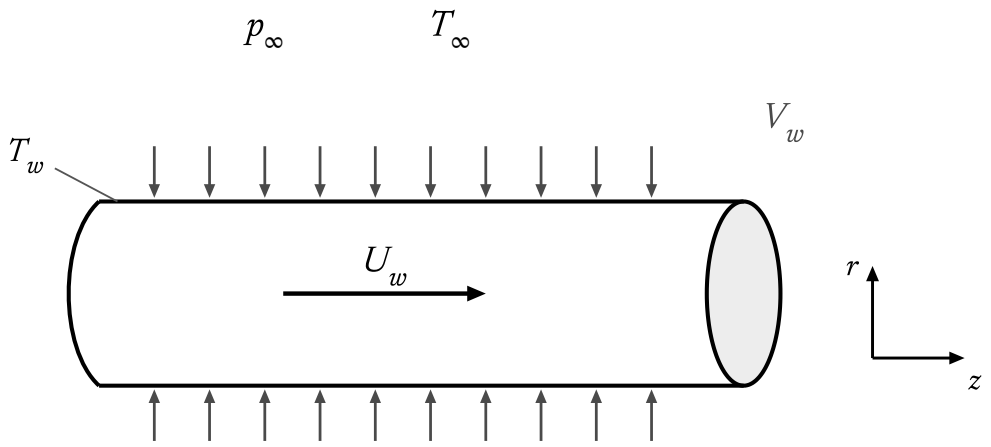
- Steady, buoyancy-free flow of a perfect liquid (constant ρ, μ, k, c) around an infinitely long cylinder: the fields depend only on r ($\partial/\partial\theta = \partial/\partial z = \partial/\partial t = 0$) and $v_\theta = 0$. No chemical reaction and no radiation.
- The wall is porous: it sucks in the volume flow $q = 2\pi R V_w$ per unit length, uniformly, so $v_r(R) = -V_w$ with $V_w > 0$, and it moves axially with U_w.
- Far from the cylinder the liquid is at rest at p_∞ and T_∞.
- Data: $[\rho,\ \mu,\ k,\ c,\ R,\ U_w,\ V_w,\ p_\infty,\ T_\infty,\ T_w,\ R_i,\ z_i]$. Unknowns: v_r, p, v_z, the trajectory and t_s, the force, $T(r)$ and q_w.
- Dimensionless numbers: the suction Reynolds number $\mathrm{Re} = \rho V_w R/\mu = \rho q/(2\pi\mu)$, the Prandtl number $\mathrm{Pr} = \mu c/k$ and the product $\Pi = \mathrm{Pr}\,\mathrm{Re} = \rho c V_w R/k$.

Symbol	Meaning	SI unit
R	cylinder radius	m
U_w, V_w	axial velocity of the cylinder, suction velocity	m/s
$q = 2\pi R V_w$	suction flow rate per unit length	m²/s
Re, Pr, Π	suction Reynolds number, Prandtl number, their product	1
T_w, T_∞	wall and ambient temperatures	K
R_i, z_i	initial radial and axial position of the particle	m

PHASE 2

Frames, Field Form and Boundary Conditions

- Cylindrical frame (r, θ, z) with z along the axis of the cylinder; the basis $\{\bar{e}_r, \bar{e}_\theta, \bar{e}_z\}$ is orthonormal and right-handed. The liquid occupies $r > R$.
- Field form: $\bar{v} = v_r(r)\,\bar{e}_r + v_z(r)\,\bar{e}_z$, $p = p(r, z)$ until determined, $T = T(r)$.



Setup of NS9 (drawn for the solution): porous cylinder of radius R at temperature T_w moving with U_w along z, suction velocity V_w ($q = 2\pi R V_w$ per unit length), fluid at rest at p_∞, T_∞ far away, axes r, z

Location	Condition	Origin
$r = R$	$v_r = -V_w$	uniform suction through the porous wall
$r = R$	$v_z = U_w$	no-slip on the moving cylinder
$r \to \infty$	$v_z \to 0, p \to p_\infty, T \to T_\infty$	fluid at rest far away
$r = R$	$T = T_w$	uniform wall temperature

PHASE 3

Step-by-Step Derivation

3.1 Continuity (item 1)

$$\frac{1}{r}\frac{d(r v_r)}{dr} + \underbrace{\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}}_{v_\theta = 0} + \underbrace{\frac{\partial v_z}{\partial z}}_{\partial_z = 0} = 0 \implies r\,v_r = \text{const} = -R V_w \implies \boxed{v_r(r) = -\frac{R V_w}{r} = -\frac{q}{2\pi r}} \quad (1)$$

3.2 Radial momentum and the pressure (item 2)

The viscous term of the r component vanishes because $r v_r$ is constant, and the θ and z terms are zero:

$$\rho\, v_r \frac{dv_r}{dr} = -\frac{\partial p}{\partial r} \;\Longrightarrow\; p + \frac{\rho v_r^2}{2} = f(z) \tag{2}$$

where integration at fixed z leaves a function of z. The condition $p \to p_\infty$ for $r \to \infty$ (where $v_r \to 0$) holds for every z, so $f(z) = p_\infty$ is a constant:

$$\boxed{p(r) = p_\infty - \frac{\rho R^2 V_w^2}{2 r^2}}, \qquad \frac{\partial p}{\partial z} = 0 \tag{3}$$

3.3 Axial momentum (item 3)

The z component of (4.41), with $\partial_z = 0$, $v_\theta = 0$, no mass force and $\partial p/\partial z = 0$, is $\rho v_r\, dv_z/dr = \mu\, r^{-1} d(r\, dv_z/dr)/dr$. Substituting (1) and multiplying by r/μ:

$$\frac{d}{dr}\left(r \frac{dv_z}{dr} \right) + \mathrm{Re}\, \frac{dv_z}{dr} = 0 \tag{4}$$

Try $v_z = r^\alpha$: $\alpha^2 + \mathrm{Re}\,\alpha = 0$, so $\alpha = 0$ or $\alpha = -\mathrm{Re}$ and $v_z = C_1 + C_2 r^{-\mathrm{Re}}$. The condition at infinity gives $C_1 = 0$ and $v_z(R) = U_w$ gives $C_2 = U_w R^{\mathrm{Re}}$:

$$\boxed{v_z(r) = U_w \left(\frac{R}{r} \right)^{\mathrm{Re}}} \tag{5}$$

3.4 Trajectory and sucking time (item 4)

The velocity of the particle is the local velocity. Radial motion: $dr/dt = -RV_w/r$, so $r\,dr = -RV_w\,dt$; integrating from R_i at $t=0$, $(r^2 - R_i^2)/2 = -RV_w t$:

$$r(t) = \sqrt{R_i^2 - 2RV_w t}, \qquad \boxed{t_s = \frac{R_i^2 - R^2}{2RV_w}} \tag{6}$$

Axial motion: $dz/dr = v_z/v_r = -\dfrac{U_w R^{\mathrm{Re}-1}}{V_w} r^{1-\mathrm{Re}}$. Integrating from R_i (where $z = z_i$), with $\int r^{1-\mathrm{Re}} dr = r^{2-\mathrm{Re}}/(2-\mathrm{Re})$ for $\mathrm{Re} \neq 2$:

$$\boxed{z(r) = z_i + \frac{U_w R}{V_w\,(2-\mathrm{Re})}\left[\left(\frac{R_i}{R}\right)^{2-\mathrm{Re}} - \left(\frac{r}{R}\right)^{2-\mathrm{Re}}\right]} \qquad (\mathrm{Re} \neq 2) \tag{7}$$

For $\mathrm{Re} = 2$, $z = z_i + (U_w R/V_w)\ln(R_i/r)$. The axial position at which the particle is sucked is (7) with $r = R$.

3.5 Force on the cylinder (item 5)

The normal pointing into the liquid is $\bar{n} = +\bar{e}_r$; the pressure and the radial normal stress act radially and integrate to zero around the circumference. The axial shear stress is $\tau'_{rz} = \mu\, dv_z/dr$, and from (5), $dv_z/dr\big|_R = -\mathrm{Re}\,U_w/R$:

$$\tau'_{rz}\Big|_R = -\mu\,\mathrm{Re}\,\frac{U_w}{R} = -\rho V_w U_w, \qquad \boxed{\bar{F} = 2\pi R\,\tau'_{rz}\Big|_R\,\bar{e}_z = -\rho\,q\,U_w\,\bar{e}_z} \tag{8}$$

The force per unit length equals minus the axial momentum given per unit time to the sucked liquid, $\rho q\, U_w$.

3.6 Temperature: the equation (item 6)

The energy equation (4.42) with $v_r = -RV_w/r$, $\partial_z = \partial_\theta = 0$ and the dissipation of the cylindrical expression (Notes.pdf) keeping the non-zero terms, $\phi_v = \mu\left\{2\left[(dv_r/dr)^2 + (v_r/r)^2\right] + (dv_z/dr)^2\right\}$:

$$\rho c\, v_r \frac{dT}{dr} = \phi_v + \frac{k}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right), \qquad \phi_v = \mu\left[\frac{4R^2V_w^2}{r^4} + \frac{\mathrm{Re}^2 U_w^2}{r^2}\left(\frac{R}{r}\right)^{2\mathrm{Re}}\right] \tag{9}$$

since $dv_r/dr = RV_w/r^2$ and $v_r/r = -RV_w/r^2$. Multiplying by r/k and using $\rho c R V_w/k = \Pi$:

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) + \Pi\,\frac{dT}{dr} = -\frac{\mu}{k}\left[\frac{4R^2V_w^2}{r^3} + \frac{\mathrm{Re}^2U_w^2R^{2\mathrm{Re}}}{r^{2\mathrm{Re}+1}}\right] \tag{10}$$

with the boundary conditions $T(R) = T_w$ and $T(\infty) = T_\infty$.

3.7 Temperature: the solution (item 6)

Write $y = dT/dr$, so that (10) is $r\,y' + (1+\Pi)\,y = -S_1 r^{-3} - S_2 r^{-2\mathrm{Re}-1}$ with $S_1 = 4\mu R^2 V_w^2/k$ and $S_2 = \mu\,\mathrm{Re}^2 U_w^2 R^{2\mathrm{Re}}/k$. The homogeneous solution is $y = D\,r^{-1-\Pi}$ (substitute r^s: $s + 1 + \Pi = 0$). For each source term try a power of the same form: $y = A_1 r^{-3}$ gives $r\,y' = -3A_1 r^{-3}$ and $(\Pi - 2)A_1 = -S_1$; $y = A_2 r^{-2\mathrm{Re}-1}$ gives $(\Pi - 2\mathrm{Re})A_2 = -S_2$. Hence, for $\Pi \neq 2$ and $\mathrm{Pr} \neq 2$,

$$A_1 = \frac{S_1}{2-\Pi}, \qquad A_2 = \frac{S_2}{\mathrm{Re}\,(2-\mathrm{Pr})}$$

Integrating y ($\int r^{-n}dr = -r^{1-n}/(n-1)$) and writing $\mu/k = \mathrm{Pr}/c$:

$$T(r) = B - a_1\left(\frac{R}{r}\right)^2 - a_2\left(\frac{R}{r}\right)^{2\mathrm{Re}} + E\left(\frac{R}{r}\right)^{\Pi}, \qquad a_1 = \frac{2\,\mathrm{Pr}\,V_w^2}{c\,(2-\Pi)}, \quad a_2 = \frac{\mathrm{Pr}\,U_w^2}{2c\,(2-\mathrm{Pr})} \tag{11}$$

For $r \to \infty$ every power of R/r vanishes, so $T(\infty) = T_\infty$ gives $B = T_\infty$; $T(R) = T_w$ gives $E = T_w - T_\infty + a_1 + a_2$:

$$\boxed{T(r) = T_\infty - a_1\left(\frac{R}{r}\right)^{2} - a_2\left(\frac{R}{r}\right)^{2\mathrm{Re}} + (T_w - T_\infty + a_1 + a_2)\left(\frac{R}{r}\right)^{\Pi}} \tag{12}$$

The terms a_1 and a_2 are the viscous heating by the radial and by the axial deformation. The cases $\Pi = 2$ and $\mathrm{Pr} = 2$ (resonance of the exponents) give logarithmic terms and are not worked out here.

3.8 Heat flux to the wall (item 7)

The heat flux from the liquid to the wall is $\bar{q}\cdot\bar{n}$ with $\bar{n} = -\bar{e}_r$: $k\,dT/dr\big|_R$. Differentiating (12) and setting $r = R$:

$$\left.\frac{dT}{dr}\right|_R = \frac{1}{R}\left[2a_1 + 2\mathrm{Re}\,a_2 - \Pi\,(T_w - T_\infty + a_1 + a_2)\right] = \frac{1}{R}\left[\Pi\,(T_\infty - T_w) + (2-\Pi)a_1 + (2\mathrm{Re} - \Pi)a_2\right]$$

with $(2-\Pi)a_1 = 2\mathrm{Pr}\,V_w^2/c$ and $(2\mathrm{Re}-\Pi)a_2 = \mathrm{Re}(2-\mathrm{Pr})a_2 = \mathrm{Pr}\,\mathrm{Re}\,U_w^2/(2c)$. Multiplying by k and using $k\,\Pi/R = \rho c V_w$, $k\,\mathrm{Pr}/c = \mu$ and $\mu\,\mathrm{Re}/R = \rho V_w$:

$$k\left.\frac{dT}{dr}\right|_R = \rho c V_w\,(T_\infty - T_w) + \frac{2\mu V_w^2}{R} + \frac{\rho V_w U_w^2}{2}$$

and, multiplying by the area per unit length $2\pi R$,

$$\boxed{q_w = 2\pi R\rho V_w\left[c\,(T_\infty - T_w) + \frac{U_w^2}{2}\right] + 4\pi\mu V_w^2} \tag{13}$$

3.9 Blowing instead of suction (item 8)

If the liquid is blown, $V_w < 0$ and $\mathrm{Re} < 0$. Continuity still gives $v_r = -RV_w/r = R|V_w|/r > 0$, and the equation for v_z is (4) with $\mathrm{Re} < 0$: $v_z = C_1 + C_2\,r^{|\mathrm{Re}|}$. The condition $v_z \to 0$ for $r \to \infty$ requires $C_2 = 0$ (the power grows) and then $C_1 = 0$, so $v_z(R) = U_w$ cannot be satisfied unless $U_w = 0$. Likewise $T = B + D\,r^{-\Pi}$ with $\Pi < 0$ is unbounded unless $D = 0$, which cannot satisfy $T(R) = T_w \neq T_\infty$. Therefore there is no

steady solution with the fluid at rest and at T_∞ far away: the blown liquid carries momentum and heat outward and they cannot be balanced by diffusion towards the cylinder.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** Suction confines the axial momentum to a layer of thickness of order R/Re and the thermal layer to R/Π: the thermal layer is thinner than the momentum layer when $\mathrm{Pr} > 1$. The cylinder pays for the entrained momentum with the force (8), and the wall receives the enthalpy, kinetic-energy and viscous-heating terms of (13).
- **Limits.** For $U_w = 0$, $a_2 = 0$ and E contains only a_1; for negligible dissipation ($a_1 = a_2 = 0$) the solution is $T = T_\infty + (T_w - T_\infty)(R/r)^{\Pi}$ and $q_w = 2\pi R \rho c V_w (T_\infty - T_w)$, the result of NS8. For $V_w \to 0$ the axial profile is uniform and there is no steady solution that vanishes at infinity.
- **Dimensional check.** $\mathrm{Re} = \rho V_w R/\mu$ is dimensionless. $\rho V_w U_w$ has units N/m^2. a_1 and a_2 have units $\mathrm{Pr}\, V_w^2/c = \mathrm{m}^2/\mathrm{s}^2/(\mathrm{J}/(\mathrm{kg\,K})) = \mathrm{K}$. In (13) every term is W/m: $\rho R V_w c\, \Delta T = (\mathrm{kg/m^3})\,\mathrm{m}\cdot\mathrm{m/s}\cdot\mathrm{J}/(\mathrm{kg\,K})\cdot\mathrm{K} = \mathrm{W/m}$ and $\mu V_w^2 = (\mathrm{Pa\,s})(\mathrm{m}^2/\mathrm{s}^2) = \mathrm{W/m}$.
- **Energy check.** A steady total-energy balance on the liquid between the wall and infinity gives the same result (13): the energy flux $\rho q\, c T_\infty$ enters at infinity; through the wall leave $\rho q\,[c T_w + (U_w^2 + V_w^2)/2]$ and the pressure work $p(R)\,q$ with $p(R) = p_\infty - \rho V_w^2/2$; the viscous stresses at the wall do the work $-4\pi\mu V_w^2 - \rho q\, U_w^2$, and the wall receives q_w. The radial kinetic energy cancels against the pressure drop, and what remains is (13).

PROBLEM NS10

Underwater Explosion Cavity

Original statement · NS10.pdf

Spherical symmetry | Null stress condition | Non-dimensionalization | Limits Re large and small

OFFICIAL PROBLEM STATEMENT

As a consequence of an underwater explosion, a spherical cavity of initial radius R_0 and initial radial velocity V_0 is created inside the ocean. The growth of the cavity induces a flow in the surrounding liquid, whose density ρ and viscosity μ can be considered constant. It is also assumed that the effect of gravity is negligible, and that the motion is sphericosymmetric, $\bar{v} = v_r(r,t)\,\bar{e}_r$, where $R_c(t) \leq r < \infty$, being $R_c(t)$ the radius of the cavity at time t. The liquid is at rest far from the cavity, $r \to \infty$, where the pressure is p_∞, while the pressure inside the cavity can be neglected compared to p_∞.

1. (1.5 points). Find the radial velocity of the liquid, $v_r(r,t)$, as a function of the radius of the cavity, $R_c(t)$, and of its associated expansion velocity, $\dot{R}_c(t) = dR_c/dt$. Show that the resulting flow is irrotational.

2. (1.5 points). Obtain the pressure field, $p(r,t)$, as a function of $R_c(t)$, $\dot{R}_c(t)$, $\ddot{R}_c(t)$, ρ and p_∞.

3. (1 point). Evaluate the resulting pressure distribution at $r = R_c(t)$, to find the ordinary differential equation (ODE) with initial conditions satisfied by $R_c(t)$, in terms of the unknown liquid pressure at the cavity boundary, $p_R(t)$.

4. (1.5 points). Find $p_R(t)$ by imposing the condition of null stress of the liquid at the cavity wall, $\bar{\bar{\tau}} \cdot \bar{n}\big|_{r=R_c(t)} = 0$, where $\bar{\bar{\tau}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}'$ is the stress tensor.

5. (1.5 points). Using the answer to question #4, write the ODE obtained in question #3 in closed form. Nondimensionalize the corresponding initial value problem, using the following dimensionless variables: $R = R_c/R_0$, $\tau = tV_0/R_0$. Show that the problem is governed by two dimensionless parameters, namely the Reynolds number, $\mathrm{Re} = \rho V_0 R_0/\mu$, and $\Lambda = p_\infty/(\rho V_0^2)$.

6. (1.5 points). Simplify the ODE in the limit $\mathrm{Re} \to \infty$. By using $z = \ln(R)$ as the independent variable and $F = (dR/d\tau)^2$ as the dependent one, show that $F(z)$ satisfies the linear ODE

$$\frac{dF}{dz} + 3F = -2\Lambda,$$

with initial condition $F(0) = 1$. Obtain $F(z)$, and use it to find $dR/d\tau$ as a function of R.

7. (1.5 points). Obtain the maximum radius of the cavity, $R_{\max}$, as a function of Λ, and show that the time $\tau_{\max}$, defined such that $R(\tau_{\max}) = R_{\max}$ is given by the integral

$$\tau_{\max}(\Lambda) = \int_1^{\left(1+\frac{3}{2\Lambda}\right)^{1/3}} \frac{dx}{\sqrt{x^{-3} + \frac{2\Lambda}{3}\left(x^{-3}-1\right)}}.$$

Provide an explicit approximation for $\tau_{\max}$ in the limit $\Lambda \ll 1$.

8. (1 additional point). Simplify the ODE found in question #5 in the limit $\mathrm{Re} \to 0$. Is it possible to satisfy the initial conditions? Find an approximate solution, and discuss its validity.

The sheet is headed "INTRODUCTION TO FLUID MECHANICS, Dec 18, 2012, PROBLEM 2 (1 hour)" and carries no figure; the spelling "sphericosymmetric" is the sheet's.

`Source: NS10.pdf (page 1 of 3: statement; pages 2 and 3: handwritten solution). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 in spherical coordinates, stress components in spherical coordinates.`

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Incompressible liquid with constant ρ and μ, spherically symmetric flow $\bar{v} = v_r(r,t)\,\bar{e}_r$ in $r > R_c(t)$, gravity neglected. Equations (4.40) and (4.41) in spherical coordinates.
- The cavity gas pressure is neglected compared with p_∞: the liquid has no traction applied at the cavity wall (null stress). The liquid is at rest and at p_∞ far away.
- Initial conditions: $R_c(0) = R_0$, $\dot{R}_c(0) = V_0$.
- Data: $[\rho,\ \mu,\ p_\infty,\ R_0,\ V_0]$. Unknowns: $v_r, p, p_R(t)$, the ODE for $R_c(t)$ and its non-dimensional limits.

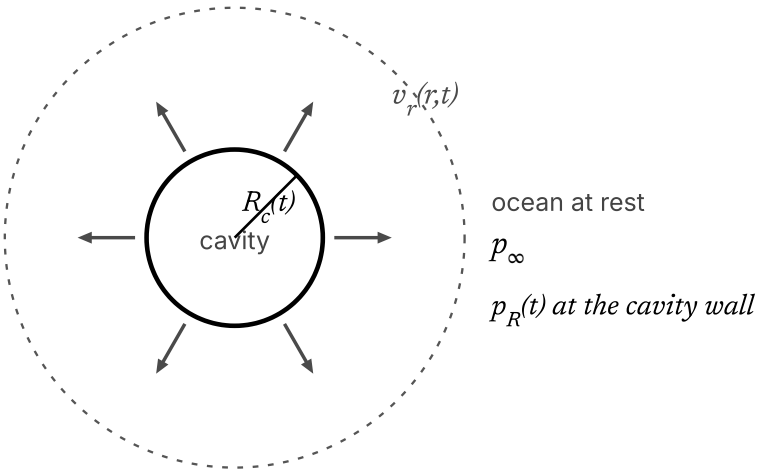
Symbol	Meaning	SI unit
$R_c(t)$, $\dot{R}_c$, $\ddot{R}_c$	cavity radius, velocity and acceleration	m, m/s, m/s^2
R_0, V_0	initial radius and initial expansion velocity	m, m/s
$p_R(t)$	liquid pressure at the cavity wall	Pa
$\mathrm{Re} = \rho V_0 R_0/\mu$, $\Lambda = p_\infty/(\rho V_0^2)$	dimensionless parameters	1

Symbol	Meaning	SI unit
$R = R_c/R_0, \tau = tV_0/R_0$	dimensionless radius and time	1

PHASE 2

Frames, Field Form and Boundary Conditions

- Spherical frame (r, θ, ϕ) centred on the cavity centre, fixed; the liquid occupies $r > R_c(t)$. The basis is orthonormal and right-handed.
- Field form: $\bar{v} = v_r(r,t)\,\bar{e}_r, p = p(r,t)$.



Setup of NS10 (drawn for the solution): cavity of radius $R_c(t)$ ($R_c(0) = R_0$, $\dot{R}_c(0) = V_0$) expanding in the ocean, radial velocity field $v_r(r,t)$, pressure $p_R(t)$ at the cavity wall and p_∞ far away

Location	Condition	Origin
$r = R_c(t)$	$v_r = \dot{R}_c$	the liquid at the cavity wall moves with it
$r \to \infty$	$v_r \to 0, p \to p_\infty$	liquid at rest far away
$r = R_c(t)$	$\bar{\bar{\tau}} \cdot \bar{n} = 0, \bar{n} = \bar{e}_r$	cavity pressure neglected, no traction
$t = 0$	$R_c = R_0, \dot{R}_c = V_0$	statement

PHASE 3

Step-by-Step Derivation

3.1 Velocity and irrotationality (item 1)

$$\frac{1}{r^2}\frac{\partial(r^2 v_r)}{\partial r} = 0 \implies r^2 v_r = A(t), \qquad v_r(R_c,t) = \dot{R}_c \implies A(t) = R_c^2 \dot{R}_c \tag{1}$$

$$\boxed{v_r(r,t) = \frac{R_c^2 \dot{R}_c}{r^2}} \tag{2}$$

The vorticity of $\bar{v} = v_r(r,t)\bar{e}_r$, from the curl in spherical coordinates, has the components $\omega_r = \frac{1}{r\sin\theta}\left[\frac{\partial(v_\phi \sin\theta)}{\partial\theta} - \frac{\partial v_\theta}{\partial\phi}\right] = 0, \omega_\theta = \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial\phi} - \frac{\partial(r v_\phi)}{\partial r}\right] = 0$ and $\omega_\phi = \frac{1}{r}\left[\frac{\partial(r v_\theta)}{\partial r} - \frac{\partial v_r}{\partial\theta}\right] = 0$, because $v_\theta = v_\phi = 0$ and v_r does not depend on θ or ϕ. Hence $\nabla \wedge \bar{v} = 0$: **the flow is irrotational**, and therefore $\nabla \wedge (\nabla \wedge \bar{v}) = 0$.

3.2 Pressure field (item 2)

The viscous force of the incompressible flow is $-\mu \nabla \wedge (\nabla \wedge \bar{v}) = 0$, so the r component of (4.41) is the Euler equation:

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r}\right) = -\frac{\partial p}{\partial r} \tag{3}$$

With $\partial v_r/\partial t = (R_c^2 \ddot{R}_c + 2R_c \dot{R}_c^2)/r^2$ (product rule on $R_c^2 \dot{R}_c$) and $v_r\, \partial v_r/\partial r = -2R_c^4 \dot{R}_c^2 / r^5$:

$$\frac{\partial p}{\partial r} = -\rho\left[\frac{R_c^2 \ddot{R}_c + 2R_c \dot{R}_c^2}{r^2} - \frac{2R_c^4 \dot{R}_c^2}{r^5}\right] \tag{4}$$

Integrating from r to ∞ with $p(\infty) = p_\infty$ (the antiderivatives are $-1/r$ and $-1/(4r^4)$, and Barrow's rule gives $1/r$ and $1/(4r^4)$):

$$\boxed{\frac{p(r,t) - p_\infty}{\rho} = \frac{R_c^2 \ddot{R}_c + 2R_c \dot{R}_c^2}{r} - \frac{R_c^4 \dot{R}_c^2}{2r^4}} \tag{5}$$

3.3 Equation at the cavity wall (item 3)

At $r = R_c(t)$, (5) gives $\dfrac{p_R - p_\infty}{\rho} = R_c \ddot{R}_c + 2\dot{R}_c^2 - \dfrac{\dot{R}_c^2}{2}$:

$$\boxed{R_c \ddot{R}_c + \frac{3}{2}\dot{R}_c^2 = \frac{p_R(t) - p_\infty}{\rho}, \qquad R_c(0) = R_0, \;\; \dot{R}_c(0) = V_0} \tag{6}$$

3.4 Pressure at the wall: null stress (item 4)

The unit normal to the cavity wall, pointing into the liquid, is $\bar{n} = \bar{e}_r$. The stress vector is $\bar{\bar{\tau}} \cdot \bar{e}_r = (-p + \tau'_{rr})\bar{e}_r + \tau'_{r\theta}\bar{e}_\theta + \tau'_{r\phi}\bar{e}_\phi$. For this flow $\tau'_{r\theta} = \tau'_{r\phi} = 0$ and (Notes.pdf, stress components in spherical coordinates)

$$\tau'_{rr} = 2\mu \frac{\partial v_r}{\partial r} = -\frac{4\mu R_c^2 \dot{R}_c}{r^3}, \qquad \tau'_{rr}\Big|_{R_c} = -\frac{4\mu \dot{R}_c}{R_c} \tag{7}$$

The null-stress condition at $r = R_c$ gives $-p_R + \tau'_{rr}(R_c) = 0$:

$$\boxed{p_R(t) = -\frac{4\mu \dot{R}_c}{R_c}} \tag{8}$$

3.5 Closed ODE and non-dimensionalization (item 5)

Substituting (8) in (6) with $\nu = \mu/\rho$:

$$R_c \ddot{R}_c + \frac{3}{2}\dot{R}_c^2 + \frac{4\nu \dot{R}_c}{R_c} = -\frac{p_\infty}{\rho} \tag{9}$$

With $R_c = R_0 R$ and $t = R_0 \tau / V_0$: $\dot{R}_c = V_0\, R'$ and $\ddot{R}_c = (V_0^2/R_0) R''$ (chain rule, $d/dt = (V_0/R_0)\, d/d\tau$, with $R' = dR/d\tau$). Substituting and dividing by V_0^2:

$$\boxed{R\,R'' + \frac{3}{2}R'^2 + \frac{4}{\mathrm{Re}}\frac{R'}{R} = -\Lambda, \qquad R(0) = 1,\;\; R'(0) = 1} \tag{10}$$

with $\mathrm{Re} = \rho V_0 R_0/\mu$ (because $\nu/(V_0 R_0) = 1/\mathrm{Re}$) and $\Lambda = p_\infty/(\rho V_0^2)$. The problem depends only on these two parameters.

3.6 Limit $\mathrm{Re} \to \infty$ (item 6)

The viscous term drops out: $R\,R'' + \frac{3}{2}R'^2 = -\Lambda$. With $z = \ln R$ and $F = (dR/d\tau)^2$, the chain rule gives $dF/d\tau = 2R'R''$ and $dz/d\tau = R'/R$, so

$$\frac{dF}{dz} = \frac{dF/d\tau}{dz/d\tau} = 2R\,R'' \tag{11}$$

and the equation becomes $\frac{1}{2}\frac{dF}{dz} + \frac{3}{2}F = -\Lambda$, that is $\frac{dF}{dz} + 3F = -2\Lambda$ with $F(0) = R'(0)^2 = 1$ (at $\tau = 0$, $R = 1$ and $z = 0$). Multiplying by the integrating factor e^{3z}: $(F\,e^{3z})' = -2\Lambda e^{3z}$, so $F\,e^{3z} = -\frac{2\Lambda}{3}e^{3z} + C$ and $F(0) = 1$ gives $C = 1 + 2\Lambda/3$:

$$F(z) = e^{-3z} + \frac{2\Lambda}{3}\left(e^{-3z} - 1\right) \tag{12}$$

With $e^{-3z} = R^{-3}$:

$$\boxed{\frac{dR}{d\tau} = \left[R^{-3} + \frac{2\Lambda}{3}\left(R^{-3} - 1\right)\right]^{1/2}} \tag{13}$$

(positive root while the cavity expands).

3.7 Maximum radius and time to reach it (item 7)

The cavity stops growing when $dR/d\tau = 0$: $R_{\max}^{-3}(1 + 2\Lambda/3) = 2\Lambda/3$, that is

$$\boxed{R_{\max} = \left(1 + \frac{3}{2\Lambda}\right)^{1/3}} \tag{14}$$

Separating variables in (13), $d\tau = dR/\sqrt{R^{-3} + \frac{2\Lambda}{3}(R^{-3} - 1)}$, and integrating from $R = 1$ at $\tau = 0$ to $R_{\max}$ at $\tau_{\max}$ (with the dummy variable x):

$$\boxed{\tau_{\max}(\Lambda) = \int_1^{(1+3/(2\Lambda))^{1/3}} \frac{dx}{\sqrt{x^{-3} + \frac{2\Lambda}{3}\left(x^{-3} - 1\right)}}} \tag{15}$$

Approximation for $\Lambda \ll 1$. Then $R_{\max} \simeq (3/(2\Lambda))^{1/3} \gg 1$. Factor x^{-3} out of the square root: the integrand is $x^{3/2}/\sqrt{(1 + 2\Lambda/3) - (2\Lambda/3)x^3}$ and, with $2\Lambda/3 \simeq R_{\max}^{-3}$ for small Λ,

$$\tau_{\max} \simeq \int_1^{R_{\max}} \frac{x^{3/2}\,dx}{\sqrt{1 - (x/R_{\max})^3}} \simeq R_{\max}^{5/2} \int_0^1 \frac{s^{3/2}\,ds}{\sqrt{1 - s^3}} \tag{16}$$

(with $x = R_{\max} s$, and the lower limit $s = 1/R_{\max} \to 0$). The substitution $u = s^3$ gives $s^{3/2}ds = \tfrac{1}{3}u^{-1/6}du$, so the integral is $\tfrac{1}{3}B(\tfrac{5}{6}, \tfrac{1}{2}) = \dfrac{\sqrt{\pi}\,\Gamma(5/6)}{3\,\Gamma(4/3)} \simeq 0.747$:

$$\boxed{\tau_{\max} \simeq 0.747 \left(\frac{3}{2\Lambda}\right)^{5/6} = 0.747 \left(\frac{3}{2}\right)^{5/6} \Lambda^{-5/6}} \qquad (\Lambda \ll 1) \tag{17}$$

Discrepancy with the official solution

The handwritten solution approximates the integrand by $x^{3/2}$ over the whole range and obtains $\tau_{\max} \simeq \frac{2}{5}(3/2)^{5/6}\Lambda^{-5/6}$, which neglects the factor $(1 - x^3/R_{\max}^3)^{-1/2}$ of (16). Integrating (15) numerically gives $\tau_{\max} = 48.3,\ 330.8,\ 2255.4$ for $\Lambda = 10^{-2},\ 10^{-3},\ 10^{-4}$, while (17) gives $48.6,\ 331.1,\ 2255.8$ and the sheet's expression $26.0,\ 177.3,\ 1208.2$. The scaling $\Lambda^{-5/6}$ is the same; the computed coefficient 0.747 is kept.

3.8 Limit $\mathrm{Re} \to 0$ (item 8)

For $\mathrm{Re} \to 0$ the viscous term dominates the inertial terms and (10) reduces to the first-order ODE

$$\frac{4}{\mathrm{Re}}\frac{R'}{R} \simeq -\Lambda \implies \frac{dR}{d\tau} = -\frac{\Lambda\,\mathrm{Re}}{4}\,R \tag{18}$$

whose solution with $R(0) = 1$ is

$$R(\tau) \simeq \exp\left(-\frac{\Lambda\,\mathrm{Re}}{4}\,\tau\right) \tag{19}$$

It is not possible to satisfy both initial conditions: a first-order ODE admits only one, and (19) has $R'(0) = -\Lambda\mathrm{Re}/4 \neq 1$. The approximation is valid only after an initial transient, of duration of the order of Re in τ, during which the inertial terms $R\,R'' + \frac{3}{2}R'^2$ cannot be neglected and the initial velocity is dissipated by viscosity. In this limit the ambient pressure p_∞ crushes the cavity, which then collapses exponentially instead of expanding.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** (6) is the Rayleigh equation for the cavity with the viscous correction (8): the cavity decelerates because of the ambient pressure p_∞ (term Λ) and the viscous normal stress (term $4/(\mathrm{Re}\,R)$). For $\mathrm{Re}\to\infty$ the cavity stops at $R_{\max}$ and then collapses; for $\mathrm{Re}\to 0$ viscosity dominates.
- **Limits.** For $\Lambda\to 0$ the right-hand side of (13) is $R^{-3/2}$ with no stopping point: the cavity expands for ever in the inviscid, pressureless case. For $\Lambda\to\infty$, $R_{\max}\to 1$: the cavity cannot expand.
- **Dimensional check.** $R_c\ddot{R}_c$ and $\dot{R}_c^2$ have units $\mathrm{m}^2/\mathrm{s}^2$, as does p_∞/ρ (Pa/(kg/m^3)). $4\nu\dot{R}_c/R_c$ has units $(\mathrm{m}^2/\mathrm{s})(\mathrm{m}/\mathrm{s})/\mathrm{m}=\mathrm{m}^2/\mathrm{s}^2$. Re and Λ are dimensionless. $p_R=-4\mu\dot{R}_c/R_c$ has units Pa.
- **Consistency.** At $R=1$ the bracket of (13) equals 1 for every Λ, so $R'(0)=1$: the solution keeps the initial velocity V_0.

PROBLEM NS11

Couette Flow between Porous Plates

Original statement · NS11.pdf

Couette flow with injection | Pathline | Wall forces and heat fluxes | Integral balances

OFFICIAL PROBLEM STATEMENT

A perfect liquid of constant properties (density ρ, viscosity μ, thermal conductivity k and specific heat c) is confined between two infinite parallel plates at a distance h, as depicted in the figure below. The lower wall is at rest, while the upper one moves parallel to itself with a constant velocity U in the positive x-direction. The working fluid is injected through the lower porous wall into the channel with a uniform velocity v_l in the y-direction, while being removed from the channel through the upper porous plate with a uniform velocity v_u, also in the y-direction.

Gravitational forces can be neglected. The velocity field, $\bar{v} = u\bar{e}_x + v\bar{e}_y$, is assumed steady and independent of the x- and z-coordinates, $\partial/\partial t = \partial/\partial x = \partial/\partial z = 0$. The pressure at the upper wall has a constant value p_u. The lower and upper walls are kept at constant temperatures T_l and T_u, respectively. Known data: $(\rho, \mu, k, c, h, U, v_l, p_u, T_l, T_u)$.

1. (1 point). Obtain the only possible value of v_u consistent with the assumption of x-independence, and find $v(y)$.

2. (2.5 points). Determine the pressure field $p(x, y)$, and the horizontal velocity field, $u(y)$. Express $u(y)$ in terms of the Reynolds number based on the injection velocity, $\mathrm{Re} = \rho v_l h/\mu$.

3. (1 point). Compute the volume flux per unit spanwise length, Q, across a surface perpendicular to the plates (z is the spanwise direction). Particularize for the limit $\mathrm{Re} \ll 1$, and explain the result.

4. (1.5 points). Obtain the trajectory of the fluid particle with initial position $(x, y) = (0, x_0)$ at $t = 0$, and the corresponding pathline. Compute the time t_s at which the particle arrives at the upper wall, and the corresponding position x_s.

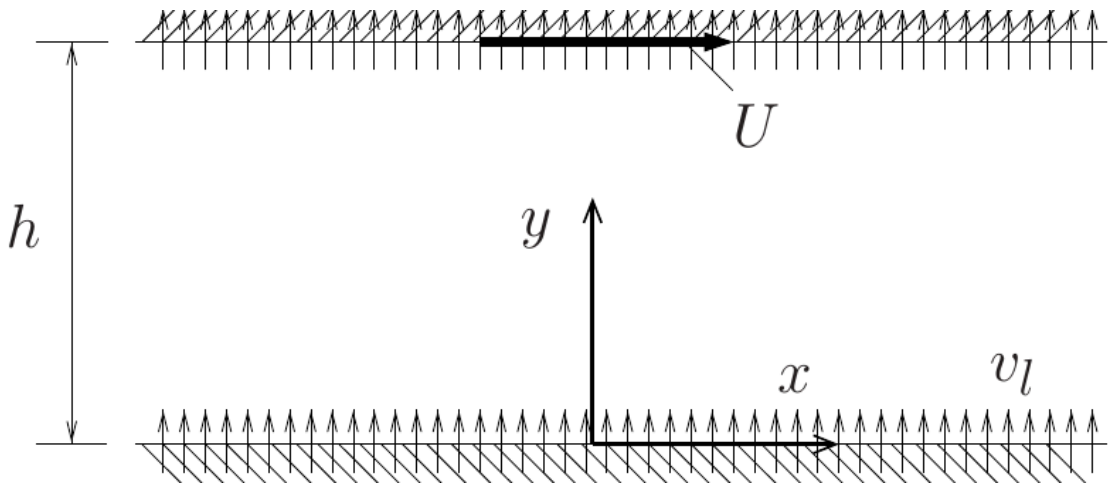
5. (1 point). Compute the viscous force per unit surface that the fluid exerts on the lower and upper walls. Particularize for the limit $\mathrm{Re} \ll 1$, and explain the result.

6. (2 points). Write the differential equation and boundary conditions that determine the temperature field, $T(y)$. Obtain $T(y)$ neglecting the effect of viscous dissipation.

7. (1 point). Compute the conductive heat flux (heat transferred by conduction p.u. surface) from the fluid to each wall.

8. (2 additional points). Using an appropriately defined control volume, perform integral momentum and energy balances, and discuss the results of questions 5 and 7 in the light of these integral balances.

The sheet is headed "INTRODUCTION TO FLUID MECHANICS, Jan 14, 2013, PROBLEM 2 (1 hour)".



Statement figure (crop of NS11.pdf, p. 1): channel of height h between two porous plates, injection velocity v_l through the lower wall at rest, upper wall moving with velocity U, axes x and y

`Source: NS11.pdf (page 1 of 3: statement; pages 2 and 3: handwritten solution). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41, energy Eq. 4.42, internal-energy balance Eq. 4.25, stress components in Cartesian coordinates.`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady planar flow of a perfect liquid (constant ρ, μ, k, c) between two porous plates, with $\partial/\partial x = \partial/\partial z = \partial/\partial t = 0$ for the velocity. Gravity neglected; no chemical reaction and no radiation.
- The lower wall ($y = 0$) is at rest at T_l and injects the liquid with the uniform normal velocity $v_l > 0$; the upper wall ($y = h$) moves with $U\bar{e}_x$ at T_u and the pressure on it is p_u. The tangential velocity of the liquid equals that of each wall (no-slip).
- Data: $(\rho,\ \mu,\ k,\ c,\ h,\ U,\ v_l,\ p_u,\ T_l,\ T_u)$. Unknowns: $v_u, v(y), p, u(y), Q$, the pathline, the wall forces, $T(y)$ and the wall heat fluxes.
- Dimensionless numbers: $\mathrm{Re} = \rho v_l h/\mu$ and $\mathrm{Pr} = \mu c/k$, so that $\rho c v_l h/k = \mathrm{Re}\,\mathrm{Pr}$.

Symbol	Meaning	SI unit
h	gap between the plates	m
U	velocity of the upper wall	m/s
v_l, v_u	injection and removal velocities	m/s

Symbol	Meaning	SI unit
p_u	pressure on the upper wall	Pa
T_l, T_u	wall temperatures	K
Re, Pr	injection Reynolds number, Prandtl number	1

PHASE 2

Frames, Field Form and Boundary Conditions

- Cartesian frame (x, y, z): x along the plates in the direction of U, y normal to the plates and upward from the lower wall, z spanwise. The basis is orthonormal and right-handed (no change of basis).
- Field form: $\bar{v} = u(y)\,\bar{e}_x + v(y)\,\bar{e}_y, T = T(y), p = p(x, y)$.
- Control volume for item 8: the box $0 < y < h$ and width Δx in x (unit spanwise length); surfaces: the lower wall Σ_l ($\bar{n} = -\bar{e}_y$), the upper wall Σ_u ($\bar{n} = +\bar{e}_y$) and the two faces Σ_1, Σ_2 ($\bar{n} = \mp\bar{e}_x$).

Location	Condition	Origin
$y = 0$	$u = 0, v = v_l, T = T_l$	lower wall at rest, injection
$y = h$	$u = U, v = v_u, T = T_u$	upper wall moving, removal
$y = h$	$p = p_u$	statement

PHASE 3

Step-by-Step Derivation

3.1 Continuity: v_u and $v(y)$ (item 1)

$$\underbrace{\cancel{\frac{\partial u}{\partial x}}}_{\partial_x = 0} + \frac{dv}{dy} = 0 \implies v = \text{const} \tag{1}$$

The constant is the injection velocity, $v(0) = v_l$, so $v = v_l$ for every y and the removal velocity must be the same:

$$\boxed{v_u = v_l, \qquad v(y) = v_l} \tag{2}$$

3.2 Pressure and horizontal velocity (item 2)

y-momentum, with $v = v_l$ constant ($dv/dy = 0$) and no gravity:

$$\rho\Big(\underbrace{u\frac{\partial v}{\partial x}}_{\partial_x = 0} + v\underbrace{\frac{dv}{dy}}_{=0}\Big) = -\frac{\partial p}{\partial y} + \mu\underbrace{\frac{d^2 v}{dy^2}}_{=0} \implies \frac{\partial p}{\partial y} = 0 \implies p = p(x) \tag{3}$$

The pressure on the upper wall is p_u for every x, so p is the constant p_u:

$$\boxed{p(x, y) = p_u \ \text{ for all } x,\ y}, \qquad \frac{\partial p}{\partial x} = 0 \tag{4}$$

x-momentum:

$$\rho\Big(\underbrace{u\frac{\partial u}{\partial x}}_{\partial_x = 0} + v_l\frac{du}{dy}\Big) = -\underbrace{\frac{\partial p}{\partial x}}_{=0} + \mu\frac{d^2 u}{dy^2} \implies \frac{d^2 u}{dy^2} - \frac{\mathrm{Re}}{h}\frac{du}{dy} = 0 \tag{5}$$

since $\rho v_l/\mu = \mathrm{Re}/h$. Trying $u = e^{\lambda y}$ gives $\lambda(\lambda - \mathrm{Re}/h) = 0$, so $u = C_1 + C_2\, e^{\mathrm{Re}\, y/h}$. The conditions $u(0) = 0$ and $u(h) = U$ give $C_1 = -C_2$ and $C_2(e^{\mathrm{Re}} - 1) = U$:

$$\boxed{u(y) = U\,\frac{e^{\mathrm{Re}\, y/h} - 1}{e^{\mathrm{Re}} - 1}} \tag{6}$$

For $\mathrm{Re} \to 0$, the Taylor expansion $e^x = 1 + x + O(x^2)$ gives $u \to U\,\dfrac{\mathrm{Re}\, y/h}{\mathrm{Re}} = U\,\dfrac{y}{h}$: plane Couette flow is recovered.

3.3 Volume flux (item 3)

The flux per unit spanwise length through a section $x = \text{const}$ is $Q = \int_0^h u\,dy$. The antiderivative of $e^{\mathrm{Re}\,y/h} - 1$ is $(h/\mathrm{Re})\,e^{\mathrm{Re}\,y/h} - y$, and Barrow's rule between 0 and h gives $(h/\mathrm{Re})(e^{\mathrm{Re}} - 1) - h$:

$$\boxed{Q = \frac{U}{e^{\mathrm{Re}} - 1}\left[\frac{h}{\mathrm{Re}}\left(e^{\mathrm{Re}} - 1\right) - h\right] = Uh\left[\frac{1}{\mathrm{Re}} - \frac{1}{e^{\mathrm{Re}} - 1}\right]} \tag{7}$$

For $\mathrm{Re} \ll 1$, $e^{\mathrm{Re}} - 1 = \mathrm{Re}\left(1 + \mathrm{Re}/2 + \mathrm{Re}^2/6 + \dots\right)$, so $1/(e^{\mathrm{Re}} - 1) = 1/\mathrm{Re} - 1/2 + \mathrm{Re}/12 + \dots$ and

$$Q \simeq Uh\left(\frac{1}{2} - \frac{\mathrm{Re}}{12}\right) \to \frac{Uh}{2} \tag{8}$$

The flux tends to that of plane Couette flow, the mean velocity $U/2$ times the gap: for small injection velocity the transverse flow barely distorts the linear profile.

3.4 Trajectory and pathline (item 4)

A particle moves with the local velocity: $dy/dt = v_l$ and $dx/dt = u(y)$. Let its initial position be (x_i, y_i) at $t = 0$. Then $y = y_i + v_l t$ and, with $dt = dy/v_l$,

$$x - x_i = \int_0^t u\,dt' = \frac{U}{v_l\,(e^{\mathrm{Re}} - 1)}\int_{y_i}^{y}\left(e^{\mathrm{Re}\,y'/h} - 1\right)dy' = \frac{U}{v_l\,(e^{\mathrm{Re}} - 1)}\left[\frac{h}{\mathrm{Re}}\left(e^{\mathrm{Re}\,y/h} - e^{\mathrm{Re}\,y_i/h}\right) - (y - y_i)\right] \tag{9}$$

which is the pathline $x(y)$; with $y = y_i + v_l t$ it is also the trajectory $x(t)$. The particle reaches the upper wall at $y = h$:

$$t_s = \frac{h - y_i}{v_l}, \qquad x_s = x_i + \frac{U}{v_l\,(e^{\mathrm{Re}} - 1)}\left[\frac{h}{\mathrm{Re}}\left(e^{\mathrm{Re}} - e^{\mathrm{Re}\,y_i/h}\right) - (h - y_i)\right] \tag{10}$$

Reading of the initial position

The sheet writes the initial position as $(x, y) = (0, x_0)$; the handwritten solution takes the particle at the lower wall, $(x, y) = (x_0, 0)$. Both follow from (9) and (10): for the printed values $(x_i, y_i) = (0, x_0)$ and for the reading of the key $(x_i, y_i) = (x_0, 0)$, which gives $t_s = h/v_l$.

$$\boxed{x - x_0 = \frac{U}{e^{\mathrm{Re}} - 1}\left[\frac{h}{\mathrm{Re}\, v_l}\left(e^{\mathrm{Re}\, v_l t/h} - 1\right) - t\right], \quad t_s = \frac{h}{v_l}, \quad x_s = x_0 + \frac{Uh}{v_l}\left[\frac{1}{\mathrm{Re}} - \frac{1}{e^{\mathrm{Re}} - 1}\right]} \qquad (y_i = 0) \tag{11}$$

Note that $x_s - x_0 = Q/v_l$: the particle crosses the channel in the time h/v_l at the mean horizontal velocity Q/h.

3.5 Viscous forces on the walls (item 5)

The normal pointing into the liquid is $+\bar{e}_y$ on the lower wall and $-\bar{e}_y$ on the upper wall; $(-p\bar{\bar{I}} + \bar{\bar{\tau}}') \cdot \bar{n}$ is the force per unit area that the liquid exerts on the wall. The viscous components are $\tau'_{xy} = \mu\, du/dy$ and $\tau'_{yy} = 2\mu\, dv/dy = 0$ (Notes.pdf, stress components, Cartesian coordinates). From (6), $du/dy = \dfrac{U\,\mathrm{Re}}{h}\dfrac{e^{\mathrm{Re}\, y/h}}{e^{\mathrm{Re}} - 1}$:

$$\bar{f}_l^{\,\mathrm{visc}} = \bar{\bar{\tau}}' \cdot \bar{e}_y = \mu \frac{du}{dy}\Big|_0 \bar{e}_x = \frac{\mu U\,\mathrm{Re}}{h\,(e^{\mathrm{Re}} - 1)}\bar{e}_x, \qquad \bar{f}_u^{\,\mathrm{visc}} = \bar{\bar{\tau}}' \cdot (-\bar{e}_y) = -\mu \frac{du}{dy}\Big|_h \bar{e}_x = -\frac{\mu U\,\mathrm{Re}\, e^{\mathrm{Re}}}{h\,(e^{\mathrm{Re}} - 1)}\bar{e}_x \tag{12}$$

$$\boxed{\bar{f}_l = \frac{\mu U\,\mathrm{Re}}{h\,(e^{\mathrm{Re}} - 1)}\,\bar{e}_x, \qquad \bar{f}_u = -\frac{\mu U\,\mathrm{Re}\, e^{\mathrm{Re}}}{h\,(e^{\mathrm{Re}} - 1)}\,\bar{e}_x} \tag{13}$$

For $\mathrm{Re} \ll 1$, $\mathrm{Re}/(e^{\mathrm{Re}} - 1) \to 1$ and $\mathrm{Re}\, e^{\mathrm{Re}}/(e^{\mathrm{Re}} - 1) \to 1$: $\bar{f}_l \to \dfrac{\mu U}{h}\bar{e}_x$ and $\bar{f}_u \to -\dfrac{\mu U}{h}\bar{e}_x$, the equal and opposite shear forces of plane Couette flow. For $\mathrm{Re} \neq 0$ they are not equal: the lower wall, where the injected liquid has no horizontal velocity, feels less shear than the upper one.

3.6 Temperature (item 6)

The energy equation (4.42) with $v = v_l$ and $\partial_x = 0$, and the boundary conditions, are

$$\rho c\, v_l \frac{dT}{dy} = \mu \left(\frac{du}{dy}\right)^2 + k\frac{d^2T}{dy^2}, \qquad T(0) = T_l, \quad T(h) = T_u \tag{14}$$

where $\phi_v = \mu(du/dy)^2$ is the viscous dissipation. Neglecting it, and dividing by k with $\rho c v_l h/k = \mathrm{Re}\,\mathrm{Pr}$:

$$\frac{d^2T}{dy^2} - \frac{\mathrm{Re}\,\mathrm{Pr}}{h}\frac{dT}{dy} = 0 \implies T = B_1 + B_2\, e^{\mathrm{Re}\,\mathrm{Pr}\, y/h} \tag{15}$$

the same equation as (5) with Re replaced by $\mathrm{Re}\,\mathrm{Pr}$. The conditions $T(0) = T_l$ and $T(h) = T_u$ give $B_1 + B_2 = T_l$ and $B_2(e^{\mathrm{Re}\,\mathrm{Pr}} - 1) = T_u - T_l$:

$$\boxed{\frac{T - T_l}{T_u - T_l} = \frac{e^{\mathrm{Re}\,\mathrm{Pr}\, y/h} - 1}{e^{\mathrm{Re}\,\mathrm{Pr}} - 1}} \tag{16}$$

3.7 Heat flux to each wall (item 7)

The heat flux from the liquid to a wall is $\bar{q}\cdot\bar{n}$ with $\bar{n}$ pointing out of the liquid and $\bar{q} = -k\nabla T$. From (16), $dT/dy = (T_u - T_l)\dfrac{\mathrm{Re}\,\mathrm{Pr}}{h}\dfrac{e^{\mathrm{Re}\,\mathrm{Pr}\, y/h}}{e^{\mathrm{Re}\,\mathrm{Pr}} - 1}$, and $k\,\mathrm{Re}\,\mathrm{Pr}/h = \rho c v_l$:

$$\boxed{q_u = -k\frac{dT}{dy}\Big|_h = -\rho c v_l\,(T_u - T_l)\,\frac{e^{\mathrm{Re}\,\mathrm{Pr}}}{e^{\mathrm{Re}\,\mathrm{Pr}} - 1}, \qquad q_l = k\frac{dT}{dy}\Big|_0 = \rho c v_l\,(T_u - T_l)\,\frac{1}{e^{\mathrm{Re}\,\mathrm{Pr}} - 1}} \tag{17}$$

For $\mathrm{Re}\,\mathrm{Pr} \to 0$, $q_l \to k(T_u - T_l)/h$ and $q_u \to -k(T_u - T_l)/h$: pure conduction between the walls.

3.8 Integral balances (item 8)

Momentum, x component (Notes.pdf, integral momentum balance for the fixed volume, steady):

$$\int_\Sigma \rho\, u\, (\bar v \cdot \bar n)\, d\sigma = -\int_\Sigma p\, n_x\, d\sigma + \int_\Sigma (\bar{\bar\tau}' \cdot \bar n)_x\, d\sigma$$

Surface	$\int \rho u\, \bar v \cdot \bar n\, d\sigma$	$-\int p\, n_x\, d\sigma$	$\int (\bar{\bar\tau}' \cdot \bar n)_x\, d\sigma$
Σ_l	0 ($u = 0$)	0 ($n_x = 0$)	$-\mu\, u'(0)\, \Delta x$
Σ_u	$\rho U\, v_l\, \Delta x$	0 ($n_x = 0$)	$+\mu\, u'(h)\, \Delta x$
$\Sigma_1 + \Sigma_2$	cancel	cancel (p uniform)	cancel ($\tau'_{xx} = 0$)

$$\rho\, v_l\, U = \mu \left[\left.\frac{du}{dy}\right|_h - \left.\frac{du}{dy}\right|_0 \right] \tag{18}$$

Using (13), this is $\rho v_l U = -(f_{u,x} + f_{l,x})$: the net horizontal force that the liquid exerts on the two walls is $-\rho v_l U$, because the liquid leaves through the upper wall carrying the horizontal momentum ρU per unit mass, which it does not have when it enters through the lower one, and the walls supply it. Equation (18) holds exactly for every Re (indeed $\mu\, U\, \mathrm{Re}/h = \rho v_l U$) and confirms (13); for $\mathrm{Re} \to 0$ the right-hand side vanishes and the two shear forces cancel.

Internal energy (Notes.pdf, Eq. 4.25, steady, no compression work, dissipation neglected): $\int_\Sigma \rho c\, T\, (\bar v \cdot \bar n)\, d\sigma = -\int_\Sigma \bar q \cdot \bar n\, d\sigma$. The faces Σ_1, Σ_2 cancel and the walls give

$$\rho c\, v_l\, (T_u - T_l)\, \Delta x = -(q_u + q_l)\, \Delta x \implies q_u + q_l = -\rho c\, v_l\, (T_u - T_l) \tag{19}$$

which agrees with (17): the sum of the heat conducted out through the two walls balances the net enthalpy flux that the liquid carries across the control volume, $\rho c v_l (T_u - T_l)$, so that the conductive heat flux balances the net internal-energy flux through the volume.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The injection convects momentum and heat across the channel: for $\mathrm{Re} > 0$ the profiles are exponential, with the velocity gradient concentrated near the upper wall (towards which the liquid is carried), and the thermal profile has the same form with $\mathrm{Re}\,\mathrm{Pr}$.
- **Limits.** For $\mathrm{Re} \to 0$: $u = Uy/h$, $Q = Uh/2$, $\bar{f}_l = -\bar{f}_u = \mu U/h\,\bar{e}_x$ and the temperature is linear (Couette flow with conduction). For $\mathrm{Re} \gg 1$, $u \simeq U\, e^{\mathrm{Re}\,(y-h)/h}$: the horizontal velocity is confined to a layer of thickness h/Re at the upper wall and $Q \simeq Uh/\mathrm{Re} \to 0$, since the injected liquid, which starts at rest, is swept to the upper wall before it can be dragged.
- **Dimensional check.** $\rho v_l h/\mu$ is dimensionless; Uh has units of a flux per unit span, m^2/s; $\mu U\,\mathrm{Re}/h$ is $\mathrm{N/m}^2$; $\rho c v_l\,\Delta T$ is $\mathrm{W/m}^2$.
- **Consistency of the limits.** At $\mathrm{Re} = 0$, (18) gives $0 = 0$ and (19) gives $q_u = -q_l$.

PROBLEM NS12

Flow between Concentric Spheres

Original statement · NS12.pdf

Dimensional analysis | Spherical coordinates | Separation of variables | Volume flux

OFFICIAL PROBLEM STATEMENT

A very viscous liquid (of constant density ρ and viscosity μ) flows steadily through the gap between two concentric spheres, as shown in the figure. The flow is driven by a given pressure difference between the entrance and exit section, $\Delta p = p_i - p_o$, and gravity effects are negligible. We wish to determine the resulting volume flux Q as a function of the known geometry data, fluid properties and pressure difference, $Q(R, \kappa, \epsilon, \rho, \mu, \Delta p)$.

1. Apply the dimensional analysis technique to write the parametric dependence of Q as a function of the minimum number of parameters.

2. Simplify the non-dimensional relation for the case of flow dominated by viscous forces, for which the liquid inertia is negligible.

3. Disregarding end effects in the entrance and exit regions, we can assume that the flow is unidirectional and symmetric, $\bar{v} = v_\theta(r, \theta)\bar{e}_\theta$ (note that $v_r = v_\phi = 0$ and $\partial/\partial\phi = 0$). With this assumption, write the simplified continuity and momentum equations in spherical coordinates, as well as the necessary boundary conditions to solve the system. Write the expression to evaluate the volume flux Q and then check that the parametric dependence stated earlier, that is $Q(R, \kappa, \epsilon, \rho, \mu, \Delta p)$, is correct.

4. Simplify the θ-momentum equation with the assumption that the velocity is small enough for the viscous terms to be dominant over the convective terms. Under this conditions, the radial pressure variations are negligible compared with Δp and we can consider that the pressure is a function of θ alone, $p(\theta)$.

5. Write the mathematical formulation (equations and boundary conditions) in non-dimensional form. Use the non-dimensional variables $\tilde{r} = r/R$, $\tilde{v}_\theta = v_\theta/V_\theta$ and $\tilde{p} = (p - p_o)/\Delta p$, where you can define V_θ such that there are no parameters in the non-dimensional equation.

6. Using the continuity equation, show that $\tilde{v}_\theta \operatorname{sen}\theta = \tilde{u}(\tilde{r})$, where $\tilde{u}(\tilde{r})$ is a function of $\tilde{r}$ to be determined.

7. Rewrite the θ-momentum equation in terms of the function $\tilde{u}(\tilde{r})$ instead of $\tilde{v}_\theta$ and separate variables to show that

$$\operatorname{sen}\theta \, \frac{d\tilde{p}}{d\theta} = C$$

$$\frac{1}{\tilde{r}}\frac{d}{d\tilde{r}}\left(\tilde{r}^2\frac{d\tilde{u}}{d\tilde{r}}\right)=C$$

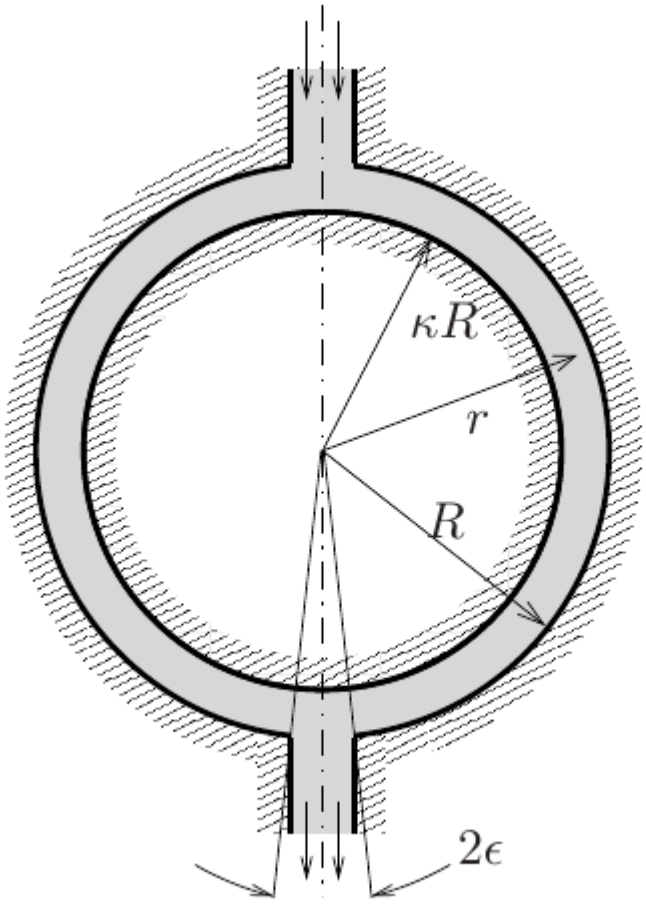
where C is a constant.

8. Integrate the ODE for the pressure and determine the value of the constant C Note that the boundary conditions for the pressure are to be applied at $\theta=\epsilon$ and $\theta=\pi-\epsilon$.

9. With the known value of C, integrate the ode for $\tilde{u}(\tilde{r})$ obtained in item and thus obtain the velocity profile $\tilde{v}_\theta(\tilde{r},\theta)$.

10. Define a proper non-dimensional form for the volume flux $\tilde{Q}$ and obtain it by integration of the velocity profile.

Transcribed as printed (exam of June 27, 2013, "PROBLEM 1 (90 minutes)"), including "Under this conditions", the missing full stop after "constant C" in item 8 and the missing item number in item 9.



Statement figure (crop of NS12.pdf, p. 1): gap between two concentric spheres of radii κR and R, entrance and exit sections of half-angle ϵ and the radial coordinate r

Source: NS12.pdf (page 1 of 4: statement; pages 2 to 4: handwritten solution). Notes.pdf, Chapter 4: continuity Eq. 4.40 and momentum Eq. 4.41 in spherical coordinates; dimensional analysis of Unit 6 is used for item 1.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady flow of a very viscous incompressible liquid (constant ρ, μ), gravity neglected. The liquid flows through the gap $\kappa R \le r \le R$ between two concentric spheres, from an entrance section (polar angle $0 \le \theta \le \epsilon$ of the outer sphere, pressure p_i) to an exit section ($\pi - \epsilon \le \theta \le \pi$, pressure p_o).
- Away from the end regions the flow is unidirectional and symmetric: $\bar{v} = v_\theta(r,\theta)\,\bar{e}_\theta$, $v_r = v_\phi = 0, \partial/\partial\phi = 0$; the model domain is $\kappa R \le r \le R, \epsilon \le \theta \le \pi - \epsilon$.
- Data: $[R,\ \kappa,\ \epsilon,\ \rho,\ \mu,\ \Delta p]$ with $0 < \kappa < 1$. Unknowns: $Q, v_\theta(r,\theta), p(\theta)$.
- The "viscous" regime is the limit in which the dimensionless group $\rho\,\Delta p\,R^2/\mu^2$ is small.

Symbol	Meaning	SI unit
R, κR	outer and inner sphere radii	m
ϵ	half-angle of the entrance and exit sections	rad
$\Delta p = p_i - p_o$	driving pressure difference	Pa
Q	volume flux through the gap	m^3/s
V_θ	velocity scale, defined in item 5	m/s
ρ, μ	density and viscosity	kg/m^3, Pa s

PHASE 2

Frames, Field Form and Boundary Conditions

- Spherical frame (r,θ,ϕ) centred on the common centre of the spheres, with the axis through the entrance and exit; the basis $\{\bar{e}_r, \bar{e}_\theta, \bar{e}_\phi\}$ is orthonormal and right-handed.
- Field form: $\bar{v} = v_\theta(r,\theta)\,\bar{e}_\theta, p = p(r,\theta)$ (reduced to $p(\theta)$ in item 4).

Location	Condition	Origin
$r = \kappa R$	$v_\theta = 0$	no-slip on the inner sphere
$r = R$	$v_\theta = 0$	no-slip on the outer sphere
$\theta = \epsilon$	$p = p_i$	entrance section

Location	Condition	Origin
$\theta = \pi - \epsilon$	$p = p_o$	exit section

PHASE 3

Step-by-Step Derivation

3.1 Dimensional analysis (items 1 and 2)

The variables of the problem are Q, R, κ, ϵ, ρ, μ, Δp. The dimensional ones, Q $[\mathrm{L^3T^{-1}}]$, R $[\mathrm{L}]$, ρ $[\mathrm{ML^{-3}}]$, μ $[\mathrm{ML^{-1}T^{-1}}]$ and Δp $[\mathrm{ML^{-1}T^{-2}}]$, are five and involve three fundamental dimensions (M, L, T), so by the Vaschy-Buckingham theorem there are $5 - 3 = 2$ dimensionless groups from them; κ and ϵ are already dimensionless. Choosing R, μ and Δp as repeated variables:

$$\Pi_1 = \frac{Q\,\mu}{\Delta p\, R^3}, \qquad \Pi_2 = \frac{\rho\,\Delta p\, R^2}{\mu^2}$$

Check: $\Pi_1 = (\mathrm{L^3T^{-1}})(\mathrm{ML^{-1}T^{-1}})/[(\mathrm{ML^{-1}T^{-2}})\,\mathrm{L^3}] = 1$ and $\Pi_2 = (\mathrm{ML^{-3}})(\mathrm{ML^{-1}T^{-2}})\,\mathrm{L^2}/(\mathrm{M^2L^{-2}T^{-2}}) = 1$. The minimum parametric dependence is

$$\boxed{\frac{Q\,\mu}{\Delta p\, R^3} = f\left(\kappa,\ \epsilon,\ \frac{\rho\,\Delta p\, R^2}{\mu^2}\right)} \tag{1}$$

The group Π_2 is a Reynolds number: with the viscous velocity $V = \Delta p\, R/\mu$, $\rho V R/\mu = \rho \Delta p R^2/\mu^2$. When viscous forces dominate and the inertia of the liquid is negligible, the density disappears from the problem and the dependence on Π_2 is lost:

$$\boxed{\frac{Q\,\mu}{\Delta p\, R^3} = f(\kappa,\ \epsilon)} \tag{2}$$

3.2 Equations and boundary conditions (item 3)

Continuity in spherical coordinates with $v_r = v_\phi = 0$ and $\partial_\phi = 0$:

$$\frac{1}{r\sin\theta}\frac{\partial(v_\theta \sin\theta)}{\partial\theta} = 0 \implies \frac{\partial(v_\theta \sin\theta)}{\partial\theta} = 0 \tag{3}$$

Momentum (steady, no mass force), the three components of (4.41) with $v_r = v_\phi = 0$. In the viscous terms of the θ component, $\frac{1}{r^2}\frac{\partial}{\partial\theta}\left[\frac{1}{\sin\theta}\frac{\partial(v_\theta\sin\theta)}{\partial\theta}\right]$ contains the left side of (3), so it vanishes; the terms with v_r, v_ϕ and ∂_ϕ vanish too. The r component keeps the centrifugal term and the ϕ component only the pressure derivative:

$$-\rho\frac{v_\theta^2}{r} = -\frac{\partial p}{\partial r} \tag{4}$$

$$\rho\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial\theta} = -\frac{1}{r}\frac{\partial p}{\partial\theta} + \mu\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial v_\theta}{\partial r}\right) \tag{5}$$

$$0 = -\frac{1}{r\sin\theta}\frac{\partial p}{\partial\phi} \implies \frac{\partial p}{\partial\phi} = 0 \tag{6}$$

Boundary conditions (table of Phase 2): $v_\theta(\kappa R,\theta) = 0$, $v_\theta(R,\theta) = 0$, $p = p_i$ at $\theta = \epsilon$ and $p = p_o$ at $\theta = \pi - \epsilon$. Writing $p' = p - p_o$ (so $dp' = dp$), these are $p'(\theta = \epsilon) = \Delta p$ and $p'(\theta = \pi - \epsilon) = 0$.

Volume flux. Through the equatorial section $\theta = \pi/2$, the annulus $\kappa R \le r \le R$ with area element $2\pi r\,dr$:

$$Q = \int_{\kappa R}^{R} v_\theta\left(r, \tfrac{\pi}{2}\right) 2\pi r\,dr \tag{7}$$

Check of the parametric dependence. The system (3)-(5) with its boundary conditions contains only ρ, μ, R, κ, ϵ and Δp, so $Q = Q(R, \kappa, \epsilon, \rho, \mu, \Delta p)$, as stated.

3.3 Viscous-dominated flow (item 4)

Estimate the sizes of the terms of (5): the pressure term is $\Delta p / R$, the viscous term is $\mu V_\theta / R^2$ and the convective term is $\rho V_\theta^2 / R$. The first two balance if $V_\theta \sim \Delta p \, R / \mu$, and the ratio of the convective term to the pressure term is then $\rho V_\theta^2 / \Delta p \sim \rho \Delta p R^2 / \mu^2 = \Pi_2 \ll 1$. The convective term is dropped. In (4) the radial pressure variation is of order ρV_θ^2, also small against Δp by the same factor, so $p = p(\theta)$ and (5) becomes

$$0 = -\frac{1}{r}\frac{dp}{d\theta} + \frac{\mu}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial v_\theta}{\partial r} \right) \tag{8}$$

3.4 Non-dimensional formulation (item 5)

With $\tilde r = r/R$, $\tilde v_\theta = v_\theta / V_\theta$ and $\tilde p = (p - p_o)/\Delta p$, the pressure term of (8) is $(\Delta p / (R\tilde r))\, d\tilde p / d\theta$ and the viscous term is $(\mu V_\theta / R^2)(1/\tilde r^2)\, \partial(\tilde r^2 \partial \tilde v_\theta / \partial \tilde r)/\partial \tilde r$. Choosing $V_\theta = \Delta p \, R / \mu$ makes the two coefficients equal, and multiplying by $\tilde r$:

$$0 = -\frac{d\tilde p}{d\theta} + \frac{1}{\tilde r}\frac{\partial}{\partial \tilde r}\left(\tilde r^2 \frac{\partial \tilde v_\theta}{\partial \tilde r} \right), \qquad \frac{\partial (\tilde v_\theta \sin\theta)}{\partial \theta} = 0 \tag{9}$$

with the boundary conditions

$$\tilde v_\theta(\kappa, \theta) = \tilde v_\theta(1, \theta) = 0, \qquad \tilde p(\epsilon) = 1, \qquad \tilde p(\pi - \epsilon) = 0 \tag{10}$$

and the flux

$$\tilde Q = \frac{Q\,\mu}{\Delta p \, R^3} = 2\pi \int_\kappa^1 \tilde v_\theta\left(\tilde r, \tfrac{\pi}{2}\right) \tilde r \, d\tilde r \tag{11}$$

The problem has no parameters except κ and ϵ (through the boundary conditions and the integration limits), so $\tilde Q = \tilde Q(\kappa, \epsilon)$, in agreement with (2).

3.5 Continuity and separation of variables (items 6 and 7)

From (9), $\partial(\tilde{v}_\theta \sin\theta)/\partial\theta = 0$, so $\tilde{v}_\theta \sin\theta$ does not depend on θ:

$$\tilde{v}_\theta \, \sin\theta = \tilde{u}(\tilde{r}) \tag{12}$$

Substituting $\tilde{v}_\theta = \tilde{u}/\sin\theta$ in the momentum equation of (9):

$$\frac{d\tilde{p}}{d\theta} = \frac{1}{\tilde{r}\sin\theta}\frac{d}{d\tilde{r}}\left(\tilde{r}^2\frac{d\tilde{u}}{d\tilde{r}}\right) \implies \sin\theta\,\frac{d\tilde{p}}{d\theta} = \frac{1}{\tilde{r}}\frac{d}{d\tilde{r}}\left(\tilde{r}^2\frac{d\tilde{u}}{d\tilde{r}}\right) \tag{13}$$

The left side depends only on θ and the right side only on $\tilde{r}$, so both are equal to a constant C (which may depend on κ and ϵ):

$$\boxed{\sin\theta\,\frac{d\tilde{p}}{d\theta} = C, \qquad \frac{1}{\tilde{r}}\frac{d}{d\tilde{r}}\left(\tilde{r}^2\frac{d\tilde{u}}{d\tilde{r}}\right) = C} \tag{14}$$

3.6 Pressure and the constant C (item 8)

From (14), $d\tilde{p}/d\theta = C/\sin\theta$. Integrating from $\theta = \epsilon$ (where $\tilde{p} = 1$) to $\theta = \pi - \epsilon$ (where $\tilde{p} = 0$):

$$0 - 1 = C\int_{\epsilon}^{\pi-\epsilon}\frac{d\theta}{\sin\theta} \equiv C\,E(\epsilon) \implies \boxed{C = -\frac{1}{E(\epsilon)}} \tag{15}$$

The antiderivative of $1/\sin\theta$ is $\ln\tan(\theta/2)$ (its derivative is $\dfrac{1}{2\sin(\theta/2)\cos(\theta/2)} = \dfrac{1}{\sin\theta}$), and Barrow's rule with $\tan\left(\frac{\pi-\epsilon}{2}\right) = \cot(\epsilon/2)$ gives

$$E(\epsilon) = \left[\ln\tan\frac{\theta}{2}\right]_{\epsilon}^{\pi-\epsilon} = \ln\cot\frac{\epsilon}{2} - \ln\tan\frac{\epsilon}{2} = -2\ln\tan\frac{\epsilon}{2} = \ln\frac{1+\cos\epsilon}{1-\cos\epsilon} \tag{16}$$

(using $\tan^2(\epsilon/2) = (1-\cos\epsilon)/(1+\cos\epsilon)$). The pressure is $\tilde{p}(\theta) = 1 - [\ln\tan(\theta/2) - \ln\tan(\epsilon/2)]\,/E(\epsilon)$.

3.7 Velocity profile (item 9)

The second equation of (14): $\dfrac{d}{d\tilde{r}}\left(\tilde{r}^2\tilde{u}'\right) = C\tilde{r}$. Integrating, $\tilde{r}^2\tilde{u}' = C\tilde{r}^2/2 + A$, so $\tilde{u}' = C/2 + A/\tilde{r}^2$, and again $\tilde{u} = C\tilde{r}/2 - A/\tilde{r} + B$. The conditions $\tilde{u}(1) = 0$ and $\tilde{u}(\kappa) = 0$ (from $\tilde{v}_\theta = 0$ at both spheres) give

$$\frac{C}{2} - A + B = 0, \qquad \frac{C\kappa}{2} - \frac{A}{\kappa} + B = 0$$

Subtracting, $\frac{C}{2}(1-\kappa) - A\left(1 - \frac{1}{\kappa}\right) = 0$, hence $A = -C\kappa/2$ and $B = A - C/2 = -C(1+\kappa)/2$:

$$\tilde{u}(\tilde{r}) = -\frac{C}{2}\left[1 - \tilde{r} + \kappa\left(1 - \frac{1}{\tilde{r}}\right)\right] = -\frac{C}{2}\,\frac{(1-\tilde{r})(\tilde{r}-\kappa)}{\tilde{r}} \tag{17}$$

and, with (15) and $\tilde{v}_\theta = \tilde{u}/\sin\theta$,

$$\boxed{\tilde{v}_\theta(\tilde{r},\theta) = \frac{1}{2E(\epsilon)\,\sin\theta}\left[1 - \tilde{r} + \kappa\left(1 - \frac{1}{\tilde{r}}\right)\right]} \tag{18}$$

The bracket equals $(1-\tilde{r})(\tilde{r}-\kappa)/\tilde{r}$, which is positive in $\kappa < \tilde{r} < 1$: the flow goes from the entrance to the exit, with the largest velocity at the equator ($\sin\theta = 1$).

3.8 Volume flux (item 10)

The non-dimensional flux is $\tilde{Q} = Q\mu/(\Delta p R^3)$, (11). At $\theta = \pi/2$, $\sin\theta = 1$ and $\tilde{v}_\theta = \tilde{u}$:

$$\tilde{Q} = 2\pi\int_\kappa^1 \tilde{u}\,\tilde{r}\,d\tilde{r} = \frac{\pi}{E(\epsilon)}\int_\kappa^1 (1-\tilde{r})\,(\tilde{r}-\kappa)\,d\tilde{r} \tag{19}$$

The integrand is $-\tilde{r}^2 + (1+\kappa)\tilde{r} - \kappa$, with antiderivative $-\tilde{r}^3/3 + (1+\kappa)\tilde{r}^2/2 - \kappa\tilde{r}$; Barrow's rule between κ and 1 gives $\dfrac{(1-\kappa)^3}{6}$ (check: $\int_a^b (b-r)(r-a)\,dr = (b-a)^3/6$). Therefore

$$\boxed{\tilde{Q} = \frac{\pi\,(1-\kappa)^3}{6\,E(\epsilon)}, \qquad Q = \frac{\pi\,\Delta p\,R^3\,(1-\kappa)^3}{6\,\mu\,E(\epsilon)}}, \qquad E(\epsilon) = \ln\frac{1+\cos\epsilon}{1-\cos\epsilon} \tag{20}$$

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The flow in the gap is a Poiseuille-like flow along the meridians: the pressure drop Δp is distributed as $\ln\tan(\theta/2)$ along θ, and across the gap the velocity has the parabola-like shape (17) that vanishes at both spheres. The flux is proportional to $\Delta p/\mu$, as every Stokes flow, and to the cube of the gap, $(1-\kappa)^3 R^3$.
- **Limits.** For $\kappa \to 1$ the gap closes and $Q \to 0$ as $(1-\kappa)^3$. For $\epsilon \to 0$ the end sections shrink and $E(\epsilon) \simeq 2\ln(2/\epsilon) \to \infty$, so Q vanishes only logarithmically. For $\epsilon \to \pi/2$ the sections cover the whole hemispheres and $E \to 0$ makes $Q \to \infty$, which is the breakdown of the unidirectional model (the end effects it neglects then control the flow).
- **Validity.** The result requires $\rho \Delta p R^2/\mu^2 \ll 1$ (item 4).
- **Dimensional check.** $\Delta p R^3/\mu$ has units $(\mathrm{N/m^2})\,\mathrm{m^3}/(\mathrm{Pa\,s}) = \mathrm{m^3/s}$, the units of Q. $V_\theta = \Delta p R/\mu$ has units m/s. $E(\epsilon)$ and $\tilde{Q}$ are dimensionless.
- **Consistency of the profile.** The flux through the cone $\theta = \text{const}$ (area element $2\pi\tilde{r}\sin\theta\,d\tilde{r}$, normal velocity $\tilde{v}_\theta$) is $2\pi\int_\kappa^1 \tilde{v}_\theta \sin\theta\,\tilde{r}\,d\tilde{r} = 2\pi\int_\kappa^1 \tilde{u}\,\tilde{r}\,d\tilde{r}$, independent of θ as continuity requires; item 10 evaluates it at $\theta = \pi/2$, where the cone is the equatorial plane.